A COMPACT ILLUSTRATED GUIDE

Emergency Nursing Fundamentals

A Compact Illustrated Guide to Rapid Assessment, Priorities and Procedures for Nursing, Paramedic, EMT and Allied Health Students

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ABOUT THIS BOOK

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Scope. The book covers basic, non-invasive emergency skills only. It gives no medicine doses (verified emergency-procedure figures, such as the rate of chest compressions, are given), and it does not teach intravenous access, electrocardiogram interpretation or advanced airway skills. Medicines are named only from a short list, and they are described only as far as helping a patient use their own prescribed device or using a product your protocol names, never with amounts. Giving medicines is outside many roles; follow your scope and medical direction.

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Illustrative cases. The cases in this book are fictional composites written for teaching. They do not describe real patients, responders or places.

Use of artificial intelligence. The text, the review questions, the captions, the glossary and the illustrations in this book were generated with artificial intelligence tools. Of the 25 figures, 15 are illustrations generated with an image tool and 10 are diagrams and flow charts written as vector drawings by an artificial intelligence tool. Labels and arrows were added to the illustrations in a separate drawing layer by the same process. The author directed the project and is responsible for what is published.

Contents

How to Use This Book

Part 1 The Emergency Approach: Safety, Triage and Assessment

1.1 Scene Safety, Personal Protection and Responder Wellbeing

1.2 Triage and Acuity

1.3 The Primary Survey

1.4 History, Vital Signs and the Secondary Survey

1.5 Communication, Consent, Scope of Practice, Handover, Reassessment and Documentation

Part 2 Airway and Breathing Emergencies

2.1 Airway Anatomy and Assessment

2.2 Airway Obstruction and Basic Airway Management

2.3 Ventilation Support: Bag-Valve-Mask and Oxygen Delivery

2.4 Respiratory Distress: Asthma, COPD and Pneumonia

2.5 Anaphylaxis

Part 3 Cardiac Emergencies and Resuscitation

3.1 The Heart in Brief: Pump, Vessels and Electrical System

3.2 Acute Chest Pain and Acute Coronary Syndrome

3.3 Recognizing Dangerous Rhythms

3.4 Cardiac Arrest, High-Quality CPR and the AED

3.5 Shock: Recognition and First Priorities

Part 4 Trauma, Burns and Environmental Emergencies

4.1 Trauma Assessment and Mechanism of Injury

4.2 Bleeding and Hemorrhage Control

4.3 Head, Spine and Chest Injuries

4.4 Fractures, Dislocations and Splinting

4.5 Burns

4.6 Heat and Cold Emergencies

Part 5 Medical Emergencies

5.1 Altered Mental Status and Hypoglycemia

5.2 Stroke

5.3 Seizures

5.4 Sepsis

5.5 Poisoning and Overdose

5.6 Acute Abdominal Pain and Gastrointestinal Bleeding

Part 6 Special Populations, Behavioral Emergencies and Safety

6.1 Children: Assessment and How They Differ

6.2 Children: Choking, Breathing Support and Resuscitation Differences

6.3 Pregnancy Emergencies and Imminent Birth

6.4 Older Adults

6.5 Behavioral and Mental Health Emergencies

6.6 Abuse, Neglect and Interpersonal Violence

Appendix A: Answers to Review Questions

Appendix B: Abbreviations

Appendix C: Reference Values

Glossary

Index

BEFORE YOU BEGIN

How to Use This Book

This guide follows a responder's order of thinking: make the scene safe, sort who cannot wait, assess in a fixed order, then deal with the problem. It is written for nursing, paramedic, EMT and allied health students in their first emergency placements or courses.

Six Parts. Part 1 teaches the order of thinking and is the one to read first. Parts 2 to 6 deal with airway and breathing, the heart and resuscitation, injury and the environment, medical emergencies, and special groups of patients. Each Part opens with what you will be able to do, gives its topics as numbered entries, and closes with an illustrative case, a summary of priorities and eight review questions. The answers are in Appendix A.

Scope of practice. What you may do depends on your license, certification, course and the rules of your service or school. Each procedure says so, and each tells you when to stop and call for help. When this book and your local protocol differ, follow your protocol.

What the boxes mean. *Safety* boxes warn of harm from a wrong action. *Red Flags* boxes list signs that call for urgent help. *Scope Note* boxes mark limits of role or training. *Remember* boxes give a memory aid. *Care Priorities* boxes close each Part with tie-breaking rules. *Illustrative Case* boxes are fictional.

What is not here. This book gives no medicine doses. It does not teach intravenous access, medicine amounts, reading heart tracings or advanced airway skills. Practice each skill on a manikin, with an instructor, first.

PART 1

The Emergency Approach: Safety, Triage and Assessment

IN THIS PART

1. Describe the scene and personal safety steps taken before patient contact.
2. Explain how triage assigns priority using a generic five-level idea.
3. Describe the order of the primary survey and the check made at each step, including level of consciousness and a blood glucose check.
4. Describe how to gather a focused history and a set of vital signs, and recognize and report a worsening trend.
5. Describe a structured handover and the reassessment and documentation that go with it, within your scope of practice.

Every emergency starts the same way. Someone arrives at a scene they did not choose, takes in far more than they can process, and has a few seconds to decide what matters first. People who handle this well have a fixed order of thinking and follow it while their pulse is racing.

This Part gives you that order. It begins with the ground you are standing on, because a responder who becomes a casualty helps no one. It then moves to sorting, the quiet decision about who can wait and who cannot. After that comes the primary survey, a short sequence of checks that finds the problems most likely to kill a patient soonest. The Part closes with the slower work that follows: gathering a history, measuring vital signs, handing the patient over clearly, and checking again.

Read it as a map for the rest of the book. Parts 2 to 6 deal with particular problems, but each of them assumes you can arrive safely, sort, assess in order and communicate. Where a skill depends on your training level, the text says so. What you may do is set by your scope of practice and local protocol, not by this book.

1.1 Scene Safety, Personal Protection and Responder Wellbeing

Before you kneel beside a patient, stop and look at the scene once more. Check for traffic, spilled fluid, loose electrical equipment, unstable structures, animals, and any person who may still be a danger. Your first duty is to arrive able to help. An injured responder is a second patient and takes a rescuer away from the first.

WHAT IT IS. Scene safety is the habit of judging risk before contact. It has three parts: the environment, the people present, and your own protection. A few deliberate seconds at the edge of a scene cost far less than treating a second patient.

WHY IT MATTERS. Emergencies create their own hazards. A crash scene may have traffic still moving. A patient who is confused or frightened may strike out without meaning to. A spilled

fluid may be slippery or harmful to skin. Because you are focused on the patient, you are the person least likely to notice these things, so the check must be a routine step you take every time.

PERSONAL PROTECTION. Standard precautions are the baseline for every patient. They include hand hygiene, personal protective equipment, respiratory hygiene and cough etiquette, safe handling of needles and sharps, and careful cleaning and handling of equipment, textiles and the surrounding environment. The guiding rule is to use protective equipment whenever you expect possible contact with infectious material, such as blood or body fluids. For most emergency contacts that means gloves first, with eye protection, a mask or a gown added when splashes or sprays are likely. Your service, school or hospital sets the exact requirements, and you follow them.

PROCEDURE: MAKE THE SCENE SAFE

Purpose: to reach the patient without becoming a second casualty.
Scope: follows your training level and local protocol.

1. Stop at the edge of the scene and look in every direction before you move in.
2. Put on gloves and any other protection your role and the situation require.
3. Name each hazard aloud, even if you are alone, so you do not skip one.
4. Remove a hazard only if it is safe and within your role; otherwise keep clear and wait for help.
5. Count the patients, and ask bystanders to keep back and, if needed, to call for more help.
6. Approach, tell the patient who you are, and begin the primary survey (Entry 1.3).

> **⚠ SAFETY**
>
> Do not enter a scene you believe is unsafe. If violence, fire, unstable ground, live electrical equipment or traffic is present and you cannot control it, stay back and call for help.

STOP AND CALL FOR HELP IF a hazard remains, if you are unsure whether it has gone, or if the number of patients is more than you can manage.

The steps work because they put thinking before action. Naming hazards aloud turns a vague feeling of unease into a list you can act on, and counting patients early tells you how much help to request while you still have a free hand to request it.

RESPONDER WELLBEING. Emergency work asks you to see injury, fear and sometimes death, and it is normal for this to leave its mark. Do not treat strong reactions after a hard call as a personal failing. Eat, drink and rest when you can, tell a trusted colleague or supervisor when something stays with you, and ask whether your service or school offers a debrief or peer support. Seeking that help is part of staying safe to work.

NURSING AND RESPONDER PRIORITIES. Check the scene before touching the patient, protect yourself with standard precautions, call for more help early if the scene or the number of patients is beyond your resources, and look after your own wellbeing as part of the job.

See also: Entry 1.3 (primary survey); Entry 1.5 (communication and handover).

1.2 Triage and Acuity

Triage is not a judgment about who deserves care. It is a way of deciding who cannot wait. A patient who walks in talking, holding a swollen ankle, can sit for a while. A patient who looks calm but is breathing far too fast cannot, and the person at the desk has only moments to notice. This entry gives you a short, repeatable way to sort patients so that the quiet emergency is not missed behind the loud inconvenience.

WHAT IT IS. Triage is the process of sorting patients by urgency so that limited people, space and equipment reach the sickest first. The word comes from the French for sorting. In a US emergency department (ED), triage is usually done by a specially trained nurse soon after the patient arrives. At a scene with many casualties, the same logic is applied quickly and with little equipment, under local protocol. Different places use different tools, and local protocol decides which one you use.

WHY IT MATTERS. Emergency departments, ambulances and scenes are often not staffed for every patient at once. When demand is higher than capacity, the order in which patients are seen is itself a treatment decision. A good sorting method protects the patient who does not complain, such as the older adult with a vague story who may be seriously ill, from being passed over in favor of someone who is louder or more distressed but more stable.

A FIVE-LEVEL IDEA. Many emergency departments in the United States sort patients into five levels of acuity, where level one is the most urgent and level five the least. You do not need the details of any particular published tool to understand how the principle works. It rests on a small number of ordered questions.

REMEMBER

Sort by danger before convenience. Look first for the patient who may die soon, then for the one who must not wait, then count what each of the others will need, and let the vital signs have the last word.

Three questions, asked in order, do the sorting. The first finds the patient who needs action at once, for example from an obstructed airway or a stopped heart. The second finds the patient who should not wait even though nothing must happen in the next few seconds. Everyone else is sorted by how many different kinds of resources they will probably need, such as laboratory tests, imaging, intravenous treatment or a procedure. Simple care such as an examination, a dressing, a splint or a medicine taken by mouth does not count. The last step is to check the vital signs against the level you chose and consider moving the patient up if they are in a danger range (Entry 1.4 gives typical resting values).

Figure 1 shows the sequence.

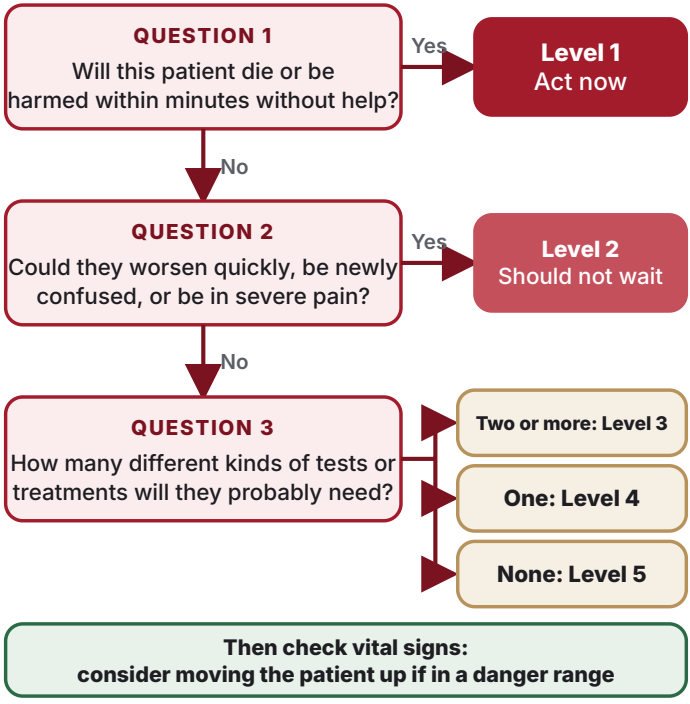


Figure 1. A generic five-level triage sequence, in which the first two questions ask about danger, not paperwork.

Five levels of acuity, in plain terms

Level	Plain description	Typical response
1	Needs immediate life-saving action to stay alive	Treated immediately
2	Could worsen quickly, newly confused, or in severe pain	Should not wait; seen as soon as possible
3	Stable, but expected to need two or more kinds of resources (for example, a blood test and imaging)	Waits in order, rechecked
4	Stable, expected to need one kind of resource	Waits in order, rechecked
5	Stable, expected to need no resources beyond an examination and simple care	Waits in order, rechecked

A level is a starting point, not a verdict. Patients move between levels as you watch them. A person given level four on arrival who becomes short of breath in the waiting room must be reassessed at once and moved to a higher level.

WHAT TRIAGE IS NOT. Triage does not diagnose. It does not say what is wrong, only how soon someone must be seen. Two patients with the same level are often seen in the order they arrived, unless one gets worse. A patient who waits longer is one whose condition allows waiting. Say this to patients and families in plain words, because waiting is easier when people understand the reason for it.

COMMON PITFALLS. Four errors appear again and again. The first is anchoring on the loudest complaint and missing the quiet one. The second is trusting a calm appearance, because patients who are very ill may be quiet, tired or confused, not agitated. The

third is using one set of vital signs as if it were permanent. The fourth is letting the label stick: a patient already sorted to a low level can be overlooked for a long time unless someone is responsible for rechecking. Build a habit of glancing at the waiting area each time you pass it.

RED FLAGS

Move the patient up a level and call for help for: difficulty speaking in full sentences, a gray or blue tint to the lips, new confusion or drowsiness, a patient who stops responding, severe bleeding, or a sudden change in how the patient looks compared with a few minutes ago.

SCOPE NOTE

Assigning a triage level is often a nursing role in an emergency department and is done under local protocol. Students and other responders observe or assist under supervision. If you notice a red flag in the waiting area, tell the responsible nurse at once; do not wait to be asked.

NURSING AND RESPONDER PRIORITIES. Sort by danger first and by resources second, recheck often, reassure waiting patients with an honest explanation, and escalate at once when a patient looks worse than their label.

See also: Entry 1.3 (primary survey); Entry 1.4 (vital signs); Entry 6.1 (children and acuity).

1.3 The Primary Survey

The primary survey is the short sequence of checks that finds the problems most likely to kill a patient soonest. It runs in a fixed order, and you treat a life-threatening problem the moment you find

it, before moving on. The order is Airway, Breathing, Circulation, Disability and Exposure, usually written ABCDE. This order is the spine of the whole book, and every later Part returns to it.

WHAT IT IS. The primary survey is a rapid, whole-patient look for immediate threats to life. It does not diagnose and it does not replace a full examination. Its aim is to keep the patient alive and improve their condition, which buys time for further treatment and for working out what is wrong. The same approach applies to every critically ill or deteriorating patient, whatever the cause.

WHY THE ORDER MATTERS. The letters follow how quickly each failure kills. A blocked airway stops oxygen reaching the lungs within moments, so nothing that comes later can help until the airway is open. Breathing comes next because an open airway is useless without air moving. Circulation follows because oxygen must be carried to the brain and heart. Disability, which means checking how the brain is working, comes after those, and Exposure, which means looking at the rest of the body, comes last. Severe bleeding is the one exception that rearranges the sequence. When blood loss is heavy and visible, many trauma protocols direct you to control it first, because there is little point in checking the airway of a patient who is bleeding to death. Entry 4.1 explains how bleeding is brought forward in trauma.

> **REMEMBER**
>
> Find it, fix it, check again. After any treatment, wait a short while before judging its effect, then repeat the survey.

THE FIRST LOOK. Before the letters, take a rapid first look at the patient in general. If the patient is awake, ask a simple question such as “How are you?” If they appear unconscious, speak to them loudly. If you can rule out a neck injury, tap the shoulders gently as you ask whether they are all right; if a neck injury is pos-

sible, do not shake the patient and keep the head and neck still. A patient who answers in a normal voice has an open airway, is breathing and is probably getting enough blood to the brain for the moment. A patient who speaks only in short sentences may have a breathing problem. A patient who does not respond is critically ill until shown otherwise. This look takes about 30 seconds and often tells you at once that the patient needs urgent help. If the patient is unresponsive and not breathing normally, call for help and start cardiopulmonary resuscitation (CPR) as set out in Entry 3.4. Occasional gasps are not normal breathing.

AIRWAY. Ask whether air can get in and out. Talking means the airway is open. Noisy breathing is a warning, because a partly blocked airway is often noisy, and a high-pitched sound such as stridor suggests a significant partial blockage. Complete blockage may give no sound at all, with the chest and abdomen moving in opposite directions as the patient strains. Clear and open the airway with the simple measures in Entry 2.2, and call for help at once.

BREATHING. Look, listen and feel. Count the breathing rate (respiratory rate, RR) carefully, for as long as your protocol directs, and watch for effort: sweating, use of the neck and shoulder muscles, belly breathing, or a gray or blue tint to the lips. A high or rising rate is a warning that the patient may deteriorate suddenly. Note how deeply the patient breathes and whether both sides of the chest move equally. If you have a pulse oximeter, record the reading, and remember what it does not tell you, as Entry 1.4 explains.

CIRCULATION. Look at the skin of the hands: is it pink, pale, blue or mottled? Feel whether the hands are warm or cool. Check the capillary refill time by pressing on a fingertip, held level with the heart, for 5 seconds, then timing how long the color takes to

return. In a healthy adult it is usually under 2 seconds, and a longer time suggests poor blood flow to the skin, though a cold room or cold hands can also lengthen it. Feel a pulse and note whether it is fast, slow, weak, strong or irregular. A bounding pulse and a barely palpable one both matter. Look for bleeding you can see, and think about where blood might be hiding, in the chest, abdomen or pelvis. Entry 3.5 explains shock in full.

DISABILITY. This step checks how the brain is working. Begin with the quickest tool, the AVPU scale. Is the patient Alert? Do they respond only to Voice? Only to Pain? Are they Unresponsive? Some services add a letter for new confusion, so that the scale reads A, C, V, P, U, and new confusion in a previously normal patient is treated as a serious sign. Where your protocol uses it, the Glasgow Coma Scale gives a fuller picture. It scores three responses: eye opening, verbal response and motor response. The three scores are added to give a total from 3 to 15, where 15 is the highest score. Lower scores mean greater impairment, and the trend over time is important in showing when new action is needed. Check the pupils for size, equality and reaction to light, and check the blood glucose, because low blood glucose can cause confusion or loss of consciousness and is treatable. Entry 5.1 covers its causes and management.

EXPOSURE. To examine a patient properly you may need to expose the body. Do it with respect for dignity, with screens or blankets where possible, and take care that the patient does not lose heat. Look for wounds, rashes, bruising, medical alert jewelry, and anything that was hidden by clothing.

PROCEDURE: THE PRIMARY SURVEY

Purpose: to find and treat immediate threats to life in the order they kill. Scope: follows your training level and local protocol.

1. Make the scene safe and put on protection (Entry 1.1).

2. Take the first look: speak to the patient and watch the response for about 30 seconds; if heavy bleeding is visible, your protocol may direct you to control it first (Entry 4.1).
3. If the patient is unresponsive and not breathing normally, call for help and begin CPR.
4. Airway: confirm the patient can speak or breathe freely; open and clear it if not.
5. Breathing: count the rate, look at the effort and chest movement, and record the oxygen saturation if a device is available.
6. Circulation: check skin color and warmth, capillary refill and pulse; control any severe bleeding.
7. Disability: check AVPU or Glasgow Coma Scale, the pupils and the blood glucose.
8. Exposure: expose as needed, protect dignity and warmth, and look for injuries.
9. Treat each life-threatening problem as you find it, call for help, and repeat the survey after every intervention.

> ⚠ **SAFETY**
>
> Treat the neck (cervical spine) with care if there is any chance of neck injury, and do not move an injured patient unless the scene is unsafe or your protocol directs a move to give life-saving care, and then keep the head, neck and body in line as far as you can.

STOP AND CALL FOR HELP IF any step finds a threat to life that you cannot correct, or if the patient does not improve after you treat it.

Each step in this procedure answers a single question, and the answers build on one another. When you finish the lettered checks, you can say whether the patient is stable, unstable, or deteriorating, and you can state in a sentence what you did and how the patient responded. That sentence becomes the core of your handover in Entry 1.5.

CHECKING BLOOD GLUCOSE. The blood glucose check is one step inside Disability, and it is a skill only where your role and protocol allow it. A meter reads a small drop of blood from a fingertip. Use the device exactly as its maker directs, wear gloves, use a new single-use lancet each time, and place the used lancet in a sharps container at once. For many people with diabetes, a reading below 70 milligrams per deciliter (3.9 millimoles per liter) is low, though the number that counts as low can differ from person to person. The number is a prompt to act under your protocol, not a diagnosis. Entry 5.1 describes the response. If your role does not include glucose testing, report that you could not check it and tell the responsible clinician.

Figure 2 shows the loop.

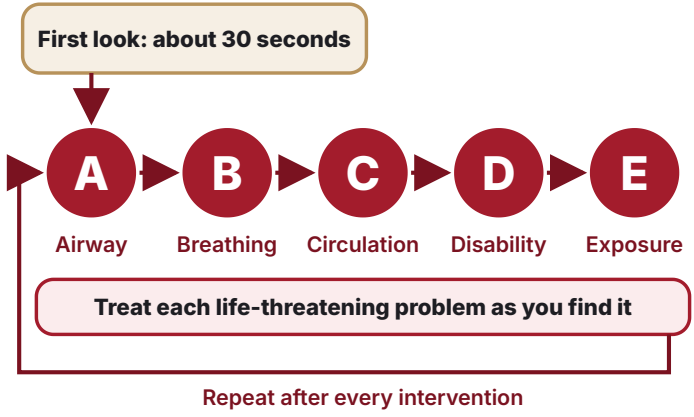


Figure 2. The primary survey as a loop, with treatment after each check and a repeat of the whole loop after every intervention.

NURSING AND RESPONDER PRIORITIES. Complete the first look quickly, work through the letters in order, treat what you find before moving on, call for help early, and repeat the sequence whenever the patient changes or after any intervention.

RED FLAGS

Unresponsiveness, absent or abnormal breathing, noisy or absent airflow, blue or gray lips, a very fast or very slow pulse, cold mottled skin, heavy bleeding, a falling level of consciousness, or a new confusion in a patient who was normal a short time ago.

See also: Entry 1.4 (vital signs); Entry 2.2 (airway); Entry 3.5 (shock); Entry 4.1 (trauma); Entry 5.1 (altered mental status).

1.4 History, Vital Signs and the Secondary Survey

Once the primary survey has found and treated immediate threats, you have time to learn more. The secondary survey is a slower, systematic look from head to toe combined with a focused history. Its job is to catch what the primary survey was not designed to find, and to give the receiving team a clear picture. It never replaces the primary survey, and you return to the primary survey whenever the patient worsens.

WHAT IT IS. The secondary survey has two strands that run together. One is the history, which asks the patient, family, bystanders and records what happened. The other is the examination, which includes a full set of vital signs and a head-to-toe check. In an unstable patient the secondary survey is usually shortened or postponed so that treatment and transport are not delayed.

GATHERING A HISTORY. A short, structured set of questions keeps the history focused. Ask what happened and when it started. Ask about the main symptom: where it is, how bad it is and what makes it better or worse. Ask about allergies, current medicines, past illnesses, the last meal or drink, and the events leading up to the problem. These questions are often remembered with a mnemonic, and your school or service will teach one. Use the mnemonic you are taught and cover the same ground in the same order every time, so that nothing important, such as an allergy or a blood-thinning medicine, is forgotten. If the patient cannot talk, turn to family, bystanders, medical alert jewelry or records, and note who gave each piece of information.

VITAL SIGNS. Vital signs are measurements of essential body functions: heart rate, breathing rate, blood pressure and temperature, usually with oxygen saturation added. Normal values change with age, sex, weight, fitness and overall health, so the ranges below describe a healthy adult at rest and should be read as a guide, not a rule.

Typical resting ranges for a healthy adult

Measure	Typical range
Pulse (heart rate)	60 to 100 beats per minute
Breathing rate	12 to 18 breaths per minute
Blood pressure	between 90/60 and 120/80 mm Hg
Temperature	97.7 to 99.1 degrees F (36.5 to 37.3 degrees C)

A reading outside its range may be abnormal for that patient. Report it, and treat a large departure, or any departure in a patient who looks unwell, as urgent.

Treat the trend as more informative than any one reading. A breathing rate that climbs from 18 to 28 over 20 minutes tells you more than a single count, and a breathing rate above about 25 breaths per minute is a marker of illness and a warning that the patient may deteriorate suddenly, so report it to your supervisor or the responsible clinician at once. Blood pressure can stay normal for a time even in shock, because the body tightens its blood vessels to compensate, so a normal reading does not by itself make a patient safe.

OXYGEN SATURATION. A pulse oximeter clips on a finger and estimates the percentage of hemoglobin carrying oxygen. It is useful, but it has limits that you must know. It does not measure how well a patient is breathing out carbon dioxide, so a patient can

have a normal reading while still ventilating poorly. Readings can also be less accurate with poor circulation, cold hands, some nail polish colors, motion and differences in skin pigmentation, and the device can be misled by carbon monoxide poisoning. Read the number together with the patient's color, effort and mental state. A reassuring number in a patient who looks ill should not reassure you.

Figure 3 shows the places to check.

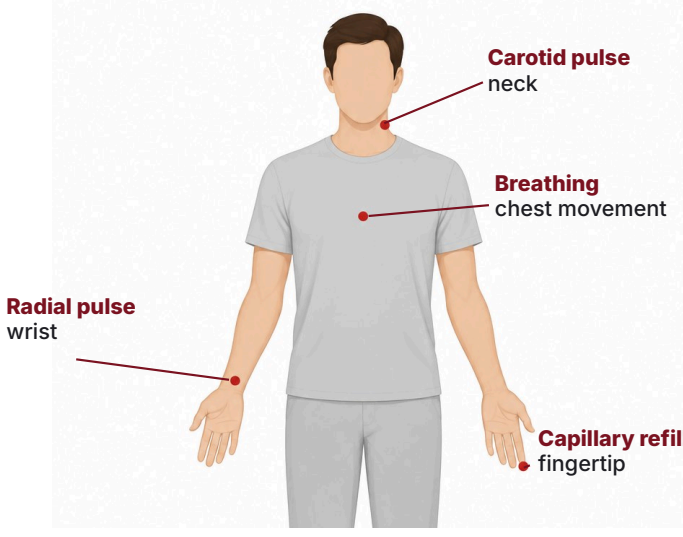


Figure 3. Where to check pulse, breathing and skin perfusion on an adult, in the same places each time.

HEAD-TO-TOE CHECK. Work from the head down: scalp and face, eyes, ears, nose and mouth, neck, chest, abdomen, pelvis, back, arms and legs. Look, then feel gently, and ask about pain. Note wounds, deformity, swelling, bruising and anything that looks wrong or smells wrong. Record the time of each set of observations.

> **SCOPE NOTE**
>
> Taking vital signs is a core skill for nursing, paramedic and emergency medical technician (EMT) students, but interpreting patterns and acting on them is done with your supervisor and under local protocol. Always tell a qualified clinician about an abnormal or worsening finding at once.

NURSING AND RESPONDER PRIORITIES. Return to the primary survey if the patient deteriorates, take a complete set of vital signs and record the time, trust trends over single numbers, ask structured history questions the same way each time, and treat reassuring numbers with caution when the patient looks unwell.

See also: Entry 1.3 (primary survey); Entry 5.4 (sepsis); Entry 6.1 (children's values).

1.5 Communication, Consent, Scope of Practice, Handover, Reassessment and Documentation

A patient cared for well by three people is easily harmed when the three do not talk to each other. This last entry covers the work that holds a response together: speaking with the patient, obtaining consent, staying within your scope, passing information on clearly, checking again, and writing down what you saw and did.

SPEAKING WITH THE PATIENT AND CONSENT. Introduce yourself, say what you are about to do and why, and ask permission before you touch. A patient who has capacity, meaning the ability to understand the information, see how it applies to their own situation, weigh the options and state a choice, may accept or refuse care, and you respect that choice, tell a clinician and record it. In a true emergency, when a patient cannot take part in deci-

sions and no one who can decide for them is available, treatment may start without prior consent, and the patient is told as soon as possible. The rules differ between states, services and roles, so follow your local protocol and the guidance of your supervisor.

SCOPE OF PRACTICE. Your scope of practice is the set of tasks you are legally authorized and trained to perform. It depends on your license or certification, your education, your role and the setting, and students work under the rules of their program and their supervisor. Do only what it allows, say plainly when a task is beyond it, and ask your supervisor before trying anything you have not been taught and cleared to do.

HANDOVER. Handover is the structured passing of a patient's information to the next team. One widely used format has four parts: Situation, Background, Assessment and Recommendation. Situation says who the patient is and what the main problem is. Background gives the history, medicines, allergies and what led up to this. Assessment gives your findings in the order of the primary survey, the vital signs and what you did. Recommendation states what you think should happen next or what you need from the receiver. A good handover is short, follows a fixed order, and labels a guess as a guess.

Figure 4 shows the four parts.

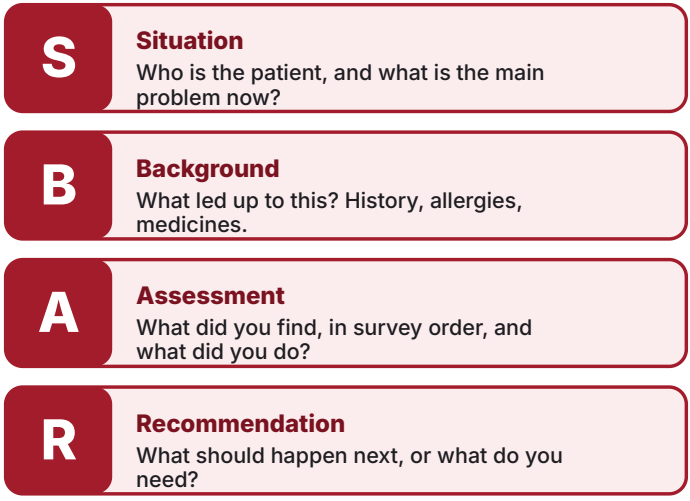


Figure 4. A four-part handover in which each part answers one question for the receiver.

REASSESSMENT. A patient who was stable a moment ago may not be stable now. Recheck the primary survey and vital signs at intervals set by the patient's condition and your protocol, after every treatment, and before and after any move.

DOCUMENTATION. Write down what you found, what you did, when you did it and how the patient responded, using the times you recorded. Record facts and observations, not opinions, and keep the record accurate and legible. A clear record lets the next team see what has already been done and when.

NURSING AND RESPONDER PRIORITIES. Explain what you are doing, obtain consent or follow the emergency rules for those who cannot give it, hand over in a fixed order, recheck often, and record accurately and promptly.

See also: Entry 1.3 (primary survey); Entry 1.4 (vital signs).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Priya was a second-year nursing student volunteering at a school sports day. She was carrying a water cooler across the field when a spectator in the stands, a man in his sixties, slid sideways on the bench. People nearby stood up and shouted. Priya stopped at the foot of the stands and looked before climbing. The bleachers were crowded and the step was wet from a spilled drink, so she slowed down, pulled on gloves from her kit bag, and asked a nearby parent to find the event first-aid lead and bring the automated external defibrillator.

She reached the man and said her name. His eyes opened when she spoke loudly, and he mumbled a few words. His breathing was quick and shallow, and he could say only two or three words at a time. His lips looked pale. She asked the parent to call emergency services and to give the stand number. Then she began the survey she had practiced in class.

His airway was open, because he was making sound. She counted his breathing carefully and found it fast. His fingertips were cool and the color took longer than usual to return after she pressed. His pulse was quick and weak. He responded to voice but not to questions, so she noted that he was not fully alert. When the first-aid lead arrived, Priya gave the handover she had rehearsed: who he was, what she had seen, her findings in order, and that he seemed to be getting worse. The lead took over and asked her to stay, record the times and keep checking his breathing until the ambulance came.

Care Priorities

CARE PRIORITIES

When two things compete, your own safety comes before the patient's need, and the patient who cannot wait comes before the patient who complains most. Treating a threat comes before measuring it: a pulse oximeter reading can wait, an obstructed airway cannot. Calling for help comes before finishing the history, the head-to-toe check or the paperwork. A plan is only as good as the next check, so put a time on every observation and decide before you leave the bedside when you will look again. If a task is outside your scope, say so at once and stay with the patient until someone who can do it arrives.

Review Questions

1. What should you do first when you arrive at an emergency scene?
   - **A)** Begin the primary survey at once.
   - **B)** Check the scene for hazards and protect yourself.
   - **C)** Take a full set of vital signs.
   - **D)** Record the time on the patient's chart.

2. Triage sorts patients mainly by which factor?
   - **A)** The order in which they arrived.
   - **B)** How loudly they complain.
   - **C)** How long they are expected to stay.
   - **D)** How soon they must be seen to stay safe.

3. A patient sorted to a low level of acuity becomes short of breath in the waiting room. What is the best response?

   A) Reassess, raise the level and call for help.

   B) Leave the level unchanged, since a nurse assigned it.

   C) Ask the patient to wait quietly for their turn.

   D) Take only their temperature.

4. A patient has a normal blood pressure, cool pale skin and fast breathing. What is the best interpretation?

   A) The patient is stable because the blood pressure is normal.

   B) The skin color is caused by cold hands and can be ignored.

   C) The patient may still be seriously ill, because blood pressure can stay normal for a time in shock.

   D) The breathing rate does not matter when the blood pressure is normal.

5. A patient responds only when you speak loudly to them. On the AVPU scale, which level describes this?

   A) Alert.

   B) Pain.

   C) Unresponsive.

   D) Voice.

6. Which of these is a limitation of a pulse oximeter?

   A) It does not show how well a patient is breathing out carbon dioxide.

   B) It cannot be used on a finger.

   C) It can only be used when the patient is awake.

   D) It replaces the need to count the breathing rate.

7. In a handover that uses Situation, Background, Assessment and Recommendation, where do the primary survey findings belong?

   A) Situation.

   B) Background.

   C) Assessment.

   D) Recommendation.

8. Short case. You find an older adult slumped on a bench in a park, breathing fast and responding only to voice. List, in order, the first four things you would do.

PART 2

Airway and Breathing Emergencies

IN THIS PART

1. Identify the structures of the upper airway and what narrows or blocks them.
2. Recognize partial and complete airway obstruction and state the response.
3. Describe basic airway positioning, suction and bag-valve-mask ventilation, and their safety limits.
4. Recognize respiratory distress and respiratory failure and state first priorities for asthma, chronic obstructive pulmonary disease (COPD) and pneumonia.
5. Recognize anaphylaxis and state the first priorities.

A patient can go for days without food, but only minutes without air reaching the lungs and the brain. That is why airway and breathing come first in the primary survey of Entry 1.3, and why this Part comes next.

The five entries follow the path that air takes and the ways that path fails. Entry 2.1 describes the passage itself and how to judge whether it is open. Entry 2.2 covers what to do when it is not, from simple positioning to clearing a choking adult. Entry 2.3 explains how to move air for a patient who cannot, with a bag-valve-mask and supplemental oxygen. Entry 2.4 turns to the lungs and the

conditions that make breathing hard. Entry 2.5 ends with anaphylaxis, which can close an airway within minutes and which a bystander may be the first to treat.

Everything here assumes the habits of Part 1: a safe scene, a first look, and a recheck after every action. As before, what you may do depends on your scope of practice and local protocol. Where a skill needs training that a book cannot give, the text says so plainly and tells you when to stop and call for help.

2.1 Airway Anatomy and Assessment

The airway is the passage that carries air from the outside to the lungs. You cannot judge whether it is open, or why it has closed, without a picture of what it is made of. This entry gives you the picture and then the short set of observations that tell you how well it is working.

WHAT IT IS. Air enters through the nose and mouth and passes into the throat, called the pharynx. Below the throat sits the larynx, or voice box, guarded by a flap of tissue called the epiglottis that covers the entrance when you swallow. Below the larynx the windpipe, or trachea, runs down into the chest and divides into the two main bronchi, which branch into ever smaller tubes inside the lungs. The part from the nose down to and including the larynx is the upper airway, and most emergency airway care is aimed there, because it is the part you can reach and open.

Figure 5 shows these structures; notice the order from nose to windpipe.

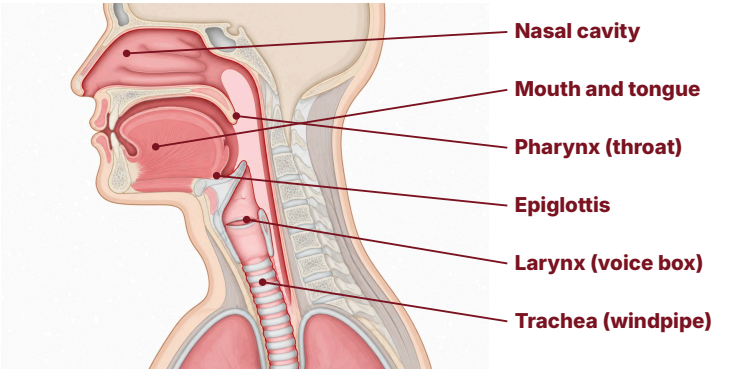


Figure 5. The upper airway from nose and mouth to the windpipe, with the main structures labeled.

WHY IT MATTERS. The upper airway is narrow, shared with the route for food and drink, and held open by muscle. Anything that weakens the muscle, adds swelling, or puts something in the way can close it. Because the brain depends on a constant supply of oxygen, a blocked airway can cause loss of consciousness and then cardiac arrest if it is not cleared.

WHAT NARROWS OR BLOCKS IT. Five causes are common. In an unconscious patient lying on the back, the tongue and soft tissues of the throat can fall backward and close the passage, and this is the most common cause of obstruction in an unconscious patient. Food or a foreign object can lodge in the throat. Fluid, vomit or blood can fill the airway. Swelling from an allergic reaction, infection, a burn or an injury can narrow it from within. A direct injury to the face or neck can distort it. Each cause points to a different first action, which is why the next entry is organized by cause and not by technique.

ASSESSING THE AIRWAY. Entry 1.3 introduced looking, listening and feeling; here each finding is tied to a cause. Ask the patient a simple question. A patient who answers in a normal voice has an open airway for the moment. Then watch the chest and abdomen, listen for sounds, and feel for air at the mouth and nose.

What airway findings suggest

Finding	What it suggests
Normal voice, full sentences	Airway open for now
Very hoarse or whispered voice	Swelling or narrowing of the throat
High-pitched sound on breathing in (stridor)	Significant partial blockage
Gurgling	Fluid in the airway
Chest and abdomen moving in opposite directions	Obstruction with strong effort to breathe
No air felt or heard at the mouth or nose	Complete obstruction or no breathing

Airway obstruction is an emergency. In a partial obstruction, air still moves but less than it should and often with noise. In a complete obstruction, no air moves, the patient cannot speak or cough, and the chest and abdomen may move in opposite directions as the patient strains. A blue or gray tint to the lips, called central cyanosis, is a late sign. Do not wait for it.

RED FLAGS

A very hoarse or whispered voice, a high-pitched sound on breathing in, drooling or inability to swallow, swelling of the face or tongue, a patient who cannot speak or cough, or a blue or gray tint to the lips. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Assess the airway first and again after every change in the patient, and treat obstruction as an emergency without waiting for a blue tint.

See also: Entry 1.3 (primary survey); Entry 2.2 (opening and clearing the airway); Entry 6.1 (child airway differences).

2.2 Airway Obstruction and Basic Airway Management

The simple measures in this entry can be done with few or no tools, and they address the most common airway problems before advanced help arrives. They follow the cause. If the tongue has fallen back, you reposition. If there is fluid, you suction. If there is a solid object, you relieve choking. If there is swelling, you call for help and treat the cause, as Entry 2.5 describes for allergic swelling.

WHAT IT IS. Basic airway management is the set of non-invasive actions that open and protect the airway without placing any device inside the patient. It includes positioning the head and jaw, clearing the mouth with suction, placing a patient who is breathing but unresponsive on their side, and relieving choking.

WHY IT MATTERS. Each measure is quick and each can be done by a student with the right training and a supervisor present. They matter because they act on the cause directly and

because they must be repeated: an airway opened once may close again as the patient's condition changes.

OPENING THE AIRWAY. The two standard positions lift the tongue away from the back of the throat. In the head-tilt chin-lift, you tilt the forehead back with one hand and lift the chin with the fingertips of the other. It is used for a patient with no suspected neck injury. In the jaw thrust, you place your fingers behind the angles of the jaw on both sides and lift the jaw forward without tilting the head. It is the choice when a neck injury is possible, because it moves the neck less. If the jaw thrust does not open the airway, your protocol may direct you to use the head tilt anyway, since an open airway comes first. Entry 4.3 returns to the neck.

Figure 6 compares the two positions; notice that the jaw thrust leaves the head in line with the body.

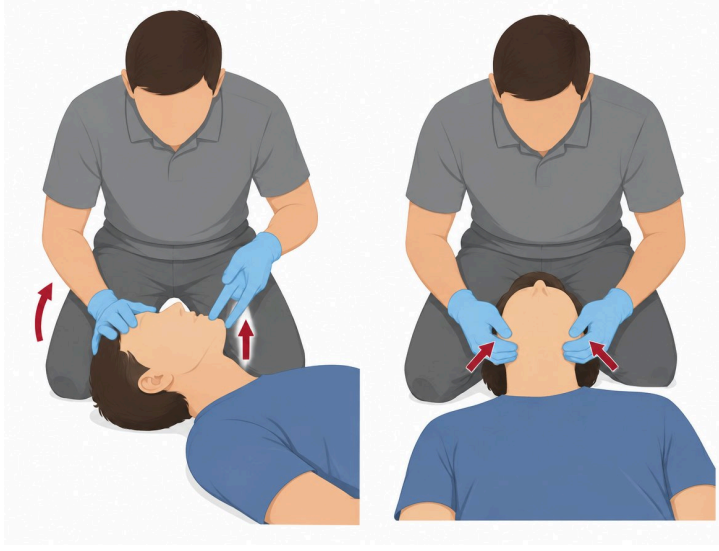


Figure 6. The head-tilt chin-lift and the jaw thrust, drawn side by side to show where the hands go.

CLEARING THE MOUTH. Suction removes liquid and small particles from the mouth and throat. Use it only when you can see or hear fluid, apply it in short bursts, and stop to check the patient's breathing and condition between bursts, because suction can lower the patient's oxygen level. Follow the device maker's instructions and your protocol for how long and how deep.

PROCEDURE: OPEN THE AIRWAY OF AN UNRESPONSIVE PATIENT

Purpose: to keep air moving past the tongue. Scope: follows your training level and local protocol.

1. Make the scene safe, put on gloves, and call for help.
2. Check for a response and for normal breathing (Entry 1.3).
3. If you suspect a neck injury, use the jaw thrust; otherwise, tilt the head and lift the chin.
4. Look in the mouth and clear anything you can see and reach, such as fluid or loose debris, with suction if it is available and you are trained.
5. Look, listen and feel for breathing again after the airway is open.
6. If the patient is breathing normally and you do not suspect an injury, place them on their side in the recovery position; if you suspect an injury, leave them in the position you found them, hold the airway open with the jaw thrust, and move them only as your protocol directs.
7. If the patient is not breathing normally, begin CPR (Entry 3.4).
8. Reassess and keep checking breathing until help takes over.

> **⚠ SAFETY**
>
> Do not put your fingers blindly into a patient's mouth. Stay with an unresponsive patient, and do not move a patient with a possible neck injury unless the scene is unsafe or your protocol directs a move to give urgent care, and then keep the head, neck and body in line as far as you can.

STOP AND CALL FOR HELP IF the airway does not open, if suction does not clear the mouth, or if the patient stops breathing.

These steps work in a fixed order because each one answers what the last one found. Positioning solves the common problem first, the fallen tongue. Clearing the mouth then deals with fluid or debris. Checking breathing then tells you whether the patient needs only an open airway or needs breathing support as well.

THE RECOVERY POSITION. A patient who is breathing normally but is unresponsive, and who has no suspected neck or spinal injury, is at risk from their own tongue and from vomit. Placing them on their side lets fluid drain from the mouth and keeps the tongue forward. Kneel beside the patient, place the arm nearest you at a right angle to the body, bring the far arm across the chest and hold the back of that hand against the near cheek, bend the far knee, and roll the patient toward you by pulling on that knee. Adjust the top leg so that the hip and knee are bent at right angles and tilt the head back slightly so the airway stays open. Check breathing again at once and keep checking. The position is for a patient who is breathing normally; a patient who is not breathing normally needs CPR, and a patient with a possible spinal injury is moved only as your protocol directs. This skill follows your training level and local protocol; call for help if the patient's breathing becomes noisy, slow or stops.

CHOKING IN AN ADULT. Choking is a solid object blocking the airway. A patient with a partial blockage can still cough, speak or breathe with effort, and you should encourage coughing and stay close and do not interfere while the cough is forceful. A patient with a severe blockage cannot speak, cough or breathe, may clutch the throat, and may turn blue or gray. That patient needs immediate help. Ask "Are you choking?" and, if they nod and cannot speak, act.

PROCEDURE: RELIEVE CHOKING IN A CONSCIOUS ADULT

Purpose: to dislodge an object that blocks the airway. Scope: follows your training level and local protocol; the exact sequence your service teaches takes precedence.

1. Ask "Are you choking?" and check whether the person can cough, speak or breathe.
2. Send someone to call emergency services, or call for help yourself as soon as you can.
3. Stand to one side and slightly behind the person, support the chest with one hand, and lean them forward.
4. Give up to 5 firm blows between the shoulder blades with the heel of your other hand.
5. If the object is still stuck, stand behind the person, place a fist just above the navel, and give up to 5 quick inward and upward abdominal thrusts.
6. Keep alternating back blows and abdominal thrusts until the object comes out or the person stops responding.
7. If the person becomes unresponsive, lower them carefully to the ground, call for help, and begin CPR, starting with chest compressions (Entry 3.4).

8. Stay with the person, keep watching them until help arrives, and have a clinician check them, even if the object comes out.

> **⚠ SAFETY**
>
> Abdominal thrusts can injure internal organs. For a person who is in the late stages of pregnancy or too large to reach around, your training may direct chest thrusts instead. Do not use abdominal thrusts on infants; Entry 6.2 covers them.

STOP AND CALL FOR HELP IF the object does not come out, the person cannot breathe or turns blue or gray, or the person becomes unresponsive.

The two techniques use different forces. Back blows strike the upper back to dislodge the object, and abdominal thrusts raise the pressure in the chest so that air pushes the object upward. Current guidance has you alternate them until the object comes out or the person becomes unresponsive.

NURSING AND RESPONDER PRIORITIES. Match the action to the cause, open the airway first, and suction only when you can see or hear fluid. Stay with an unresponsive patient and recheck the airway after every move.

See also: Entry 1.3 (primary survey); Entry 3.4 (CPR); Entry 4.3 (spinal motion restriction); Entry 6.2 (infant and child choking).

2.3 Ventilation Support: Bag-Valve-Mask and Oxygen Delivery

Sometimes the airway is open and the patient still is not moving enough air. Breathing may be too slow, too shallow or absent. At that point a trained responder moves air for the patient with a bag-valve-mask, often connected to oxygen. This entry explains

the device, the steps, and the oxygen devices you may meet. It names the device types and when each is used; your protocol sets the flow.

WHAT IT IS. A bag-valve-mask, written BVM, has three parts: a soft face mask that seals over the nose and mouth, a one-way valve, and a self-refilling bag that you squeeze to push air into the lungs. Released, the bag refills from room air or from an attached oxygen supply. It is used on patients who are not breathing or are breathing too weakly, and it works best with two trained people, one to hold the mask and one to squeeze the bag.

WHY IT MATTERS. Moving air by hand is a skill that takes practice. A poor seal lets air escape. Squeezing too hard or too fast can push air into the stomach which raises the risk of vomiting. A technique that moves the chest well on a manikin may fail on a patient with a beard or a swollen face. Practice on manikins, under supervision, comes first.

USING A BAG-VALVE-MASK. Choose a mask that covers the nose and mouth without pressing on the eyes. With two people, one holds the mask with both hands, keeping the airway open with the jaw thrust or head tilt, while the other squeezes the bag. With one person, the single hand and the bag are harder to manage, and the seal is the usual point of failure. Squeeze gently and just enough to make the chest rise, then release and let the chest fall. Watching the chest tells you whether air is going in.

Figure 7 shows the two-hand mask seal; notice how the fingers lift the jaw toward the mask.



Figure 7. A two-hand mask seal with a second person squeezing the bag (eye protection not drawn).

For an adult who has a pulse but is not breathing normally, a common rule is 1 breath about every 6 seconds, or about 10 breaths a minute. Your protocol sets the rate you follow, and the rate for a child or infant differs.

PROCEDURE: ASSIST BREATHING WITH A BAG-VALVE-MASK

Purpose: to move air into the lungs of a patient who is not breathing adequately. Scope: follows your training level and local protocol; usually done under supervision.

1. Make the scene safe, put on gloves and eye protection, and call for help.
2. Open the airway and clear the mouth (Entry 2.2).

3. Check for breathing and a pulse; if there is no pulse, begin CPR (Entry 3.4).
4. Choose a mask that covers the nose and mouth, and attach the bag and, if available, the oxygen supply.
5. Hold the mask on the face with both hands, with the jaw lifted, so that no air leaks.
6. Have the second person squeeze the bag gently until the chest begins to rise, then release.
7. Give breaths at the rate your protocol sets, watching the chest rise and fall.
8. Keep the airway open, reassess often, and stay with the patient until help takes over.

> ⚠ **SAFETY**
>
> Do not over-squeeze or give breaths too fast. If the stomach swells or the patient vomits, turn the patient on the side if you can, clear the mouth, and call for help. Never use oxygen near flame, sparks or smoking.

STOP AND CALL FOR HELP IF the chest does not rise, the patient vomits, or you cannot make a seal.

IS THE VENTILATION WORKING? The chest should rise gently and fall fully as the bag is released. You should see no air escaping around the mask. A good response may include improved skin color and, in a patient with a pulse oximeter, a rising reading, though that takes time. If the chest does not rise, reposition the head and jaw, reseat the mask, and check for an obstruction before squeezing harder. Squeezing harder risks inflating the stomach.

OXYGEN DELIVERY DEVICES. Oxygen can be given by several devices, chosen by how much the patient needs and by the patient's ability to breathe. A nasal cannula is a soft tube with two short prongs that rest in the nostrils, used for patients who are breathing well and need a small boost. A simple face mask covers the nose and mouth and is used when a moderate increase is needed. A non-rebreather mask has a bag, called a reservoir, that fills with oxygen so that the patient breathes in a high concentration of oxygen, and it is used for the sickest breathing patients. A BVM connected to oxygen, with a reservoir attached where one is fitted, delivers the highest concentration to a patient who cannot breathe for themselves. Your protocol and medical direction set the choice, the flow, and the target for each patient. Oxygen is treated as a medicine in the United States. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Open and clear the airway before you ventilate, make a firm seal, and give gentle breaths until the chest rises.

See also: Entry 2.2 (airway); Entry 3.4 (CPR); Entry 6.1 (child sizing and rate).

2.4 Respiratory Distress: Asthma, COPD and Pneumonia

Breathing difficulty is one of the most common reasons people call for help, and its causes range from a minor chest cold to a life-threatening attack. This entry gives you a way to judge how serious it is and covers three common causes.

WHAT IT IS. Respiratory distress is difficult or labored breathing. Respiratory failure is the stage at which breathing can no longer meet the body's needs, and it can follow distress quickly,

particularly in a tiring patient. Signs of distress include sweating, using the neck and shoulder muscles to breathe, and belly breathing. A rate that is high or rising is a warning, and Entry 1.4 gave typical resting values. Signs of failure include drowsiness or confusion, weak or tiring breathing effort, a very quiet chest in a patient who was wheezing, and blue lips.

WHY IT MATTERS. A patient who is working hard to breathe can tire, and a tiring patient can stop. The signs of tiring are subtle: the patient becomes calmer and quieter, which is easy to mistake for improvement. Reassessing breathing rate, effort and mental state at short intervals catches the change.

REASSESSING A BREATHING PATIENT. Because the patient's condition can change quickly, set a rhythm for rechecking. Each time, note the breathing rate, how many words the patient can say at once, whether the neck and shoulder muscles are working, the color of the lips and the level of alertness. Write each set down with the time, as Entry 1.5 describes. A patient who needs more help than before is worsening even if each single number looks acceptable. A patient who is tiring may change position, lean forward on outstretched arms and start to say fewer words.

ASTHMA. Asthma is a long-term condition in which the airways narrow, tighten and become inflamed in response to triggers such as allergens, infection, exercise or cold air. In an attack, the patient may wheeze, cough, feel tightness in the chest, and struggle to breathe. Many patients carry a reliever inhaler containing albuterol (salbutamol). A severe attack is shown by speaking only a few words at a time, a high breathing rate, visible effort, a very quiet chest, drowsiness, or no improvement after the patient's usual reliever. Sit the patient upright, keep them calm, call for

help, and help them use their own inhaler as they have been taught and as your protocol allows. Giving medicines is outside many roles; follow your scope and medical direction.

COPD. COPD is a long-term lung disease, often caused by smoking, in which the airways are narrowed and the lungs are damaged. Patients are short of breath on effort, cough, and bring up mucus, and they have flare-ups called exacerbations. Oxygen needs special care here. In some patients with COPD, a high concentration of oxygen can slow breathing and let carbon dioxide build up, so these patients need a lower oxygen target set by protocol or medical direction. Give oxygen only as your scope and medical direction allow, and watch breathing closely; do not withhold oxygen from a patient who needs it. Giving medicines is outside many roles; follow your scope and medical direction.

PNEUMONIA. Pneumonia is an infection of the lungs. It causes fever, cough, sometimes chest pain on breathing in, and a fast breathing rate. In older adults or very ill patients it may cause confusion without much cough or fever. Pneumonia can lead to sepsis, which Entry 5.4 covers. Confusion with poor breathing is a warning of its own, and Entry 5.1 returns to it. First priorities are an upright or comfortable position, oxygen as directed, early reassessment, and a call for help if breathing is hard, the patient is confused, or the lips are blue.

Figure 8 sets the same priorities out as a flow: judge severity first, and note that the severe branch calls for help before anything else.

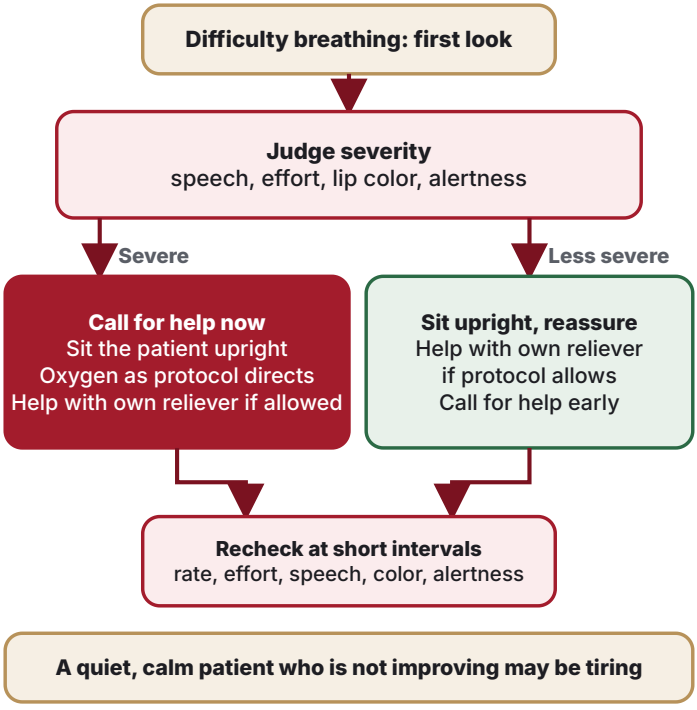


Figure 8. A first-priority flow for a patient with difficulty breathing: judge the severity, call for help, position the patient, and recheck.

RED FLAGS

Speaking only a few words at a time, a very fast breathing rate, weak or tiring effort, blue or gray lips, drowsiness or new confusion, a chest that has gone quiet, or a patient who becomes calmer without improving. Call for help at once.

> **SCOPE NOTE**
>
> Helping a patient use their own prescribed inhaler, and giving oxygen, depend on your role, your protocol and medical direction. If you are not sure whether you may, ask a supervisor, and always call for help in a severe attack.

NURSING AND RESPONDER PRIORITIES. Judge severity by effort, speech, color and mental state, call early for help, and reassess often because a quiet patient may be a tiring one.

See also: Entry 1.3 (primary survey); Entry 2.3 (oxygen devices); Entry 5.1 (confusion with poor breathing); Entry 5.4 (sepsis); Entry 6.1 (children).

2.5 Anaphylaxis

Anaphylaxis is a severe allergic reaction that affects the whole body and can be fatal. It tends to arrive fast, often within seconds or minutes of contact with the trigger, and it can close the airway, drop the blood pressure, or both. A bystander with no medical training may be the first person to treat it.

WHAT IT IS. In anaphylaxis, the immune system reacts to a substance it has become sensitive to, such as an insect sting, a food or a medicine, and releases chemicals that cause the airways to tighten and the blood vessels to widen. Some people have a reaction with no known cause.

WHY IT CAN KILL. Two things fail in anaphylaxis, often together. Swelling and tightening narrow the airway, so air cannot get in. Widened blood vessels leak fluid and drop the blood pressure, so the heart cannot keep the brain supplied. Either can cause

collapse. Because the reaction can recur after it seems to have settled, a patient who improves still needs to be assessed by a clinician.

RECOGNITION. Symptoms can involve several body systems at once. They include hives, itching or redness of the skin, swelling of the face, eyes or tongue, difficulty breathing, wheezing, a tight chest, a hoarse or whispered voice, difficulty swallowing, abdominal pain, vomiting, dizziness or fainting, and anxiety. A warning sign of dangerous throat swelling is a very hoarse voice or coarse noise on breathing in. When a person has a known allergy and develops breathing difficulty, throat swelling or faintness soon after exposure, treat it as anaphylaxis.

RED FLAGS

A very hoarse or whispered voice, noisy breathing in, swelling of the tongue or face, difficulty breathing, faintness or collapse, or symptoms that follow a known trigger within minutes. Call emergency services at once.

FIRST PRIORITIES. Anaphylaxis is an emergency. Call emergency services immediately. Check airway, breathing and circulation, and be ready to start CPR. Remove the trigger if you can, for example by scraping off a bee stinger. If the person has a prescribed epinephrine (adrenaline) auto-injector, help them use it at once; the first-line treatment is epinephrine, and delay is dangerous. Giving medicines is outside many roles; follow your scope and medical direction.

PROCEDURE: HELP A PATIENT USE THEIR OWN EPINEPHRINE AUTO-INJECTOR

Purpose: to help the patient receive their own prescribed, premeasured epinephrine as the device instructions direct. Scope: fol-

lows your training level, state law and local protocol.

1. Call emergency services, or ask someone to call, and stay with the patient.
2. Ask whether they carry an auto-injector, and find it; if there is none, do not delay the call, and keep checking breathing and consciousness.
3. Read the device label and follow its printed instructions exactly; do not guess or improvise.
4. Help the patient or, where your role and the law allow, use the device on the patient as the label shows.
5. Note the time the injection was given and tell the responders.
6. Place the patient in the position that suits their breathing: lying flat with legs raised if they are faint, sitting up if breathing is hard, or on the side if vomiting.
7. Keep checking breathing and consciousness, and be ready to start CPR.
8. Even if the patient improves, they need to be assessed by a clinician; symptoms can return.

> ⚠ **SAFETY**
>
> Do not give anything by mouth to a person who has trouble breathing. Keep your fingers away from the needle end of the device, and put the used device in a sharps container.

STOP AND CALL FOR HELP IF breathing becomes difficult, the voice changes, swelling spreads, or the patient becomes drowsy or faint.

The steps follow the logic of the problem. Anaphylaxis is a race between a reaction that is accelerating and a treatment that works quickly, so the first moves are to bring help and the treatment together. Entry 3.5 explains how a fall in blood pressure becomes shock. Position protects the airway and the blood pressure, and the recorded time lets the next team judge the response.

NURSING AND RESPONDER PRIORITIES. Recognize the pattern early, call for help at once, help the patient use their own auto-injector as the device directs, and keep reassessing until a clinician takes over.

See also: Entry 1.3 (primary survey); Entry 2.2 (airway); Entry 3.5 (shock); Entry 6.1 (children).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Marcus was a first-year paramedic student helping at a weekend youth camp with the camp nurse, Dana. During an afternoon game, a 14-year-old camper named Leo sat down on the grass and said he could not catch his breath. Marcus reached him first, checked that the area was clear, and put on gloves.

Leo was sitting forward, resting his hands on his knees. He could say only a few words between breaths and was breathing fast. Marcus could hear a high wheeze. Leo's lips looked normal, and he was alert. Leo said he had asthma and that his inhaler was in his bag at the cabin, and that he had been running hard in the cold wind.

Marcus told Dana what he saw, and she sent a counselor for the inhaler and called emergency services at once. He kept Leo sitting upright, spoke slowly, and counted his breathing rate at the start and again 2 minutes later. It had not changed. The inhaler arrived, Dana helped Leo use it as he had been taught, and she told Marcus to keep watching his color and how much he could say. When Leo managed a full sentence, they relaxed a little. When the paramedics arrived, Dana handed over what Marcus had counted and the time of the inhaler. Leo was taken in for an assessment, because his breathing had not settled.

Care Priorities

CARE PRIORITIES

When two airway or breathing problems compete, treat the one that stops air first: an obstruction before a wheeze, and a wheeze before a low reading. Use your hands before equipment, because repositioning is quicker than fetching equipment. Treat effort as an early warning, since a patient who has stopped struggling may be tiring rather than recovering. Ask for help before the patient needs it, and say in one sentence what you see, what you have done and what you need. Decide before you start a skill what you will do if it fails, so that a failure does not surprise you at the bedside.

Review Questions

1. A patient's chest and abdomen move in opposite directions and no air can be felt at the mouth. What does this suggest?

   A) Normal breathing.
   B) Complete airway obstruction.
   C) A mild chest cold.
   D) A strong cough reflex.

2. Which technique is preferred for opening the airway of an unresponsive patient with a possible neck injury?

   A) Jaw thrust.
   B) Sitting the patient upright.
   C) Head-tilt chin-lift.
   D) Rolling the patient face down.

3. When should you use suction?

   A) At all times during airway care.
   B) Only after the patient wakes up.
   C) Before checking for a response.
   D) When you can see or hear fluid in the mouth or throat.

4. A conscious adult cannot speak, cough or breathe and points to the throat. Which action is correct?

   A) Ask the person to lie down and rest.
   B) Wait for the person to improve.
   C) Follow your training for back blows and abdominal thrusts.
   D) Begin chest compressions at once.

5. Which is a risk of squeezing a bag-valve-mask too hard or too fast?

   A) Air can enter the stomach which raises the risk of vomiting.
   B) The oxygen will run out.
   C) The mask will fall off.
   D) The patient will start breathing normally at once.

6. A patient with severe asthma becomes quieter and calmer but still has blue lips and can say only a word. What does this suggest?

   A) The attack is improving.
   B) The patient needs only reassurance.
   C) Nothing needs to change.
   D) The patient is tiring and may be failing.

7. After a bee sting, a person develops hives, a hoarse voice and difficulty breathing. What is the first priority?

   A) Wash the area and watch for an hour.
   B) Call emergency services and help a patient who carries one use their own epinephrine auto-injector.
   C) Give a drink to restore strength.
   D) Wait to see whether the symptoms pass.

8 Short case. A patient with COPD has become short of breath at home and is drowsy. List, in order, the first four things you would do and explain why oxygen needs care in this patient.

PART 3

Cardiac Emergencies and Resuscitation

IN THIS PART

1. Describe the pumping and electrical functions of the heart in plain terms.
2. Recognize acute chest pain patterns and acute coronary syndrome, and state first priorities.
3. Recognize a short list of dangerous rhythm patterns and what they mean for the patient.
4. Describe high-quality adult CPR and the use of an automated external defibrillator (AED), step by step.
5. Recognize shock and state the common first priorities.

Some patients arrive with a warning: a pressure in the chest, a sweat that does not fit the weather, a sense that something is wrong. Others arrive with no warning at all, collapsed on a floor. Both stories live in the heart, and this Part follows them in order, from the quiet signs of a heart in trouble to the loud emergency of a heart that has stopped.

The Part begins with the heart itself, because the signs make sense only once you know what the organ does. It then covers chest pain and acute coronary syndrome, a common emergency that threatens the heart muscle, followed by a short guide to the rhythms you may see on a monitor or be told about. The longest entry is cardiac

arrest, high-quality CPR and the AED, since these skills matter in almost every setting where you may work. The Part closes with shock, the state in which the circulation fails to supply the body, whatever the cause.

As in every Part, what you may do depends on your scope of practice and local protocol. Where a task is outside your role, the right action is to call for help and begin what you are trained to do while you wait.

3.1 The Heart in Brief: Pump, Vessels and Electrical System

The heart is a muscular pump about the size of a closed fist, lying in the chest between the lungs. It does two jobs. It moves blood to the lungs to collect oxygen, and it moves oxygen-rich blood to every organ, including the heart muscle itself.

WHAT IT IS. The heart has four chambers. The two upper chambers, the atria, receive blood. The two lower chambers, the ventricles, push it out. The right side sends blood to the lungs, and the left side sends it to the rest of the body. Valves between the chambers keep blood moving in one direction. The heart muscle has its own blood supply through the coronary arteries, which run over its surface. If one of them becomes blocked, the part of the muscle it feeds is starved of oxygen.

Figure 9 shows the four chambers and the two coronary arteries; notice that the arteries run over the outside of the muscle they feed.

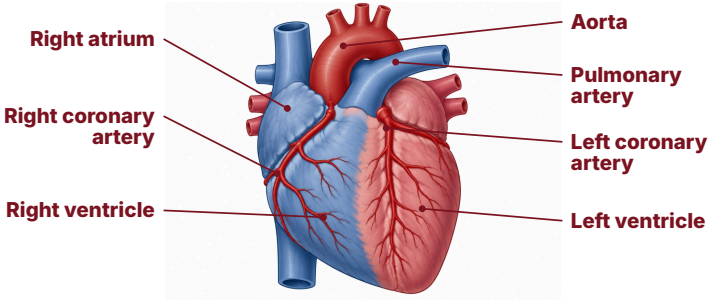


Figure 9. The heart and its coronary arteries, seen from the front.

THE ELECTRICAL SYSTEM. Each heartbeat is triggered by a small electrical signal that starts in the upper chambers and spreads through the muscle in a set order, so that the chambers squeeze in sequence. When the signal is disturbed, the heart may beat too fast, too slow, irregularly, or not effectively at all. A monitor draws that signal as a line, and Entry 3.3 explains what a few of the patterns mean.

HEART FAILURE. The heart can also weaken over time, a condition called heart failure, in which the pump no longer moves blood as well as it should. Patients are often short of breath, tire easily and may have swollen ankles. A flare-up can make breathing hard, particularly when lying flat, so the patient may be more comfortable sitting upright. Treat a worsening breathing problem as an emergency, sit the patient upright, call for help and use the distress priorities of Entry 2.4.

WHY IT MATTERS. A responder who knows that the heart both pumps and conducts can reason about what fails. A blocked coronary artery damages the muscle. A disturbed signal changes the

rhythm. A weak pump lowers the flow of blood.

NURSING AND RESPONDER PRIORITIES. Keep the two functions in mind, the pump and the electrical signal, and ask of any cardiac problem which of them has failed.

See also: Entry 3.2 (chest pain); Entry 3.3 (rhythms); Entry 3.4 (cardiac arrest).

3.2 Acute Chest Pain and Acute Coronary Syndrome

After injuries, chest pain is the second most common reason that adults come to a United States emergency department, and most chest pain turns out not to be a heart attack. The difficulty is that a heart attack can look like many harmless things, so the safe approach treats every serious chest pain as cardiac until a clinician shows otherwise.

WHAT IT IS. Acute coronary syndrome, written ACS, is the name for a group of conditions in which blood flow to part of the heart muscle drops suddenly. The best-known is the heart attack, or myocardial infarction (MI), in which a section of muscle begins to die. Cardiac arrest is a different event: it is the stopping of an effective heartbeat, and it may follow a heart attack, but the two are not the same.

HOW A HEART ATTACK DEVELOPS. Most heart attacks begin when a fatty deposit in a coronary artery cracks and a blood clot forms on it, cutting off flow. The longer the muscle goes without blood, the more of it dies, which is why the call for help comes before any further assessment.

RECOGNITION. The classic picture is pressure, tightness or squeezing in the center of the chest that may spread to the arm, jaw, neck, back or upper abdomen. It may come with shortness of breath, sweating, nausea, light-headedness or a sense of dread. It may start slowly or suddenly, may come and go, and does not ease with rest. The picture is not always classic. Women, older adults and people with diabetes more often have less typical symptoms, such as shortness of breath, nausea, unusual tiredness or discomfort in the back or jaw, with little or no chest pain. A patient who looks calm and says "it's probably indigestion" may still be having a heart attack.

RED FLAGS

Chest pressure or pain lasting more than a few minutes or returning, with shortness of breath, sweating, nausea, faintness, a pale or gray color, or collapse. Call emergency services at once and do not let the patient drive.

WHO IS AT HIGHER RISK. Risk rises with age, smoking, high blood pressure, high cholesterol, diabetes, being overweight, little physical activity and a family history of heart disease. A patient with several of these deserves extra caution, but a patient with none can still have a heart attack, so a risk list is no reason to hold back.

ANGINA AND HEART ATTACK. Angina is chest discomfort that appears when the heart muscle needs more oxygen than a narrowed artery can deliver, usually during effort or stress, and it eases within minutes of rest or of the patient's prescribed treatment. A heart attack happens when a blocked artery stops blood flow long enough for heart muscle to begin to die, so the discomfort does not settle with rest. You will not always know which you are looking at. A patient with known angina whose pain is

stronger than usual, comes on at rest, or lasts longer than their usual attacks should be treated as having a possible heart attack and given the same urgent response.

WHAT TO ASK. Use short, direct questions. When did it start? What were you doing? What does it feel like, and does it spread? Is it constant or does it come and go? Have you had it before, and is it different this time? Do you have heart disease, high blood pressure, diabetes or allergies? Which medicines do you take, and have you taken anything for this? Write the answers down with the times, because the next team will want them and the patient may not be able to repeat them.

OTHER CAUSES. Chest pain also arises from the lungs, the muscles and bones, the food pipe and the large vessels. Some of these are as dangerous as a heart attack. A sudden tearing pain that spreads to the back, pain with a sudden shortness of breath, or pain after an injury deserves the same urgent response. You do not have to decide the cause. Your task is to recognize the pattern that needs urgent help and to act on it.

FIRST PRIORITIES. Call emergency services. Help the patient rest in the position that is most comfortable and keep them still and reassured. Check airway, breathing and circulation and take vital signs. Use the questions above, and ask about allergies before any aspirin is offered. Where protocol allows, a patient who has no allergy to aspirin and has not been told to avoid it may be helped to chew aspirin. Giving medicines is outside many roles; follow your scope and medical direction. Do not give anything by mouth if the patient is faint or has trouble swallowing. Be ready to begin CPR and to use an AED if the patient collapses and stops breathing normally.

SCOPE NOTE

Decisions about medicines for chest pain belong to clinicians acting under medical direction. A student or lay responder calls for help, positions and reassures the patient, monitors, and follows protocol if instructed to assist.

REASSESSMENT. A patient with chest pain can change in minutes. Ask again about the pain at short intervals and record the time and the answer. Repeat the vital signs and note the color and sweating. Report any change in the pain, new breathlessness, faintness or a change in the pulse at once, since each may mean the heart is under greater strain. A patient who was comfortable and becomes pale and sweaty has deteriorated, whatever the numbers say.

NURSING AND RESPONDER PRIORITIES. Treat serious chest pain as cardiac, call early, keep the patient still, monitor closely for deterioration, and be ready for cardiac arrest.

See also: Entry 1.3 (primary survey); Entry 3.4 (cardiac arrest); Entry 6.4 (chest pain in older adults).

3.3 Recognizing Dangerous Rhythms

A heart monitor draws the electrical signal of the heart as a moving line. You do not need to interpret it as a specialist would. You need to recognize a small number of patterns that tell you how worried to be, and to know what to do when a monitor shows one.

WHAT IT IS. A rhythm is the pattern of the heart's electrical activity over time. The normal pattern shows a regular series of small and large waves with a steady rate. Abnormal rhythms may

be too fast, too slow, irregular, or chaotic. A rhythm is dangerous when it prevents the heart from filling or emptying, so that blood pressure falls and the brain and heart go short of oxygen.

WHY IT MATTERS. A monitor tracing is only a picture of electricity, and it can mislead. A patient can have a normal tracing and no effective pulse, and a patient can have an abnormal tracing from a loose lead or movement and feel quite well. Read the tracing together with the patient's responsiveness, breathing and skin color.

Figure 10 shows four of these patterns, simplified; notice that the regular line differs from the chaotic one in spacing as well as in shape.

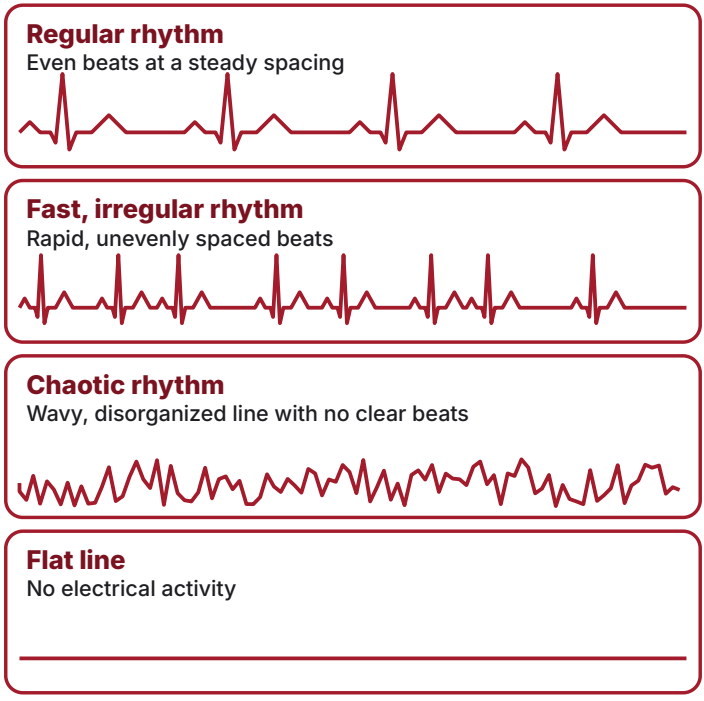


Figure 10. Simplified drawings of a regular rhythm, a fast irregular rhythm, a chaotic rhythm and a flat line.

Rhythm patterns and what to do

Pattern	What it looks like	What it means
Regular rhythm	Even beats at a steady spacing	Usually reassuring if the patient is well
Fast or irregular rhythm	Rapid or uneven beats	May cause faintness or chest pain; call for help
Chaotic rhythm	A wavy, disorganized line with no clear beats	The heart is quivering and not pumping; if the patient is also unresponsive and not breathing normally, this is cardiac arrest
Slow rhythm	Widely spaced beats	May cause faintness, confusion or collapse; call for help if the patient has symptoms
Flat line	No electrical activity	Check the leads and the patient, then treat as cardiac arrest if the patient is unresponsive

TAKING A PULSE. Entry 1.4 described the pulse check. With a monitor attached, compare your count with the monitor's rate; if the two do not match, report the difference and trust your assessment of the patient. If you are a trained health care professional, check for a pulse at the neck of an unresponsive patient for no more than 10 seconds; if you are not sure you feel one, begin compressions.

NAMING THE PATTERNS. Clinicians give these patterns names. The chaotic pattern is usually ventricular fibrillation, and a flat line is called asystole. A very fast, regular pattern can also stop the heart from pumping, which is why a fast rhythm in a faint patient needs urgent help. An AED can shock a chaotic pattern but does not shock a flat line, so when it advises no shock you keep doing compressions. A slow rhythm can be as dangerous as a fast one because too little blood leaves the heart. Atrial fibrillation, a common irregular rhythm in older adults, is not always an emergency by itself, but you report it and watch the patient.

WHAT TO DO. Look at the patient first. A flat line on a monitor in a patient who is awake and talking is most likely a loose lead or an equipment problem, so check the lead and the monitor. A chaotic pattern in an unresponsive patient who is not breathing normally is cardiac arrest, and the response is CPR and defibrillation as soon as a defibrillator is ready, as Entry 3.4 describes. A fast or irregular rhythm in a patient who is faint, has chest pain, or is short of breath needs urgent help from a clinician. Report what you see: the rate, whether the beats are regular, and how the patient looks.

RED FLAGS

Faintness, chest pain, shortness of breath or confusion with a fast, slow or irregular rhythm, or any monitor change in a patient who is unresponsive and not breathing normally. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Read the patient before the screen, and start CPR for an unresponsive patient who is not breathing normally.

See also: Entry 3.1 (electrical system); Entry 3.4 (CPR and AED).

3.4 Cardiac Arrest, High-Quality CPR and the AED

Cardiac arrest is the sudden stop of effective heartbeat, and it is the emergency in which a bystander's actions matter most. A patient in cardiac arrest is unresponsive and is not breathing normally. Blood stops flowing to the brain, and the patient collapses and loses consciousness. Chest compressions keep a little blood moving, and the AED is the tool that can correct a dangerous rhythm so that the heart can resume pumping.

WHAT IT IS. Cardiac arrest happens when the heart stops pumping blood effectively. Causes include a heart attack, a dangerous rhythm, severe bleeding, a blocked airway, drowning, poisoning and injury. It is different from a heart attack, in which the heart usually keeps beating. Cardiopulmonary resuscitation (CPR) combines chest compressions with rescue breaths. An AED is a portable device that analyzes the heart rhythm and, if it finds a rhythm that can be corrected by a shock, tells the rescuer to deliver one.

WHY IT MATTERS. Early CPR and prompt defibrillation are the most important interventions for survival, and a bystander decides how soon they begin. Every minute without CPR and defibrillation lowers the chance of survival, which is why the first person on the scene must start at once, before the cause is known. Good compressions are hard work and can be done poorly, so quality and the swapping of rescuers matter as much as speed of starting.

RECOGNITION. Check for a response by tapping the shoulders and shouting. Look for breathing for no more than 10 seconds. Occasional gasps are not normal breathing and are a sign of cardiac arrest. If the patient is unresponsive and is not breathing, or

is only gasping, treat it as cardiac arrest. If you are a trained health care professional, you may check for a pulse during the same 10 seconds; if you are not sure you feel one, begin compressions.

RED FLAGS

Unresponsive, no breathing or only occasional gasps, no response to voice or touch, a gray or blue color, or a collapse with no warning. Call emergency services, send for an AED and begin compressions at once.

THE SEQUENCE. Press hard and fast, at a rate of 100 to 120 compressions a minute, to a depth of at least 2 inches (5 centimeters) but not more than about 2.4 inches (6 centimeters) in an adult, and let the chest return fully between compressions. Keep interruptions as short as you can. If you are trained and willing to give breaths, give 2 breaths after every 30 compressions, each lasting about 1 second and just enough to make the chest rise. If you are not trained, continue compressions alone. If you are alone, put the phone on speaker, call emergency services, begin compressions, and use an AED as soon as one is within reach, as your training directs.

Figure 11 shows the hand and body position.

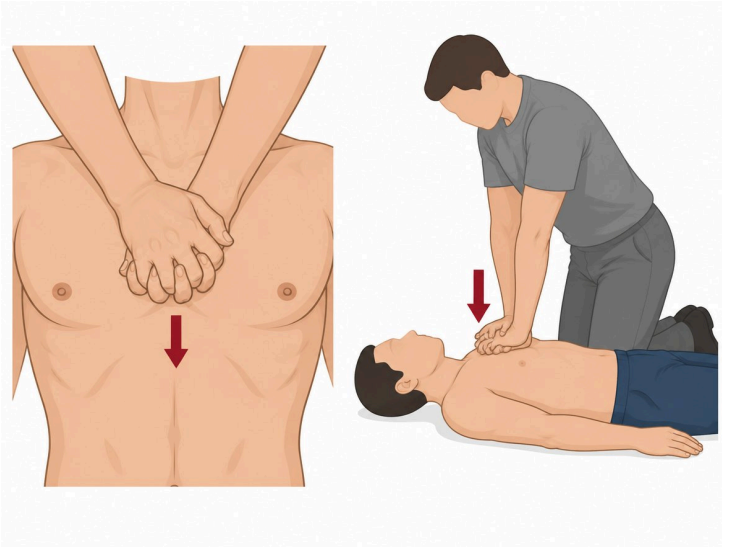


Figure 11. The hand position and the straight-armed body position for chest compressions on an adult.

PROCEDURE: ADULT CPR WITH AN AED

Purpose: to keep blood moving and to deliver a shock when the AED advises one. Scope: follows your training level and local protocol.

1. Check that the scene is safe, tap the shoulders and shout; if there is no response, call for help.
2. Send someone for an AED and ask them to call emergency services.
3. Look for breathing for no more than 10 seconds; if the patient is not breathing normally, begin compressions.

4. Place the heel of one hand on the lower half of the breastbone, with the other hand on top, arms straight.
5. Press at 100 to 120 a minute to a depth of at least 2 inches (5 centimeters), letting the chest rise fully each time; if you are trained, give 2 breaths after every 30 compressions.
6. When the AED arrives, turn it on and follow its voice prompts; attach the pads as shown on them.
7. Make sure no one is touching the patient while the AED analyzes and again before a shock is given.
8. Resume compressions immediately after the shock or if no shock is advised, and swap rescuers about every 2 minutes.
9. Continue until emergency responders take over, the patient moves or breathes normally, or you are too exhausted to continue.

Figure 12 shows the whole sequence as one flow, from collapse to handover.

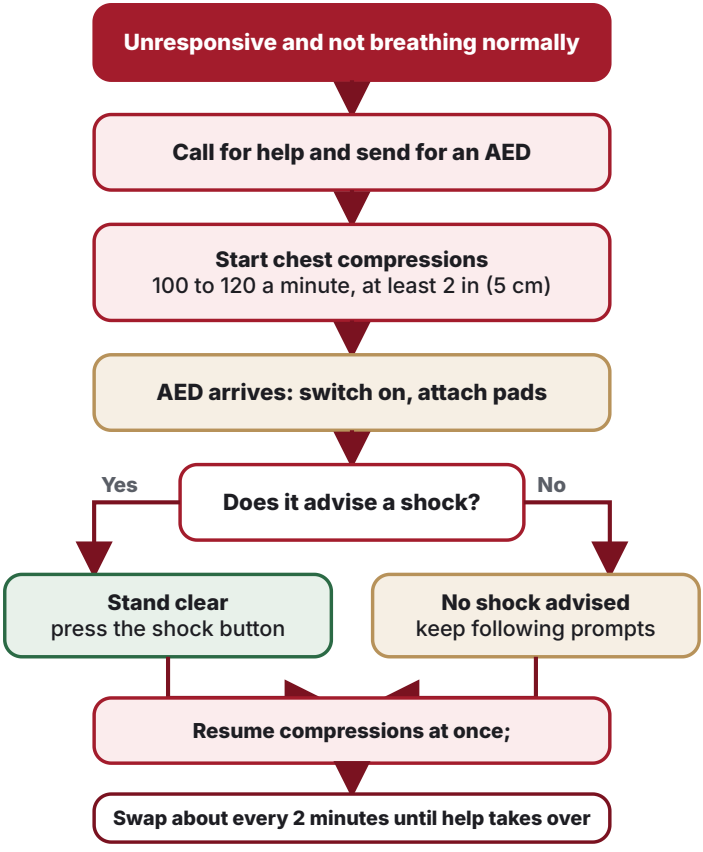


Figure 12. The adult cardiac arrest response from recognition to handover.

> **⚠ SAFETY**
>
> Do not touch the patient while the AED analyzes or delivers a shock. Dry a wet chest and move the patient away from standing water before attaching the pads. Do not delay compressions to look for the cause.

CALL FOR HELP AT ONCE IF the patient is unresponsive and not breathing normally, and ask someone to take over if you cannot continue.

The steps follow the physiology. Compressions create a small flow of blood to the brain and heart, so they begin first and are interrupted as little as possible. The AED is used as soon as it arrives, and compressions continue until it is switched on, the pads are attached and it is ready to analyze the rhythm. Swapping rescuers about every 2 minutes keeps the quality of compressions from falling as arms tire.

USING THE AED. An AED is designed for people with no training. When you open it or switch it on, it talks you through every step. Bare the chest, dry it, and place the two sticky pads as the picture on the pads shows. The device then reads the rhythm. Do not touch the patient while it does. If it advises a shock, check that everyone is clear and press the shock button. If no shock is advised, resume compressions. Either way, leave the pads on and keep following its prompts. For AED use, adult pads and the adult steps are used for a child aged about 8 years or older. Entry 6.2 covers CPR and AED differences in infants and younger children.

REMEMBER

Hard, fast, clear and continuous: press hard, press fast, let the chest rise, and keep interruptions short. If the patient is unresponsive and you are unsure whether the breathing is normal, start compressions.

COMPRESSION QUALITY. Compressions are weakened by pressing too shallowly or too deeply, pressing too slowly, too fast or unevenly, and leaning on the chest between compressions so that it cannot recoil. Keep your shoulders over your hands and your arms straight so that the weight of your upper body does the work. A firm surface under the patient helps. Counting aloud keeps the rate steady. Compressions can break ribs, and this is not a reason to stop; the priority is to keep blood moving.

SPECIAL SITUATIONS WITH THE AED. Remove any medication patch from the chest, with a gloved hand if you have one, and wipe the skin before attaching the pads. Do not place a pad directly over a lump under the skin where a pacemaker or similar device sits; place it a short distance away. Chest hair rarely stops pads from sticking, so do not delay the shock to remove it. If the patient is a child younger than about 8 years, use pads or a setting for children if the device has them, and use the adult pads if it does not. Follow the device's voice prompts first.

HANDS-ONLY CPR. A bystander who has no training, or who is unwilling or unable to give breaths, can still help by giving continuous chest compressions at the rate and depth described above until the AED arrives or help takes over. Compressions alone are far better than waiting. A trained responder gives breaths as described, particularly for patients whose arrest follows drowning, a drug overdose or a breathing problem.

AFTER THE PATIENT RESPONDS. If the patient begins to breathe normally and responds, keep the airway open, keep the patient warm and still, leave the AED pads in place, and keep watching. Keep checking that the patient is still breathing normally. Be ready to resume compressions at once and report to the paramedics exactly what you did and when.

PREGNANCY. In late pregnancy, the enlarged uterus can press on the large vessels and reduce the blood returning to the heart. Entry 6.3 describes the adaptation. Begin CPR without delay; Entry 6.3 describes how the uterus is moved to the left by hand while compressions continue.

NURSING AND RESPONDER PRIORITIES. Recognize the arrest at once, call for help and an AED, press hard and fast with few interruptions, and follow the AED until help takes over.

See also: Entry 1.3 (primary survey); Entry 2.2 (choking); Entry 6.2 (child and infant); Entry 6.3 (pregnancy).

3.5 Shock: Recognition and First Priorities

Shock is not an emotional state. It is a failure of the circulation to deliver enough blood and oxygen to the organs, and it can have several causes. It is hard to spot in its early stages because the body compensates, and a patient who is in shock may look calm for a time.

WHAT IT IS. The circulation needs four things: enough blood volume, a pump that works, vessels that are the right size, and a clear path for the blood to flow. Shock occurs when one of them fails. The blood volume may fall from bleeding or fluid loss. The pump may fail, as in a heart attack. The vessels may widen abnormally, as in a severe infection or an allergic reaction. Something

may also block the flow, as with air trapped around a lung or fluid around the heart. In each case the organs receive too little, and without treatment the failure spreads.

FOUR BROAD TYPES. Grouping shock by what has failed helps you recognize the cause and anticipate the first actions.

Four types of shock

Type	What has failed	Examples
Low blood volume	Too little blood or fluid in the circulation	Severe bleeding, burns, severe vomiting or diarrhea
Pump failure	The heart cannot move blood effectively	Heart attack, dangerous rhythm
Widened vessels	Vessels dilate and blood pressure drops	Severe infection, anaphylaxis
Blocked flow	Something prevents blood from moving	Collapsed lung under pressure, fluid around the heart

WHY IT MATTERS. The body protects the brain and heart by narrowing the vessels elsewhere and speeding the heart. In this early, compensated stage the blood pressure may stay normal, so a normal pressure does not exclude shock. When the compensation runs out, the stage is called decompensated, the pressure falls suddenly and the patient can deteriorate within minutes. The aim is to recognize shock before that point.

EARLY ACTIONS BY TYPE. The first actions depend on the cause. For heavy bleeding, control it (Entry 4.2). For anaphylaxis, call for help and, where protocol allows, help with the patient's own epinephrine auto-injector. Giving medicines is outside many

roles; follow your scope and medical direction. For a suspected heart cause, call for help, keep the patient still and be ready to start CPR. For a suspected infection, call for help and report the signs, because early treatment is given by clinicians. In every case, keep the patient warm and recheck often. Lay the patient flat unless breathing is difficult or the cause is a heart problem or chest pain, when the most comfortable position is safer.

RECOGNITION. Look for a fast heart rate, a fast breathing rate, cool, pale or mottled skin, a slow capillary refill, a weak pulse, new confusion, anxiety or drowsiness, and thirst. The pressure may be normal early and low later. Ask what has happened: bleeding, an injury, vomiting, an infection, an allergic reaction or a heart problem. Different causes lead to different first actions, which are described in their home entries.

RED FLAGS

Cool, pale or mottled skin, a fast weak pulse, fast breathing, new confusion or drowsiness, a low blood pressure, or a patient who looks worse than the numbers suggest. Call for help at once.

FIRST PRIORITIES. Call for help. Control any visible severe bleeding first, if your protocol directs it. Lay the patient flat unless they are struggling to breathe or have chest pain, and in that case let them take the most comfortable position; keep them warm, and do not give food or drink. Give oxygen as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction. Recheck the vital signs at short intervals and write down the times. Entry 4.2 covers bleeding control, Entry 2.5 covers anaphylaxis, and Entry 5.4 covers sepsis.

REASSESSING SHOCK. Shock changes quickly, so recheck at short intervals. A heart rate that is falling toward normal, a rising blood pressure, warmer skin and clearer thinking suggest the pa-

tient is responding. A rising heart rate, a falling blood pressure, cooler skin and more drowsiness suggest the patient is worse, and you should tell the clinician at once. Record each set of observations with the time, so that the trend is visible to the team that takes over.

NURSING AND RESPONDER PRIORITIES. Recognize shock from the whole pattern as well as the pressure, call for help, treat the cause you can see, and reassess often.

See also: Entry 1.4 (vital signs); Entry 2.5 (anaphylaxis); Entry 4.2 (bleeding); Entry 5.4 (sepsis).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Nadia worked the front desk at a community gym and was also a first-year EMT student. One evening a regular member, Frank, a man in his fifties, sat down on a bench after a warm-up and said he felt a pressure in his chest and a bit sick. He looked gray and was sweating. Nadia asked him to stay seated, called the emergency number, and asked a colleague to bring the AED from the wall near the desk.

While she waited, she checked his breathing and his pulse and kept him calm. A few minutes later, Frank slumped sideways and stopped answering. Nadia lowered him to the floor, shouted his name, and saw that he was not breathing normally; he gave an occasional gasp. She called to another member to tell the dispatcher that Frank had collapsed and was not breathing normally, and began chest compressions at the center of the chest, pressing hard and counting aloud to keep the pace.

The colleague returned with the AED. Nadia kept pressing while he opened it and placed the pads as the picture showed. The machine said it was analyzing the rhythm and told everyone to stand clear. Nadia lifted her hands. It advised a shock, and the colleague pressed the button after they all checked they were clear. Nadia resumed compressions at once. At the next rhythm check, Frank gave a deep breath. When the paramedics arrived, Nadia handed over what she had seen and the times.

Care Priorities

CARE PRIORITIES

When two cardiac problems compete, rank them by how soon each could end in an arrest. The patient who is unresponsive and not breathing normally comes first, ahead of every question you were about to ask. Next comes the awake patient whose chest pain is changing, because the trend decides how urgent the call becomes. Behind both stands the patient whose numbers look acceptable but whose skin, color or thinking has changed; trust that change over the number. Before a shift begins, find out where the nearest AED is kept and who is expected to fetch it, because a search made during an arrest spends minutes that compressions need. When you hand over, give the time of the collapse and the time of the first compression and the first shock, so the next team can judge how long the heart has gone without help.

Review Questions

1. Which statement distinguishes a heart attack from cardiac arrest?
   - **A)** They are the same event with two names.
   - **B)** A heart attack always causes cardiac arrest.
   - **C)** In a heart attack the heart muscle is injured, while in cardiac arrest the heart has stopped pumping effectively.
   - **D)** Cardiac arrest only happens in older adults.

2. A man in his fifties has had pressure in the center of his chest for 10 minutes, with sweating and nausea. What is the best first action?

   A) Call emergency services and keep him still and comfortable.
   B) Wait to see whether the pain eases.
   C) Have a family member drive him to the hospital.
   D) Let him rest and recheck in an hour.

3. On a monitor, a flat line appears in a patient who is awake and talking. What is the most likely explanation?

   A) Cardiac arrest.
   B) A loose lead or electrode.
   C) A normal rhythm.
   D) A heart attack.

4. For an adult in cardiac arrest, what rate of chest compressions is recommended?

   A) 60 to 80 a minute.
   B) 80 to 100 a minute.
   C) 120 to 160 a minute.
   D) 100 to 120 a minute.

5. What should you do when an AED advises a shock?

   A) Check that everyone is clear, then press the shock button.
   B) Keep pressing on the chest so that no time is lost.
   C) Turn the AED off and wait for the paramedics.
   D) Remove the pads and start again.

6. Which finding is an early sign of shock?

   A) A very high blood pressure.
   B) A strong pulse with warm hands.
   C) A fast heart rate with cool, pale skin.
   D) A very slow breathing rate.

7. Which statement about blood pressure in shock is correct?

   A) A normal blood pressure rules out shock.

   B) Blood pressure falls first, before any other sign.

   C) Blood pressure is the only reliable sign of shock.

   D) Blood pressure may stay normal for a time as the body compensates.

8. Short case. You are visiting a neighbor at home when he says his chest feels tight. A moment later he slumps in his chair, does not answer when you shout, and only gasps now and then. An AED hangs in the hallway. List, in order, the first five things you would do and say why compressions come before anything else you could do.

PART 4

Trauma, Burns and Environmental Emergencies

IN THIS PART

1. Use the mechanism of injury to raise suspicion of serious injury.
2. Describe the order of external bleeding control (pressure, packing, tourniquet) and the steps of each.
3. Recognize head, spine and chest injuries and state first priorities, including the principles of spinal motion restriction.
4. Describe basic splinting and the checks that go with it.
5. Estimate burn size and state first priorities.
6. Recognize heat and cold emergencies.

Injury is the emergency where what you see is often the least of it. A small wound may sit over a large bleed. A person who walks away from a crash may have an unstable spine. A burn that looks minor may cover a large share of the body. This Part teaches you to read the story of the injury before you decide what to do, and then to act on the problems that kill fastest.

The six entries move from the general to the particular. Entry 4.1 sets out how to size up a trauma patient and what the mechanism of injury tells you, and Entry 4.2 deals with bleeding, the one

trauma problem that a bystander can control in minutes. Injuries to the head, spine and chest come next, followed by broken bones and splinting. Burns have their own entry, and the Part closes with the emergencies caused by heat and cold.

Everything here builds on the safe scene and the primary survey of Part 1. The skills in this Part are basic and non-invasive, and each is taught in a course before it is done on a patient. Your scope of practice and local protocol decide what you may do.

4.1 Trauma Assessment and Mechanism of Injury

A trauma patient may look much better or much worse than they are. The mechanism of injury, meaning how the injury happened, is your best early clue to how worried to be.

WHAT IT IS. Trauma assessment applies the primary survey of Entry 1.3 to an injured patient, with two changes. First, as Entry 1.3 noted, life-threatening external bleeding is controlled before the airway, because a patient can lose a dangerous amount of blood in minutes; your protocol sets the exact order. Second, you consider the mechanism of injury from the first look and keep it in mind, so that the examination looks for the injuries that the mechanism makes likely.

WHY IT MATTERS. The body hides serious injury. A patient, and a child in particular, can keep a normal blood pressure for a time while bleeding inside, as Entry 3.5 explained for shock. A patient with a distracting injury, such as a broken leg, may not mention neck pain. If you wait for the patient to complain of every problem, you will miss some of them. The mechanism tells you where to look before the patient can.

HIGH-RISK MECHANISMS. Treat these as a warning that serious injury may be present even when the patient looks well: a fall from a height of more than about 10 feet (3 meters), a serious vehicle crash (for example, one in which the patient was thrown out, the passenger space was pushed in, or the patient had to be cut free), a rider thrown from a motorcycle or a similar vehicle, a crash in which another person in the same vehicle died, a pedestrian or cyclist thrown, run over or hit hard by a vehicle, a penetrating wound to the head, neck or trunk, and a crush injury. An injury that does not fit the story given, or repeated injuries in a child, may point to abuse, which Entry 6.6 covers. Age and condition change the risk. Older adults and people on blood-thinning medicines can bleed seriously after an apparently minor fall, and children may look well until they suddenly deteriorate.

> **REMEMBER**
>
> Read the scene before you read the patient: a smashed windshield, a bent steering wheel, a cracked helmet or a ladder beside the person tells you the mechanism before the patient can.

HIDDEN BLEEDING. Blood can collect where you cannot see it: in the chest, the abdomen, the pelvis and around a broken thigh bone. A patient with a tender, swollen or rigid abdomen, a painful pelvis, or a thigh that is swollen and deformed may be losing a great deal of blood inside the body. Do not press on the pelvis to test it, and do not rock it. Leave the examination of these areas to the clinicians, report what you see, and treat the signs of shock as they appear.

THE TRAUMA SURVEY. Begin with a safe scene and a look for bleeding. Control severe bleeding at once. Then work through airway, breathing, circulation, disability and exposure, using the jaw thrust in preference to the head tilt if a spinal injury is possible, as

Entry 2.2 described. Expose the body enough to look for injuries and cover the patient again. Take vital signs and write down the times. Call for help early, since a patient with a significant mechanism and any sign of instability needs a trauma center, and transport decisions belong to the responding clinicians.

RED FLAGS

Severe bleeding, a patient who is confused or drowsy after an injury, difficulty breathing, a rigid or swollen abdomen, a deformed or unstable pelvis, a fast weak pulse with pale cool skin, or a high-risk mechanism with any of these. Call for help at once.

KEEPING THE PATIENT WARM. An injured patient loses heat fast, especially when wet or lying on the ground, and a cold patient may clot blood less well. Cover the patient with a blanket or coat, put something between the patient and the ground, and remove wet clothing where you can do so without moving an injured limb. Do this while you assess, because the habit protects the patient from the start. Take care not to overheat a patient in a hot setting.

REASSESSING AFTER INJURY. The first findings in an injured patient are only the first. Repeat the primary survey at short intervals and after any move. Recheck bleeding, since a dressing that was dry may be wet, and recheck the level of consciousness, since a patient with a head injury can worsen after an alert interval. Record each set of findings with the time and tell the team that takes over what changed.

NURSING AND RESPONDER PRIORITIES. Control life-threatening bleeding first, as your protocol directs, then work through the primary survey with the spine in mind, and call early for any high-risk mechanism.

See also: Entry 1.3 (primary survey); Entry 3.5 (shock); Entry 4.2 (bleeding); Entry 4.3 (spine); Entry 6.1 (children); Entry 6.6 (abuse).

4.2 Bleeding and Hemorrhage Control

Severe bleeding can kill within minutes. It is a leading cause of preventable death after injury, and a bystander can act on it before help arrives; a short course teaches the skills in this entry.

WHAT IT IS. Hemorrhage is severe bleeding. External hemorrhage is the kind you can see and reach. The three techniques are direct pressure, wound packing and a tourniquet, and they are used in escalation: pressure for most wounds, packing for deep wounds where pressure alone is not enough, and a tourniquet for life-threatening bleeding from an arm or a leg that pressure does not control. When a limb is bleeding massively, as after an amputation, a tourniquet may be applied first, as your training directs.

WHY IT MATTERS. The adult body holds a limited amount of blood, and a heavy bleed can drain it fast. Bleeding that spurts, pools on the ground, soaks through clothing, or does not stop with firm pressure is life-threatening. The signs of shock from blood loss are those of Entry 3.5. The best moment to act is before they appear.

DIRECT PRESSURE. Press firmly on the wound with a clean cloth, gauze or a dressing, using both hands. Do not lift the dressing to look. If blood soaks through, add more material on top and keep pressing. Hold pressure until help arrives or until the bleeding stops. This is the first step for most bleeding.

WOUND PACKING. When the wound is deep and the bleeding does not stop under pressure, pack the wound. Push gauze, or gauze that contains a clotting agent if you have it, deeply into the wound and onto the bleeding source until the wound is filled, then press firmly over it with both hands and keep pressing. Pressure over a clotting-agent dressing is held for at least 3 minutes, or as your protocol directs.

TOURNIQUETS. A tourniquet is a band that is tightened around an arm or leg above a wound to stop blood flow. It is for life-threatening bleeding from a limb that has not been controlled by other measures. It works, and it hurts. Apply it between the wound and the heart, about 2 to 3 inches (5 to 7.5 centimeters) above the edge of the wound and not over a joint. If the wound lies just below a joint such as the knee or elbow, place the tourniquet above the joint, not over it. If the source of the bleeding cannot be found or it is unsafe to look, apply it as high on the limb as possible. Tighten it until the bleeding stops, and note the time. Do not loosen it; leave that to clinicians.

WOUNDS WHERE A TOURNIQUET DOES NOT FIT. A tourniquet works only on a limb. Bleeding from the groin, the armpit, the shoulder or the neck is too close to the trunk for an ordinary limb tourniquet, and pressure and packing are the tools there; trained clinicians may use a special device at the groin or armpit. Press hard and keep pressing, and call for help at once. A nosebleed is usually managed by sitting upright, leaning forward and pinching the soft part of the nose, but a nosebleed after a head injury needs medical help.

LIMB LOSS AND AMPUTATIONS. A partly or fully severed limb can bleed heavily. Control the bleeding first with firm pressure, and use a tourniquet if the bleeding is life-threatening and

pressure does not control it. Wrap any severed part in clean gauze or cloth, place it in a plastic bag and keep the bag cool, never with the part itself directly on ice, and send it with the patient.

TOURNIQUET PITFALLS. A tourniquet that is too loose can make bleeding worse, because it blocks the veins that carry blood away while the arteries still push blood in, so tighten until the bleeding stops. A tourniquet is painful, and the patient may beg you to remove it; explain why it stays on. If one tourniquet does not stop the bleeding, a second may be placed next to it, closer to the heart, as your training directs. Leave the tourniquet visible so that no one overlooks it.

Figure 13 shows direct pressure and packing; notice that the gauze fills the wound and the hands press over it.

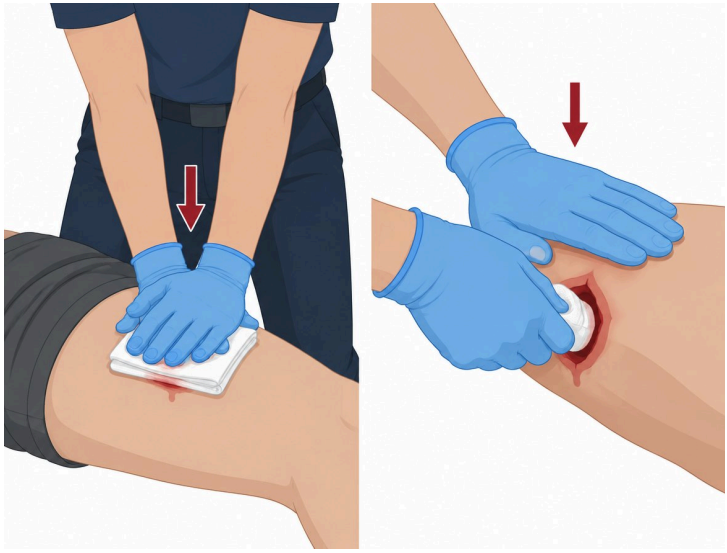


Figure 13. Direct pressure over a wound and gauze packed into a deep wound.

Figure 14 shows a tourniquet; notice that it sits between the wound and the heart, above the knee and not over the joint.



Figure 14. A tourniquet applied between a bleeding wound and the heart, above the knee and not over the joint.

PROCEDURE: CONTROL SEVERE BLEEDING

Purpose: to stop life-threatening bleeding while help comes. Scope: follows your training level and local protocol.

1. Make the scene safe, put on gloves, and call for help.
2. Find the bleeding and expose the wound.
3. Press firmly on the wound with a clean dressing or cloth, and keep pressing.
4. If the wound is deep and bleeding continues, pack it with gauze and press over the packing.

5. If bleeding from an arm or leg is severe and not controlled (or is massive, as after an amputation, and your training directs a tourniquet first), apply a tourniquet between the wound and the heart, 2 to 3 inches (5 to 7.5 centimeters) above the wound and not over a joint (or above the joint if the wound is just below it), and tighten it until the bleeding stops.
6. Write down the time the tourniquet was applied and tell the responders.
7. Keep the patient lying down, warm and calm, and watch for signs of shock.
8. Do not lift the dressings or loosen the tourniquet; if blood soaks through, add more material on top.

> ⚠ **SAFETY**
>
> Wear gloves and avoid contact with blood. Do not remove an object that is stuck in a wound; pad around it and press around it. A tourniquet must not be placed over a joint. Do not use a tourniquet for bleeding from the neck, chest or abdomen.

STOP AND CALL FOR HELP IF bleeding does not slow, the patient becomes pale, confused or faint, or you cannot reach the wound.

The sequence follows what works. Pressure acts on the vessel directly and is available to everyone. Packing reaches a bleeding source that pressure on the surface cannot. A tourniquet stops flow to the whole limb, which is why it is held back for the bleeding that other methods have failed to control. Writing down the time matters because the receiving team must know how long the limb has been without blood.

RED FLAGS

Blood that spurts or pools, a dressing that soaks through, pale or cold skin, faintness or confusion, a fast weak pulse, or bleeding from the neck, groin or armpit. Call for help at once and keep pressing.

NURSING AND RESPONDER PRIORITIES. Control life-threatening bleeding before the airway, as your protocol directs: press, pack deep wounds, use a tourniquet for limb bleeding that stays uncontrolled, and record the time.

See also: Entry 1.3 (primary survey); Entry 3.5 (shock); Entry 4.1 (trauma); Entry 6.1 (tourniquets in children).

4.3 Head, Spine and Chest Injuries

These injuries threaten the organs that control breathing and thinking, and they often occur together. Your task is to recognize them, to avoid making them worse, and to get help.

HEAD INJURIES. A blow to the head can cause a bruise, a concussion, bleeding inside the skull, or a skull fracture. The patient may have a headache, nausea, confusion, drowsiness, a change in the pupils, or a seizure. A patient who becomes more drowsy, more confused or who vomits repeatedly after a head injury is worsening and needs urgent help. Check the level of consciousness with the scale described in Entry 1.3 and recheck it, since the trend tells you most. Scalp wounds can bleed heavily and are controlled with firm pressure, but if the skull may be broken (a dent or soft area, visible bone fragments, or clear fluid from the nose or ears), do not press hard on the wound; cover it with a clean dressing and press only around it.

SPINAL INJURIES. Suspect a spinal injury after a high-risk mechanism, after a head injury, or when the patient has neck or back pain, numbness, weakness or tingling, is confused or intoxicated, or has another very painful injury that could hide neck pain. The aim is to limit movement of the spine until the patient is assessed. If the airway needs opening, use the jaw thrust, which moves the neck less than a head tilt; Entry 2.2 described it. Spinal motion restriction means keeping the head, neck and trunk in line and minimizing movement during care and transfer. It does not mean tying a patient rigid to a board in every case, and practice has changed in recent years, so follow your local protocol. In the first moments, a responder can hold the head gently in line with the body and ask the patient not to move.

FACE AND NECK INJURIES. Facial injuries can bleed heavily and bring blood and swelling into the airway. If the patient is awake and there is no spinal concern, let the patient take the position that lets blood drain from the mouth best, usually sitting up and leaning forward, and have suction ready if you are trained and your protocol allows. A wound to the front of the neck can bleed heavily and can injure the airway. Apply direct pressure with a clean dressing, enough to control the bleeding without squeezing the airway, and call for help.

HELMETS AND OTHER EQUIPMENT. A helmet protects the head and can hide a face or neck injury. Remove it only if your training and protocol direct, and if you must, do it with a second person who holds the head and neck in line. Do not remove a helmet that fits well if the patient is breathing without difficulty and help is coming. Leave in place any object that is stuck in a wound.

CONCUSSION. A concussion is a brain injury from a blow or a jolt. The patient may be dazed, forget what happened, have a headache, feel dizzy or nauseous, or be sensitive to light and noise.

A concussion can be present with no loss of consciousness. Any patient with a suspected concussion should be assessed by a clinician, should not return to sport that day or until a healthcare provider clears it, and should be watched for the warning signs in the Red Flags box.

CHEST INJURIES. Chest injuries include broken ribs, a bruised lung, a collapsed lung and bleeding into the chest. Recognize them by pain on breathing, fast shallow breathing, difficulty breathing, bruising, an abnormal movement of the chest wall, or the sound of air entering a wound. They need urgent help. This book teaches recognition only. Give oxygen as directed, let the patient find the position of greatest comfort, and cover an open chest wound as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction.

Figure 15 shows the chest wall and lungs; notice that the ribs surround the lungs on every side and that the diaphragm sits below them.

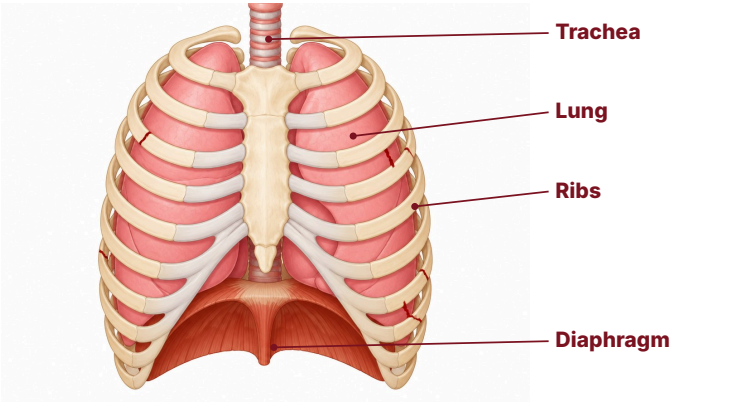


Figure 15. The chest wall and lungs, with the ribs, trachea and diaphragm labeled.

RED FLAGS

Worsening drowsiness or confusion, repeated vomiting, a seizure, unequal pupils, clear fluid from the nose or ears, weakness or numbness in the limbs, severe neck pain, difficulty breathing, or a wound that sucks air. Call for help at once.

SAFETY

Do not move a patient with a possible spinal injury unless the scene is unsafe or your protocol directs. If you must move the patient, keep the head, neck and body in line. Do not remove a helmet unless your training and protocol direct it.

STOP AND CALL FOR HELP IF the airway or breathing is threatened, the patient cannot be kept still, or the scene becomes unsafe.

SCOPE NOTE

Spinal motion restriction techniques are taught and practiced with supervision. A student or lay responder keeps the patient still, holds the head gently in line if trained, and waits for the responders.

NURSING AND RESPONDER PRIORITIES. Suspect the spine after a high-risk mechanism, limit movement, open the airway with a jaw thrust, and watch the level of consciousness for change.

See also: Entry 2.2 (jaw thrust); Entry 4.1 (mechanism); Entry 6.1 (children and the spine).

4.4 Fractures, Dislocations and Splinting

A broken bone is painful and often obvious, but the injury that matters most may be to the nerves and blood vessels beside it. Splinting protects them.

WHAT IT IS. A fracture is a break in a bone. A dislocation is a joint forced out of its normal position. A sprain is a stretched or torn ligament. The three can look alike. Treat any painful, swollen or deformed limb after an injury as a possible fracture, since you cannot tell by looking.

CHECKING CIRCULATION, SENSATION AND MOVEMENT. These three checks are repeated before and after a splint. Circulation: feel for a pulse beyond the injury, press a fingertip or toe and watch the color return, the capillary refill check of Entry 1.3, and compare the limb with the other side. Sensation: touch a fingertip or toe lightly and ask whether the patient can feel it. Movement: if it causes no pain, ask the patient to wiggle the fingers or toes beyond the injury; do not ask the patient to move the injured part itself. A limb that loses any of the three after splinting is a sign that the splint is too tight or that the injury has worsened.

RED FLAGS

A limb that is pale, blue, cold or without a pulse, an open wound with bone showing, a deformed thigh or pelvis, or signs of shock. Call for help at once.

PELVIS, HIP AND THIGH. Fractures of the pelvis and the thigh bone can cause heavy bleeding and are emergencies. A patient with a hip fracture often cannot stand, and the leg may look short-

ened and turned outward. Do not attempt to splint a pelvis yourself without training; keep the patient flat and still, treat for shock, and call for help.

DISLOCATIONS AND SPRAINS. A dislocated joint looks out of place and the patient guards it. Do not try to put it back. Support it in the position of comfort with padding and splint it as found. A sprain is treated the same way until a clinician has examined the limb. The familiar advice of rest, a cold pack wrapped in cloth, light compression and elevation fits a minor injury, and only a clinician can say that an injury is minor.

RECOGNITION. The signs include pain, swelling, bruising, deformity, an inability to use the limb, and sometimes a grating sensation. An open fracture has a wound with bone exposed or nearby, and it is more urgent. A limb that is pale, cold, blue or without a pulse beyond the injury is an emergency.

PROCEDURE: SPLINT A LIMB

Purpose: to keep a broken limb still and protect it from further injury. Scope: follows your training level and local protocol.

1. Make the scene safe and treat any severe bleeding first.
2. Check the circulation, sensation and movement below the injury and note what you find.
3. Cover any open wound with a clean dressing.
4. Support the limb in the position found, without forcing it straight.
5. Place a splint that holds the joint above and the joint below the injury still.
6. Pad the splint and secure it firmly without cutting off the circulation.

7. Check the circulation, sensation and movement again after splinting, and note the time.
8. Raise the limb if you can, apply a cold pack wrapped in cloth, and keep the patient comfortable.

Figure 16 shows a splinted limb; notice that the splint reaches past the joint above and the joint below the injury.



Figure 16. A splint that holds the joint above and the joint below a broken limb.

⚠ SAFETY

Do not try to push a bone back or straighten a badly deformed limb unless your protocol directs. Do not apply a cold pack directly to the skin. If the limb loses its pulse or becomes pale or blue, loosen the splint and call for help.

STOP AND CALL FOR HELP IF the limb is pale, blue, cold or pulseless, the bone is exposed, or the patient has signs of shock.

Holding the joint above and the joint below stops the broken ends from moving when the muscles pull, and padding spreads the pressure so that the splint does not press on skin and vessels. The checks before and after are the heart of the skill. A splint that is too tight can cut off blood flow and cause serious harm, and the only way to know that it has not is to compare what you found before and after.

NURSING AND RESPONDER PRIORITIES. Check circulation, sensation and movement before and after splinting the joint above and below, and call for help for a limb that is pale, cold or pulseless.

See also: Entry 4.1 (trauma); Entry 4.2 (bleeding).

4.5 Burns

A burn is damage to the skin and the tissues beneath from heat, chemicals, electricity or radiation. The size and depth decide how serious it is, and burns to the face, hands, feet, genitals and joints are more serious than their size suggests.

WHAT IT IS. Burns are grouped by depth. A superficial burn reddens the skin and is painful. A partial-thickness burn blisters. A full-thickness burn may look white, leathery or charred and may be painless because the nerves are destroyed. A burn can deepen over hours.

ESTIMATING SIZE. The size is stated as a percentage of the total body surface area (TBSA). In adults, the rule of nines assigns about 9 percent to the head and neck, 9 percent to each arm, 18

percent to each leg, 18 percent to the front of the trunk, 18 percent to the back of the trunk and 1 percent to the genital area. Each of the head, arm and leg shares is split evenly between the front and the back. Count only burns that blister or are deeper; simple redness is not counted. A quick alternative is that the palm of the patient's hand, with the fingers, is about 1 percent of the body surface.

Figure 17 shows the body divided for the rule of nines; notice that the front and back views each carry half of the head, arm and leg shares.

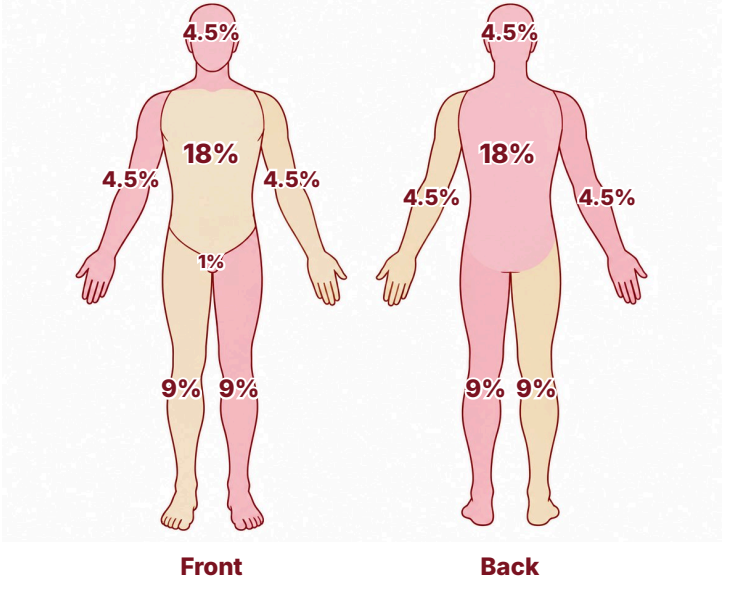


Figure 17. The adult body in front and back views, with each region's share of the body surface.

COOLING LARGE BURNS. Cooling reduces damage and eases pain, but a patient with a large burn can become dangerously cold. Cool for the time your protocol directs (for most burns from heat,

at least 10 minutes and ideally 20 minutes, and not beyond 40 minutes), then cover the burn loosely and keep the rest of the body warm. Do not cool with ice or very cold water, and watch for signs of getting too cold, such as heavy shivering or confusion; if they appear, stop cooling and keep the patient warm, as your protocol directs. For a burn that covers a large area, call for help before cooling and do not let cooling delay transport.

CHEMICAL, ELECTRICAL AND SMOKE INJURIES. Chemical burns continue as long as the chemical is on the skin. Brush off any dry powder first, then flush with large amounts of running water, and take care that the runoff does not reach you or another person. An electrical injury can leave a small entry burn with serious damage beneath: do not touch the patient until the current is off, then call for help. Hoarseness, soot around the mouth or nose, or singed facial hair after smoke or flame points to an airway burn, which is an airway emergency.

CHILDREN AND OLDER ADULTS. Children have thinner skin, so the same exposure burns deeper, and the head is a larger share of their body surface, so a burn to the head covers a larger percentage. Older adults also have thin skin and often several illnesses. Burns in both groups need earlier help, and a burn in a child with an unclear history can be a sign of abuse, which Entry 6.6 covers.

FIRST PRIORITIES. Stop the burning process and make sure the scene is safe. Remove the patient from the source. Remove rings and tight clothing near the burn before swelling begins, unless stuck to the skin. Cool the burn with cool running water for at least 10 minutes and ideally 20 minutes, or as your protocol directs. Cover it loosely with a clean, non-stick dressing. Keep the patient warm elsewhere, since large burns cause heat loss. Check the

airway if the burn is on the face or was caused by smoke or flame in a closed space, because swelling can close it. Do not apply creams, ice or butter, and do not break blisters.

RED FLAGS

Burns to the face, hands, feet, genitals or joints, burns that go all the way around a limb, electrical or chemical burns, burns with smoke inhalation, hoarseness or soot around the mouth, and burns in children and older adults. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Stop the burning, cool the burn as directed, cover it loosely, protect the airway, keep the patient warm, and call for help for any significant burn.

See also: Entry 2.1 (airway); Entry 6.1 (burns in children).

4.6 Heat and Cold Emergencies

The body works only in a narrow temperature range. Heat or cold can push it past that range, and the most severe forms are emergencies.

HEAT ILLNESS. Heat exhaustion causes heavy sweating, weakness, headache, nausea and a fast pulse. Heat stroke is the emergency form. The body temperature rises to a dangerous level, and the patient becomes confused, faints, has a seizure or collapses. Move the patient to a cool place, remove excess clothing, call for help, and cool the body quickly with whatever method your protocol directs, such as cool water or wet cloths with fanning. Give cool drinks only to a patient who is fully alert.

COLD ILLNESS. Hypothermia is an abnormally low body temperature. Early signs are shivering, clumsiness and confusion. As it worsens, shivering stops, the patient becomes drowsy and the

pulse slows. Move the patient from the cold, handle them gently, replace wet clothing with dry, insulate from the ground and the air, and call for help. Do not rub the limbs or give alcohol.

WHO IS AT RISK. Infants, older adults, people who work or exercise outdoors, people with long-term illness, and people taking some medicines are at higher risk of heat illness. Cold illness is more likely in people who are wet, tired, hungry, intoxicated, or ill, and it can occur at temperatures that are above freezing. Ask how long the patient has been exposed and what they were wearing.

COOLING METHODS. In heat stroke, speed matters more than the method. Immersion in cold water cools fastest where it is available and safe, and ice packs in the armpits, neck and groin or wet cloths with fanning are alternatives. Keep the airway and consciousness under watch during cooling, and stop when the patient is cool enough as your protocol directs, since overcooling is also possible.

FROSTBITE. In frostbite, the tissue itself freezes, most often in the fingers, toes, ears and nose. The skin may look white or waxy and feel numb. Move the patient to shelter, remove wet or tight clothing and rings, and cover the area loosely with dry material. Do not rub the area, and do not rewarm it if there is a chance it could freeze again before the patient reaches care.

Figure 18 shows how heat and cold illness progress; notice that the signs and the first actions change at each stage.

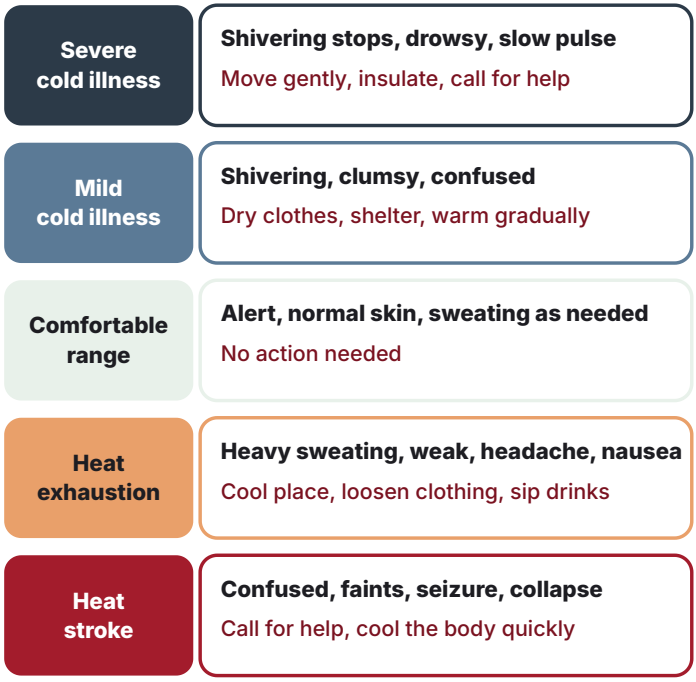


Figure 18. A continuum from cold illness to heat illness with the signs at each stage.

RED FLAGS

Confusion, fainting, a seizure or collapse in the heat, or shivering that stops with drowsiness or a slow pulse in the cold. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Recognize confusion in a hot or cold setting as an emergency, cool or warm the patient as directed, handle gently, and call for help.

See also: Entry 6.4 (older adults).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Tomas, a paramedic student, was cycling on a trail when he came across a rider who had fallen on a rocky section. The rider, Ellie, sat beside her bike, holding her thigh. There was blood running through her fingers onto the ground. Tomas stopped, looked up and down the trail, and put on the gloves from his bag. He put his phone on speaker beside him, called the emergency number and gave the location.

He pressed a folded spare shirt from his pack firmly against the wound with both hands. Blood soaked through, so he added more on top and kept pressing. It did not slow. He packed the wound with gauze from his kit and pressed hard over it, but the blood kept coming. Ellie went pale and said she felt cold. Tomas reached for the tourniquet in his kit, applied it on the thigh a few inches above the wound, between the wound and her body, and tightened it until the bleeding stopped. He wrote the time on her arm with a pen. When Ellie winced and tried to reach for the band, he told her it would hurt but was holding the bleeding back, and he left the dressing in place. He kept her lying down, covered her with his jacket, and watched her color and her thinking while he waited. He had checked that the band sat above the wound and not over the knee. When the paramedics arrived, he told them what he had seen on the trail, the time on her arm and that he had not loosened the band.

Care Priorities

CARE PRIORITIES

When injuries compete, rank them by how soon each can kill. Life-threatening bleeding that you can stop goes first, once the scene is safe and you can reach the patient, as your protocol directs. Airway and breathing follow, with the spine protected while you work, because a spinal injury is something you guard against and do not clear yourself. A dramatic wound is not always the quickest threat, so look past it for the quiet signs of shock before you settle on a plan. Cold and wet make every injury worse, so shelter the patient from the weather from the first minute. At handover, describe what you saw at the scene as well as what you found on the patient, because the mechanism tells the team where to look for injuries you could not see.

Review Questions

1. In a trauma patient, which problem is controlled before the airway, as most protocols direct?
   - **A)** A deformed wrist.
   - **B)** Severe external bleeding.
   - **C)** A bump on the head with a headache.
   - **D)** A superficial burn on a finger.

2. Which is a high-risk mechanism of injury?
   - **A)** A short trip on level ground.
   - **B)** A cut finger from a kitchen knife.
   - **C)** A twisted ankle while walking.
   - **D)** A fall from a height.

3. What is the first technique for controlling bleeding from a wound?

   A) Apply direct pressure with a clean dressing.
   B) Apply a tourniquet.
   C) Wash the wound.
   D) Lift the dressing often to check.

4. Where is a tourniquet applied on a bleeding limb?

   A) Directly over the wound.
   B) Over a joint.
   C) Between the wound and the heart, a few inches above the wound.
   D) Below the wound.

5. What is the aim of spinal motion restriction?

   A) To straighten the neck so that injuries can be seen.
   B) To move the patient to the ambulance as fast as possible.
   C) To keep every patient lying flat on a board.
   D) To keep the head, neck and trunk in line and limit movement.

6. When splinting, what should you check before and after?

   A) How far the limb can be straightened without pain.
   B) Circulation, sensation and movement below the injury.
   C) Whether the patient can walk on the limb.
   D) Whether the patient has eaten recently.

7. About what share of the body surface is the palm of the patient's hand, with fingers?

   A) 9 percent.
   B) 5 percent.
   C) 1 percent.
   D) 18 percent.

8 Short case. On a hot afternoon a worker who has been digging outdoors becomes confused and then faints. His skin is hot and flushed. List, in order, the first four things you would do and say why this is an emergency.

PART 5

Medical Emergencies

IN THIS PART

1. Describe an approach to a patient with altered mental status and the reversible causes to check, such as low blood glucose.
2. Recognize stroke with a standard stroke screen and state why time matters.
3. Describe the response to a seizure.
4. Recognize sepsis and state first priorities.
5. Describe first priorities in poisoning and overdose.
6. Recognize acute abdominal emergencies and gastrointestinal bleeding.

Not every emergency begins with an injury or a collapse. Many begin with a change that is easily explained away: a patient who is a little confused, a limb that feels odd, a fever that will not settle, a stomach ache that has lasted too long. The skill of this Part is to take these small changes seriously and to know which of them can wait and which cannot.

The Part follows the problems you are most likely to meet. Entry 5.1 gives a way to approach any patient whose thinking or alertness has changed and starts with the cause that can be checked and reversed in minutes, low blood glucose. Entry 5.2 covers stroke, where time decides how much of the brain can be saved.

Entry 5.3 covers seizures, 5.4 covers sepsis, 5.5 covers poisoning and overdose, and 5.6 covers abdominal pain and bleeding from the gut.

These entries rely on Part 1 more than the others do. The primary survey, vital signs and history are how you find these problems, and the same order applies. Your scope of practice and local protocol decide what you may do, and the medicines used in these conditions are given by trained clinicians under medical direction.

5.1 Altered Mental Status and Hypoglycemia

A change in how a person thinks, speaks or responds is one of the most important signs in emergency care, and it has many causes. The aim is to check the causes that kill quickly and can be reversed, and to call for help while you do.

WHAT IT IS. Altered mental status means that a patient is less alert, less oriented or behaving differently than usual. It runs from mild confusion to unresponsiveness. Entry 1.3 gave you the AVPU scale and the Glasgow Coma Scale to describe it, and the trend over time is important in showing when new action is needed. A patient who was alert 10 minutes ago and is drowsy now is worsening, whatever the cause.

WHY IT MATTERS. The brain is sensitive to its supply of oxygen and glucose and to what is in the blood. A fall in either, an infection, a poison, a head injury or a stroke can change behavior long before other signs appear. Patients who are confused may be unable to give a history, may refuse help, or may be mistaken for intoxicated. Do not assume that confusion is caused by alcohol or by old age until the reversible causes have been excluded.

REVERSIBLE CAUSES. Work through a short list: the airway and oxygen supply, the blood glucose, signs of stroke, head injury, infection, poisoning or a recent seizure, extreme heat or cold, and low blood pressure. Each of these is covered in this book: oxygen in Part 2, shock in Entry 3.5, head injury in Entry 4.3, heat and cold in Entry 4.6, and the others in this Part.

Figure 19 sets out the approach as a flow; notice that airway, breathing and glucose come before the search for other causes.

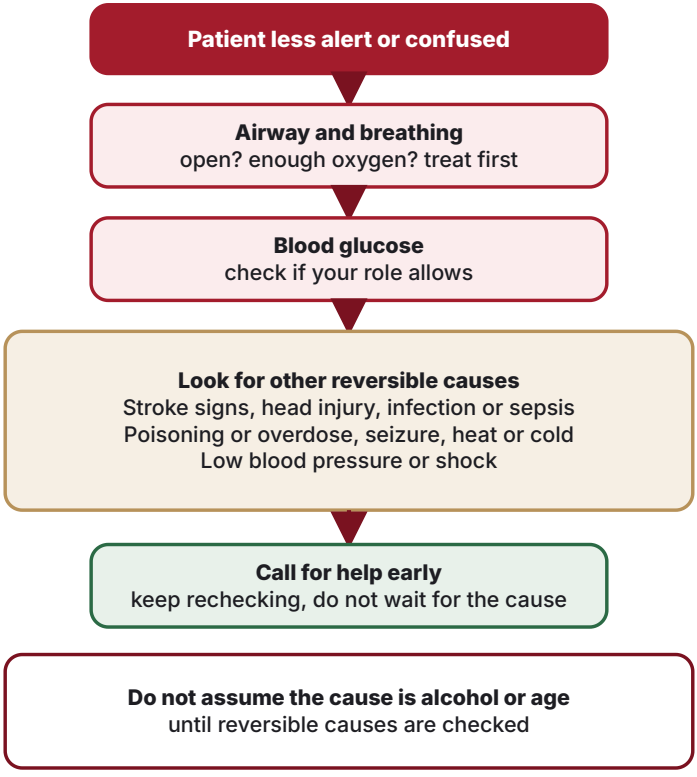


Figure 19. An approach to a patient with altered mental status, from airway and breathing to the reversible causes.

TALKING WITH A CONFUSED PATIENT. Introduce yourself by name, speak slowly and in short sentences, and ask one question at a time. Ask the patient's name, where they are and what day it is, and note the answers. Do not argue with what a confused person says. Ask family, caregivers or bystanders what the patient is usually like and when the change began, because the answer tells you whether this is new. A medical alert bracelet or card may give the diagnosis and the medicines.

ALCOHOL AND OTHER SUBSTANCES. A patient who smells of alcohol can still have low blood glucose, a head injury, a stroke or a serious infection, and drowsiness in an intoxicated patient can hide all of them. Treat altered mental status in a patient who has been drinking as a medical emergency, check the same causes, and keep the patient on their side if drowsy, because vomiting is a risk.

LOW BLOOD GLUCOSE. Hypoglycemia is low blood glucose, and it is the cause that is quickest to check and quickest to reverse. Entry 1.3 gave the usual low reading, 70 milligrams per deciliter (mg/dL) or 3.9 millimoles per liter, which differs from person to person. It most often occurs in people who take certain diabetes medicines that lower blood glucose, and it can follow missed meals, extra exercise or alcohol. Early signs include shakiness, sweating, hunger, headache and anxiety. Later signs include confusion, odd behavior, slurred speech, drowsiness, seizures and loss of consciousness. A high reading can also cause drowsiness: a person with diabetes who is very unwell with deep, fast breathing, thirst and vomiting may have diabetic ketoacidosis, so report any high reading and call for help. Slow breathing, very small pupils and unresponsiveness suggest an opioid overdose, which Entry 5.5 covers.

PROCEDURE: HELP A CONSCIOUS PATIENT WITH LOW BLOOD GLUCOSE

Purpose: to restore the patient's blood glucose with a fast-acting sugar. Scope: follows your training level and local protocol; the product and the amount are set by your protocol and medical direction.

1. Make the scene safe and carry out the primary survey (Entry 1.3).
2. Check the blood glucose with a meter if your role allows it.
3. Confirm that the patient is awake, can sit up, and can swallow safely.
4. Give a fast-acting sugar that your protocol names, as it directs.
5. Keep the patient seated and calm, and stay with them.
6. Recheck the blood glucose after about 15 minutes, and if it is still low, follow your protocol and call for help.
7. When the patient has recovered, tell the responders what happened and when, because the cause still needs to be found.

> **⚠ SAFETY**
>
> Give nothing by mouth to a patient who is drowsy, confused to the point of not following commands, has a seizure, or cannot swallow safely. Call for help. Place a patient who is unconscious but breathing in the recovery position (Entry 2.2) unless a spinal injury is suspected, and stay with the patient. Giving medicines is outside many roles; follow your scope and medical direction.

STOP AND CALL FOR HELP IF the patient cannot swallow, becomes drowsier or has a seizure, does not improve after treatment, or the blood glucose cannot be measured and the

patient is worsening.

The steps work in this order because swallowing is the safety check. A patient who is awake and swallowing normally can be helped by mouth, and a patient who is not can choke on a drink. Rechecking the reading tells you whether the treatment worked and whether the problem has returned.

RED FLAGS

A patient who is hard to wake or cannot follow commands, new confusion with a very fast or slow pulse, a seizure, signs of stroke, a head injury, slow breathing, or a low glucose reading that does not improve. Call emergency services at once.

NURSING AND RESPONDER PRIORITIES. Check airway, breathing and glucose early, and call for help for any patient whose cause is not clear.

See also: Entry 1.3 (primary survey and glucose check); Entry 2.2 (recovery position); Entry 5.2 (stroke screen); Entry 6.1 (low glucose in children).

5.2 Stroke

Stroke is damage to part of the brain from a blocked or burst blood vessel. It is a time-critical emergency. The treatments that limit damage work best when they are started early, so recognizing a stroke quickly and calling for help is the most useful thing a bystander can do.

WHAT IT IS. Most strokes occur when a clot blocks an artery that supplies the brain. The brain tissue beyond the block is starved of oxygen and begins to die. A smaller number occur when a vessel bursts and bleeds into the brain. A transient ischemic at-

tack, sometimes called a mini-stroke, is a brief blockage that clears on its own. It is a warning and needs urgent assessment because a full stroke may follow.

Figure 20 shows a simplified side view of the brain; notice that the front, middle and rear regions are supplied mainly by different arteries, which is why different strokes cause different signs.

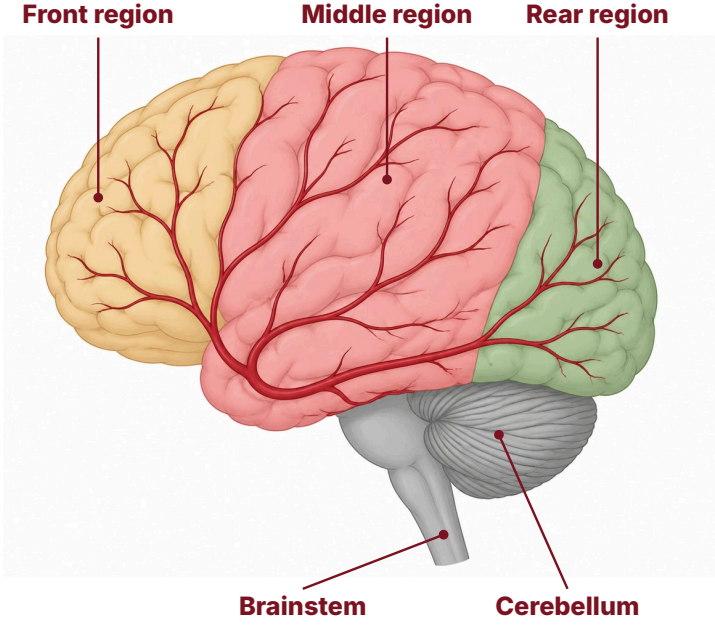


Figure 20. A simplified side view of the brain with the front, middle and rear regions shaded; each region is supplied mainly by a different main artery.

RECOGNITION. The signs begin suddenly. The face droops on one side. An arm or leg becomes weak or numb, usually on one side. Speech is slurred or confused, or the patient cannot find words or understand. Other signs include sudden loss of vision, double vision, sudden severe headache, dizziness, and loss of bal-

ance. A short screening tool checks the face, arms and speech and asks the time. The common version, FAST, is set out in the Remember box. Some versions add Balance and Eyes, because a stroke in the back of the brain can cause sudden dizziness, loss of balance or loss of vision without a drooping face. Ask the patient to smile, to raise both arms and hold them, and to repeat a short sentence. If any of these is abnormal, treat the patient as having a stroke and note the time.

Figure 21 shows the signs on a face and on the arms; notice the droop on one side of the face and the arm that drifts lower.

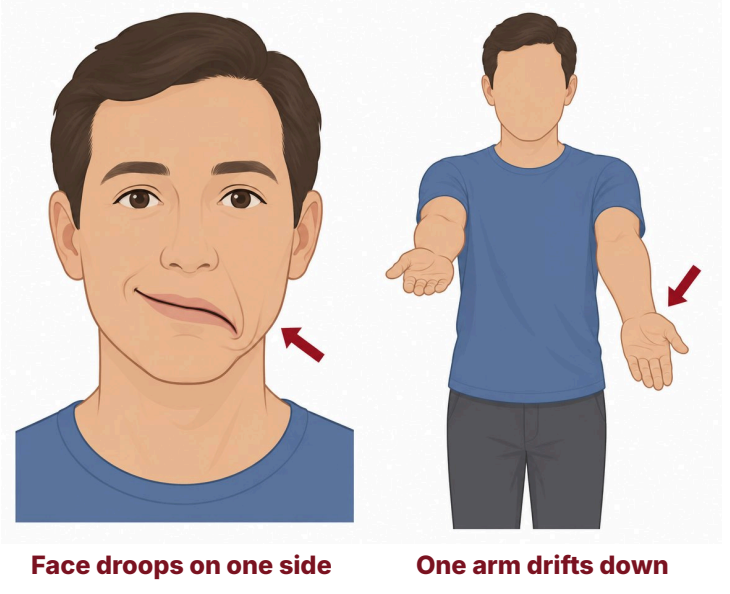


Figure 21. Facial droop and arm weakness on one side, two signs of the stroke screen.

TWO KINDS OF STROKE. A clot and a bleed need opposite treatments, which is why a scan must come first and why you do not give a medicine that affects clotting on your own judgment.

You cannot tell the two apart at the bedside. A bleed more often starts with a sudden severe headache, vomiting or a rapid fall in alertness, but a clot can look the same. Both need the same response from you: call, protect the airway and write down the times.

CONDITIONS THAT LOOK LIKE STROKE. Low blood glucose, seizures with weakness afterward, migraine and some infections can mimic a stroke. You do not need to decide which is which. Treat sudden one-sided weakness, facial droop or speech trouble as a stroke until a clinician shows otherwise, and check the blood glucose while you wait.

AFTER A BRIEF EPISODE. Symptoms that appear and then go away within minutes or hours may be a transient ischemic attack. The patient may feel well and refuse help. Explain that this is a warning and that the risk of a full stroke is highest in the first hours and days afterward. Treat it as an emergency and call emergency services even though the symptoms have gone.

> **REMEMBER**
>
> FAST: Face drooping, Arm weakness, Speech trouble, Time. The T is a reminder to call at once and to record when the symptoms began or when the patient was last known to be well.

WHY TIME MATTERS. The longer the brain goes without blood, the more of it dies, and the available treatments have time limits that are measured in hours. The most useful fact you can give the receiving team is the last time anyone saw the patient well. This may be earlier than the time the symptoms were noticed or the patient was found. A patient who wakes with symptoms was last known well at bedtime, not at waking, and the same applies to someone found on the floor: count from the last time anyone saw

them well, and ask family, neighbors or caregivers when that was. Write down how you know, because the receiving team will ask how the time was worked out.

RED FLAGS

Sudden facial droop, arm or leg weakness, trouble speaking or understanding, sudden loss of vision, sudden severe headache, or loss of balance. Call emergency services at once and note the time the patient was last known to be well.

FIRST PRIORITIES. Call emergency services without delay, and say that you suspect a stroke. Note the time the symptoms began or the patient was last well. Check airway, breathing and circulation and the blood glucose, because low glucose can look like a stroke, but do not let the glucose check delay the call. Let the patient rest in the position of greatest comfort, or on their side if drowsy. Give nothing by mouth, because swallowing may be impaired. Do not give aspirin or any medicine unless medical direction tells you to, since some strokes are caused by bleeding. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Screen with a stroke tool, call early, and hand over the last known well time.

See also: Entry 1.3 (primary survey); Entry 5.1 (altered mental status); Entry 6.4 (older adults).

5.3 Seizures

A seizure is a burst of abnormal electrical activity in the brain. It can look dramatic, and the most useful thing a responder can do is also the simplest: stay calm, protect the patient from injury, and watch the clock.

WHAT IT IS. Seizures take many forms. A generalized tonic-clonic seizure causes loss of consciousness, stiffening and rhythmic jerking of the limbs, and is followed by a period of drowsiness and confusion. Other seizures cause staring, brief lapses, or movements of part of the body. Seizures have many causes, including epilepsy, low blood glucose, head injury, fever in young children, alcohol withdrawal, poisoning and stroke.

FIRST AID DURING A SEIZURE. Note the time it begins. Ease the patient to the floor if they are standing, move hard objects away, and cushion the head. Loosen tight clothing at the neck. Turn them onto their side once the jerking stops, or sooner if they vomit, and keep the airway open. Entry 2.2 described the recovery position.

> ⚠ **SAFETY**
>
> Do not restrain the patient. Do not put anything in the mouth. Do not give food or drink until the patient is fully alert. Stay with them until they recover.

LONG AND REPEATED SEIZURES. A seizure that lasts longer than 5 minutes is a medical emergency, as is a series of seizures with no recovery between them. Treatment is given by clinicians. Your role is to protect the patient, keep the airway open as well as you can, and tell the dispatcher how long the seizure has lasted. Look for a medical alert bracelet or card and ask whether the patient has a seizure plan.

FEVER AND SEIZURES IN CHILDREN. Young children can have a seizure when they have a fever. It is frightening and usually brief, but the first one needs assessment by a clinician, and Entry 6.1 returns to children.

WHAT TO WATCH. Notice what the clinicians cannot: which part of the body moved first, whether the eyes turned to one side, how long it lasted, whether the patient answered during it, and how long the drowsiness lasted. Not every seizure convulses. If a patient stares, makes chewing movements or wanders, stay close, guide them away from hazards, do not restrain them, and speak calmly. Tell the dispatcher what you saw.

AFTER THE SEIZURE. The patient will be drowsy and confused, and may not remember. Check breathing, speak calmly, and tell them where they are. Look for injuries. Check the blood glucose if your role allows.

RED FLAGS

A seizure that lasts longer than 5 minutes, repeated seizures without recovery between them, difficulty breathing afterward, an injury during the seizure, a first seizure, a seizure in water, a seizure in pregnancy, or a seizure in a patient with diabetes. Call emergency services at once.

NURSING AND RESPONDER PRIORITIES. Note the time, protect the patient, and call for help for any red flag.

See also: Entry 2.2 (recovery position); Entry 5.1 (altered mental status); Entry 6.1 (fever seizures in children).

5.4 Sepsis

Sepsis is the body's life-threatening response to an infection. The infection may be in the lungs, the urinary tract, the skin or the abdomen, and the response injures the organs. Sepsis can progress within hours, and the earlier it is recognized, the sooner treatment can start.

WHAT IT IS. Sepsis occurs when the body's response to an infection goes out of control and begins to damage its own organs. When it causes dangerously low blood pressure that does not recover, it is called septic shock. Anyone can develop sepsis, and risk is higher in older adults, infants, people with weakened immune systems, people with long-term illness, and people with recent surgery or wounds.

RECOGNITION. Suspect sepsis in a patient who has or may have an infection and who also has any of these signs: a high or a low temperature, a fast heart rate, a fast breathing rate, new confusion or drowsiness, a low blood pressure, cool or mottled skin, or a patient who looks very unwell. In older adults a high temperature is less common and confusion may be the main sign; in infants, poor feeding, unusual sleepiness or a low temperature may be the signs. Early warning scores, which combine several vital signs into a single number, are used in many settings to flag deterioration, and your service will teach the one it uses. A patient who has had an infection and then becomes more unwell rather than better should be assessed again.

Figure 22 shows the pathway from suspicion to action; notice that any one warning sign is enough to reach the call for help.

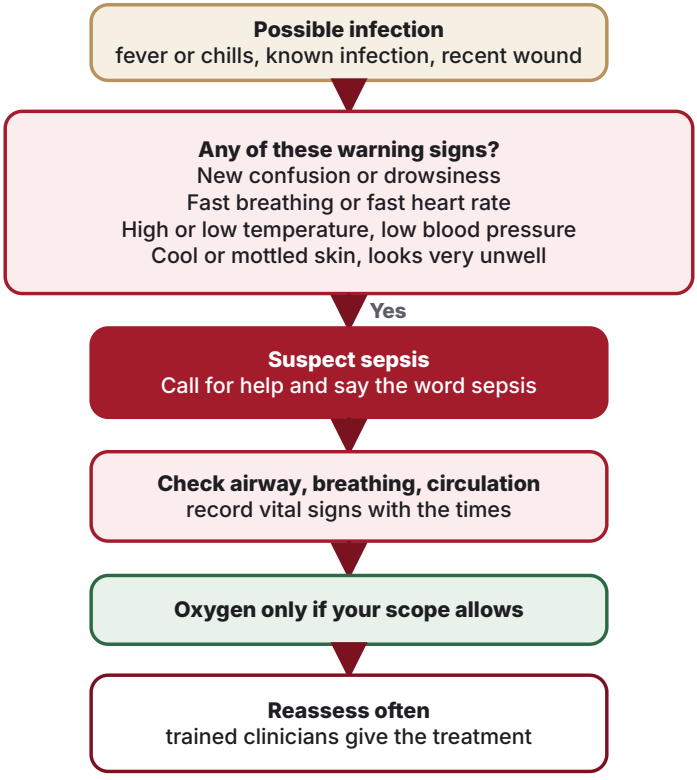


Figure 22. A sepsis recognition pathway from a suspected infection to a call for help.

RED FLAGS

A suspected infection with new confusion, a fast breathing rate, a fast heart rate, a low blood pressure, cool or mottled skin, or a patient who looks very unwell. Call for help at once and say that you suspect sepsis.

WHY SCREENING USES SEVERAL SIGNS. No single sign proves sepsis, and a normal temperature does not exclude it. A fast heart rate, a fast breathing rate or a fever each occur in many harmless illnesses, so services screen for a combination and for change, often by adding the signs into an early warning score. A score is a prompt to call for help and is not a diagnosis. Entry 1.4 gave typical resting values; here the question is whether the numbers are moving in the wrong direction between sets.

SEPTIC SHOCK. Entry 3.5 explained shock from widened vessels. In sepsis the vessels both widen and leak, so the circulation is short of volume although no blood has been lost, and the blood pressure falls. Early on the skin may be warm and flushed; later it turns cool and mottled. A patient who is cool, mottled and newly confused is deteriorating.

HANDOVER. Entry 1.5 set out the structured handover. For sepsis, make sure it includes the suspected source of infection, the vital signs with their times, any new confusion, and the word sepsis.

FIRST PRIORITIES. Call for help early and say the word sepsis. Check airway, breathing and circulation, and take a full set of vital signs with the times. Give oxygen as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction. Treatment is given by trained clinicians under medical direction, so your part is to recognize the problem, record the times and bring the team.

NURSING AND RESPONDER PRIORITIES. Suspect sepsis in any unwell patient with a possible infection, check and record vital signs with times, call for help early, and reassess often.

See also: Entry 1.4 (vital signs); Entry 3.5 (shock); Entry 6.1 (children); Entry 6.4 (older adults).

5.5 Poisoning and Overdose

A poison is any substance that harms the body when it is swallowed, breathed in, injected or absorbed through the skin. Poisonings include medicines taken in the wrong amount or by the wrong person, household products, plants, alcohol, drugs of misuse, carbon monoxide and chemicals. Many are accidental, and some are intentional, which Entry 6.5 covers from the mental health side.

WHAT IT IS. Overdose is a poisoning by a medicine or a drug. The effects depend on the substance, the amount, the time and the person. Common patterns include drowsiness and slow breathing, agitation and a racing heart, vomiting and abdominal pain, seizures and confusion. Signs of some poisons take time to develop, so a patient who looks well may still be in danger.

WHY IT MATTERS. The history is the most useful tool. The substance, the amount, the time and the person's weight and age guide the experts who advise on treatment. Poisoning can also look like a stroke, a seizure or a head injury, and the primary survey and the blood glucose check will find the problems that need action right away.

FIRST PRIORITIES. Make the scene safe. The substance may harm you too, as carbon monoxide or a chemical spill can. Remove the patient from exposure if it is safe to do so. Check airway, breathing and circulation. Call emergency services and, in the United States, the national Poison Help line at 1-800-222-1222, which is staffed by experts who give advice for any poisoning. Collect the container, the package or a sample of what the patient took, and bring it. Do not make the patient vomit and do not give anything to drink unless the experts direct you.

COMMON POISONINGS. Household products, medicines left within reach of children, plants, alcohol and drug use account for many poisonings. Carbon monoxide, a gas without smell or color from faulty heaters and engines, causes headache, dizziness and confusion, and everyone in the same space may be affected, including pets. If it is safe, get everyone into fresh air and call emergency services; Entry 1.4 noted that a pulse oximeter can read normally in this poisoning, so a normal reading does not rule it out. Alcohol poisoning causes drowsiness, vomiting and slow breathing, and the patient should be kept on their side.

OPIOID OVERDOSE. Opioids are a class of drugs that include some pain medicines and illegal drugs. An overdose causes slow or absent breathing, unresponsiveness and very small pupils. Check for a response and for breathing, call emergency services, and give breaths or compressions if the patient is not breathing normally, as Entries 2.3 and 3.4 describe. Naloxone is a medicine that reverses the effects of an opioid overdose. Where naloxone is available and you are trained and permitted to use it, use it as the device directs, and stay with the patient, because the effects of naloxone can wear off before the opioid does and the patient can stop breathing again. Giving medicines is outside many roles; follow your scope and medical direction.

⚠ SAFETY

Protect yourself from exposure and from needles. Wear gloves. If the scene is unsafe, such as a closed space that may hold poisonous gas, stay out and call for help. Do not give anything by mouth to a patient who is drowsy.

RED FLAGS

Drowsiness or slow breathing, a seizure, chest pain, a very fast or slow pulse, severe vomiting, burns around the mouth, or any poisoning in a child or an older adult. Call emergency services at once, and call the Poison Help line.

NURSING AND RESPONDER PRIORITIES. Protect yourself, call early, gather the container, and support airway and breathing.

See also: Entry 1.3 (primary survey); Entry 5.1 (altered mental status); Entry 6.1 (children); Entry 6.5 (self-harm).

5.6 Acute Abdominal Pain and Gastrointestinal Bleeding

Abdominal pain has dozens of causes, and most are mild. A few are emergencies, and they can be hard to tell apart from the others at the bedside. The safest rule is to look for features that make the pain dangerous.

WHAT IT IS. Acute abdominal pain is new pain in the belly. Causes include a bowel infection, appendicitis, gallstones, a blocked or perforated bowel, an ulcer, and bleeding from a vessel. Gastrointestinal bleeding is bleeding from the digestive tract. Bleeding from the upper gut, in the stomach or food pipe, can cause vomiting of blood, which may be bright red or look like coffee grounds. Bleeding from the lower gut usually causes bright red or maroon blood in the stool, while black, tarry, foul-smelling stools usually point to bleeding higher up in the gut.

RECOGNITION. Ask where the pain began, where it is now, how severe it is, and whether it has changed. Ask about vomiting, bowel changes, fever, recent injury, pregnancy, and medicines. Look at the patient's color and look at the abdomen without pressing hard. A

rigid, board-like abdomen, pain that gets worse with any movement, or a patient who curls up and will not let anyone touch the belly suggests an irritated lining, which is an emergency. Signs of shock with abdominal pain suggest bleeding.

RED FLAGS

Severe or worsening pain, a rigid abdomen, vomiting blood, black tarry stools, fainting, a fast pulse with cool pale skin, pain after an injury, or pain in a pregnant patient or an older adult. Call for help at once.

APPENDICITIS AND OTHER SURGICAL CAUSES. Appendicitis often begins as vague pain near the navel that moves to the lower right of the abdomen over hours, with loss of appetite, nausea and low fever. The pain worsens with movement, coughing or sneezing. Other emergencies that need surgery include a blocked or perforated bowel, a leaking blood vessel and an inflamed gallbladder. In any patient who could be pregnant, abdominal pain with a missed period or vaginal bleeding may mean an ectopic pregnancy, which can bleed heavily, and needs immediate help.

FIRST PRIORITIES. Check airway, breathing and circulation. Let the patient find the position of greatest comfort; if they are faint or pale, lay them down and, if your protocol allows, raise the legs. Give nothing by mouth, because the patient may need surgery or may vomit. Take vital signs with the times, and watch for shock. Wear gloves, record what the patient has vomited or passed, and keep a sample if the clinicians ask. Do not give painkillers or medicines for the stomach unless medical direction directs. Giving medicines is outside many roles; follow your scope and medical direction.

NURSING AND RESPONDER PRIORITIES. Look for the dangerous features, give nothing by mouth, support the position of comfort, watch for shock, and call for help early.

See also: Entry 3.5 (shock); Entry 4.1 (abdominal injury); Entry 6.3 (pregnancy); Entry 6.4 (older adults).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Grace, a nursing student on a home-care placement, arrived for a visit and found Walter, an older man who lived alone, sitting in his kitchen. His daughter had told the agency that he seemed more confused than usual that morning. He did not know what day it was, and his speech was slow. Grace checked the room for hazards and put on gloves.

She worked through the survey she had practiced. His airway was open and his breathing was a little fast. His skin was cool and clammy. He was alert enough to say his name, but not to answer other questions. His face and arms looked symmetrical and he could squeeze her hands equally, so she saw no sign of a stroke. She phoned her supervisor, who asked for an ambulance to be sent and told her to check his blood glucose. It was low. She asked his daughter whether he took medicine for diabetes; the daughter said he did and that he had eaten nothing that morning. After checking that he was awake and able to swallow safely, she was told to follow the protocol for a conscious patient. Grace helped him take the sugar as directed, stayed with him, and rechecked the reading after about 15 minutes. He was clearer, and his speech improved.

Grace noted the time and the readings and left the ambulance call in place, because she did not know why his glucose had dropped. The paramedics arrived, and she handed over her findings and the times.

Care Priorities

CARE PRIORITIES

In medical emergencies the cause is usually unknown at the bedside, so the priorities have to work before the diagnosis does. First, protect the airway and breathing whatever the cause, because a patient who is drowsy and unexplained can still be kept safe. Second, look for the cause that is quick to check and can be reversed, and do that before the causes that only a hospital can sort out. Third, treat the change over time as more important than any single reading: write each observation with its time, because the trend is the part the receiving team cannot see for themselves. Give weight to what family or caregivers say is different from usual, since a confused patient cannot report their own baseline. When the story changes, start again from the airway. Any time that arrives with the patient, such as the start of symptoms, the last time anyone saw the patient well or the last meal, belongs on paper before you leave the bedside. An unexplained patient who has become quiet is not a settled patient, so keep rechecking, keep them where you can see them, and ask for help while the patient can still talk to you.

Review Questions

1. A patient is newly confused, sweaty and cool, and your role allows you to check blood glucose. Why is the glucose checked early?

   A) The reading decides whether you call for help.

   B) Low blood glucose is a common cause that can be reversed within minutes.

   C) Confusion is always caused by a glucose problem.

   D) The airway cannot be assessed until the glucose is known.

2. A man with diabetes is shaky and sweating. He is awake, sits up and swallows normally, and his reading is low. Which action fits Entry 5.1?

   A) Lay him flat and give nothing by mouth.

   B) Wait for the ambulance before doing anything.

   C) Give plain water and recheck in an hour.

   D) Give the fast-acting sugar your protocol names, then recheck the reading after about 15 minutes.

3. A woman wakes with a drooping face and a weak left arm. Which time matters most to the receiving team?

   A) The last time she was known to be well.

   B) The time you arrived.

   C) The time she told someone.

   D) The time the call was connected.

4. Why should a responder not give aspirin to a patient with a suspected stroke unless medical direction says so?

   A) It works too slowly to help.

   B) It hides the facial droop.

   C) Some strokes are caused by bleeding, and aspirin could make bleeding worse.

   D) It is used only for chest pain.

5. A woman has a generalized seizure on a gym floor. Which action is correct?

   A) Hold her arms and legs still.

   B) Cushion her head, move hard objects away and note the time.

   C) Place a folded cloth between her teeth.

   D) Pour water on her face.

6. Which patient should be treated as possible sepsis?

   A) A patient with a mild cough, normal breathing and normal alertness.

   B) A patient with a sprained wrist and a fast heart rate from pain.

   C) A patient with a headache that eased after a drink of water.

   D) A patient with a skin infection who is newly drowsy, breathing fast, with cool, mottled skin.

7. An adult is found unresponsive with slow, shallow breathing and very small pupils. Which first step is correct?

   A) Make the scene safe, call emergency services, support breathing and, if your role allows and a device is available, use naloxone as directed.

   B) Make the patient vomit.

   C) Give a drink of water to wake the patient.

   D) Let the patient sleep and recheck in an hour.

8. Short case. A 70-year-old woman has had severe belly pain for 3 hours. She lies very still, her abdomen is rigid, and her skin is cool and pale with a fast pulse. List, in order, the first four things you would do.

PART 6

Special Populations, Behavioral Emergencies and Safety

IN THIS PART

1. Describe the structured pediatric assessment and how children differ from adults.
2. Describe how choking relief, breathing support and CPR differ in infants and children.
3. Recognize pregnancy-related emergencies, describe safe positioning, and state when to call for help for an imminent birth.
4. Describe how assessment changes in older adults.
5. Describe a safe response to behavioral and mental health emergencies using safe-messaging language.
6. Recognize signs of abuse or neglect and state reporting duties in general terms.

Everything so far has used one patient as the model: an adult of average build, healthy until the emergency. Real patients vary. A child is not a small adult, a pregnant patient has two lives to protect, and an older adult may show a serious illness in an unex-

pected way. A person in distress of mind needs a different kind of care than a person in distress of body, and any patient may be a victim of harm that nobody has named.

This Part adapts the habits of Parts 1 to 5 to these patients. Entry 6.1 gives a way to assess a child and shows how children differ. Entry 6.2 covers the changes to choking relief, breathing support and CPR for infants and children. Entry 6.3 deals with pregnancy. Entry 6.4 covers older adults, Entry 6.5 covers behavioral and mental health emergencies, and Entry 6.6 covers abuse, neglect and violence.

Each entry adds only what differs. Where the adult method still applies, the text points back to the home entry and does not repeat it. What you may do still depends on your scope of practice and local protocol.

6.1 Children: Assessment and How They Differ

Children get sick differently from adults, and they compensate until late and can then deteriorate quickly. A child who is breathing hard and still playful is in a very different place from a child who is breathing hard and has gone quiet. This entry gives you a structured way to look at a child before you touch them, and then lists the main ways they differ.

WHAT IT IS. The pediatric assessment is a rapid, hands-off look at three things: the child's appearance, the work of breathing, and the circulation to the skin. It takes a few seconds, and it works from the doorway. Because it needs no equipment, it can be done with a frightened child still in a parent's arms. After it, you carry out the primary survey of Entry 1.3 in the same order as for an adult.

Figure 23 shows the three parts of the assessment; notice that the look is made from a distance, before you touch the child.

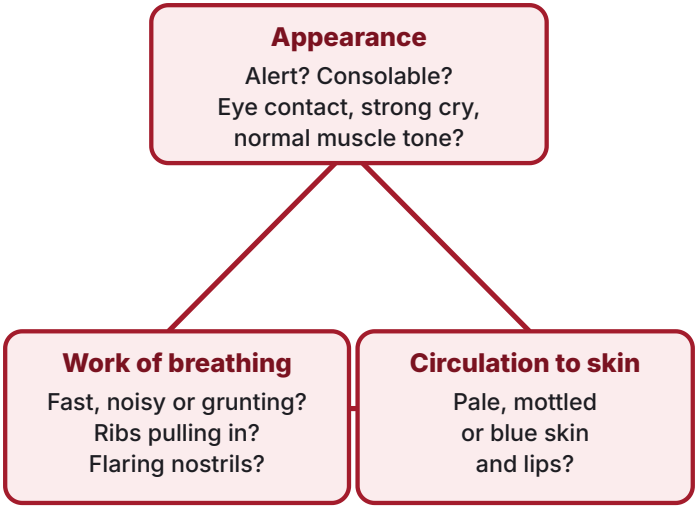


Figure 23. The three parts of the pediatric assessment: appearance, work of breathing and circulation to the skin.

APPEARANCE. Appearance reflects, among other things, whether the brain is getting enough oxygen and blood. Notice whether the child is alert, consolable and interested, whether they make eye contact, whether the cry is strong, and whether they move normally. A child who is limp, glassy-eyed, hard to wake or does not react to a parent is seriously unwell.

WORK OF BREATHING. Watch the child breathe. Look for fast or slow breathing, noisy breathing, flaring of the nostrils, grunting, pulling in of the skin between the ribs, and a child who sits in a tripod position, leaning forward on their hands.

CIRCULATION TO THE SKIN. Look at the color of the skin and lips. Pale, mottled or blue skin suggests poor circulation or poor oxygen supply. Compare it with the child's normal color.

TALKING WITH CHILDREN AND PARENTS. Get down to the child's level, use a calm voice, and let a small child stay in a parent's arms for as long as it is safe, since a frightened child is harder to assess. Ask the parent what is usual for the child, how long the illness has lasted, how much the child has drunk, how many wet diapers there have been, and what medicines and allergies the child has. A parent who says the child is not acting like themselves should be taken seriously.

DEHYDRATION. Children can lose fluid quickly through vomiting, diarrhea and fever. Signs include a dry mouth, few or no tears when crying, few wet diapers, sunken eyes, drowsiness and fast breathing. A child who is drowsy, cool and mottled with fast breathing is in shock and needs urgent help.

HOW CHILDREN DIFFER. Children differ from adults in several ways that matter in an emergency. Their airways are smaller and narrower, so a little swelling or mucus blocks a larger share of the passage. Their tongues are larger in proportion to their mouths. Their heads are large for their bodies, so a child who lies flat may have the neck bent forward, and a small pad under the shoulders may be needed to keep the airway in line. Entry 2.2 described the positions for adults. Children also breathe faster and have a faster heart rate than adults, and both change with age. The table below gives typical resting ranges by age.

Figure 24 compares the airway of a child with that of an adult; notice the size of the head and tongue relative to the airway.

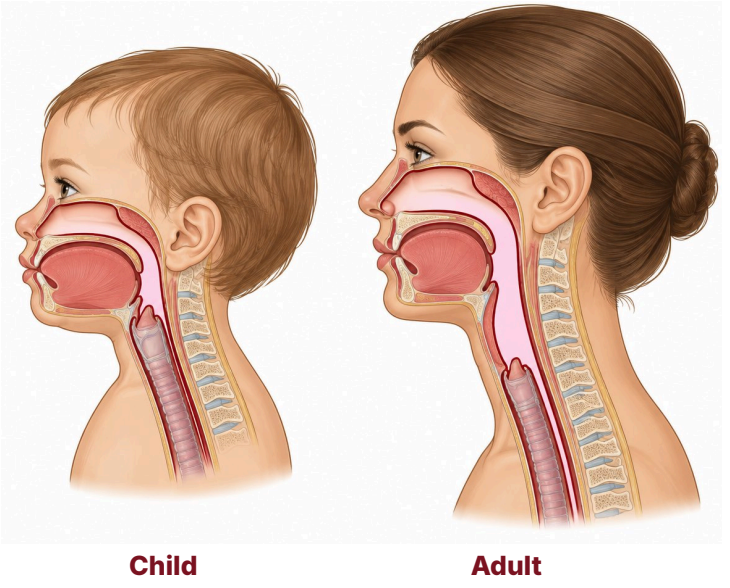


Figure 24. A child's airway compared with an adult's, drawn side by side.

Typical resting ranges in children

Age	Breathing rate (breaths per minute)	Heart rate awake (beats per minute)
Infant (1 month to 1 year)	30 to 53	100 to 190
Toddler (1 to 2 years)	22 to 37	98 to 140
Preschool (3 to 5 years)	20 to 28	80 to 120
School age (6 to 11 years)	18 to 25	75 to 118
Adolescent (12 to 15 years)	12 to 20	60 to 100

Sources differ, and newborns in the first month have their own ranges; follow the values in your protocol.

Children compensate for illness and injury by speeding the heart and tightening the vessels, and may keep a normal blood pressure until they are close to collapse. A fast heart rate and a slow capillary refill (Entry 1.3) are earlier signs than a low pressure. A child who becomes sleepy, breathes slowly or with less effort while still looking unwell, or whose heart rate falls is in danger; these are late and serious signs. Fever is common in children and by itself is usually not a danger sign, except in a baby under 3 months, where a temperature of 100.4 degrees F (38 degrees C) or higher needs urgent evaluation. A fever with a stiff neck, a rash that does not fade under pressure or purple bruise-like spots, extreme sleepiness, trouble breathing or poor feeding needs urgent care.

Several earlier entries gain one child-specific point. Asthma (Entry 2.4) shows the same signs in a child, who may tire sooner. A child has a small blood volume, so bleeding that would be minor in an adult can be serious, and an adult-sized tourniquet (Entry 4.2) may not fit a small limb; call for help early. Small children who are ill are at higher risk of low blood glucose, so a glucose check belongs in the assessment of any drowsy child, if your scope includes it (Entry 5.1). A first seizure with fever in a young child needs assessment (Entry 5.3), and any injury that does not fit the story needs attention (Entry 6.6).

URGENCY AND ALLERGY. Entry 1.2 set out the five levels of acuity. For a child the same levels apply, with one addition: a child who is limp, hard to wake or inconsolable is sorted higher whatever the numbers show, and a parent who says the child is not acting like themselves counts as a finding. Anaphylaxis (Entry 2.5) in a child may begin with hives, vomiting or a cry that turns hoarse rather than with obvious breathing trouble, so treat these after a likely trigger as an emergency. In sepsis (Entry 5.4) an infant may show only poor feeding, floppiness or an unusual cry, with or without fever.

RED FLAGS

A child who is limp, hard to wake or inconsolable, grunting or struggling to breathe, blue or gray, with a rash that does not fade, a stiff neck, repeated vomiting, or a fever in an infant under about 3 months. Call for help at once.

SCOPE NOTE

Children's equipment and medicine amounts are sized by weight and age and are outside the scope of this book. Giving medicines is outside many roles; follow your scope and medical direction. Use the sizing aids and protocols of your service, and call for help early for any seriously unwell child.

NURSING AND RESPONDER PRIORITIES. Look first from a distance, use the three-part assessment, involve the parent, and call early.

See also: Entry 1.3 (primary survey); Entry 2.4 (asthma); Entry 6.2 (child CPR); Entry 6.6 (abuse).

6.2 Children: Choking, Breathing Support and Resuscitation Differences

The principles of Parts 2 and 3 still apply, but the techniques change with the size of the child. This entry covers the differences, using the age groups that rescuers use: an infant is under 1 year, and a child is from 1 year to the start of puberty.

CHOKING IN AN INFANT. A choking infant has a weak or silent cry, a weak or silent cough, or cannot breathe. If the infant is coughing hard or crying strongly, do not give blows or thrusts; stay close and watch. Do not use abdominal thrusts on an infant.

PROCEDURE: RELIEVE CHOKING IN AN INFANT

Purpose: to clear a blocked airway in an infant under 1 year who cannot cry, cough or breathe. Scope: follows your training level and local protocol; the exact method your course teaches takes

precedence.

1. Confirm that the infant cannot cry, cough or breathe, and shout for someone to call emergency services.
2. Lay the infant face down along your forearm, resting on your thigh, with the head lower than the chest.
3. Support the head and jaw with your hand, and keep your fingers off the soft part of the throat.
4. Give up to 5 firm back blows between the shoulder blades with the heel of your other hand.
5. If the object has not come out, turn the infant face up along your other forearm or on your thigh, with the head supported, and give up to 5 chest thrusts with the heel of one hand on the middle of the breastbone, just below the nipple line.
6. Look in the mouth after each round, and remove an object only if you can see it.
7. Repeat steps 2 to 6 until the object comes out or the infant becomes unresponsive.
8. If the infant becomes unresponsive, lay the infant on a firm surface and begin infant CPR (procedure below).

> **⚠ SAFETY**
>
> Do not use abdominal thrusts on an infant. Do not sweep the mouth blindly, because you may push the object deeper. Support the head throughout and keep it lower than the chest.

STOP AND CALL FOR HELP IF the object does not come out after several rounds, the infant turns blue or gray or goes limp, or the infant becomes unresponsive.

Back blows jolt the object loose, and chest thrusts raise the pressure inside the chest so that air pushes the object toward the mouth. Abdominal thrusts are avoided in infants because their organs sit low and are poorly protected by the ribs. Alternating the two gives each technique a chance to work.

CHOKING IN A CHILD. For a child who is conscious and cannot cough, speak or breathe, the sequence follows the adult method of Entry 2.2: alternate up to 5 back blows and up to 5 abdominal thrusts, with a force suited to the child's size. If the child is coughing hard, give no blows or thrusts and stay close; call for help at once if the child cannot cough, speak or breathe. If the child becomes unresponsive, lower them to the ground, call for help, and begin CPR.

PROCEDURE: GIVE RESCUE BREATHS TO AN INFANT OR CHILD

Purpose: to give breaths to an infant or child who has a pulse but is not breathing normally. Scope: follows your training level and local protocol.

1. Make the scene safe, check for a response, and call for help; send someone for an AED.
2. Open the airway with the head in line with the body or tilted back only slightly (Entry 2.2).
3. Check for breathing, and for a pulse if your role includes it, for no more than 10 seconds.
4. Seal a mask of the right size over the nose and mouth (Entry 2.3 described the seal).
5. Give 1 breath every 2 to 3 seconds (20 to 30 breaths a minute), each lasting about 1 second, only enough to make the chest rise.

6. Recheck breathing and pulse about every 2 minutes, and start compressions if there is no pulse or you are not sure.

> **⚠ SAFETY**
>
> Give gentle breaths. Small lungs are easily over-filled, and air forced into the stomach can cause vomiting. Stop if the chest does not rise and recheck the airway.

STOP AND CALL FOR HELP IF the chest does not rise after two attempts, there is no pulse, or the child is not improving.

Children most often stop breathing before the heart stops, so breaths alone can restore a failing circulation if they start early. The rate is slower than the adult rate because the lungs are smaller, and the rise of the chest tells you the breath was enough.

PROCEDURE: CPR FOR AN INFANT OR CHILD

Purpose: to keep blood moving in a child or infant who is unresponsive and not breathing normally. Scope: follows your training level and local protocol; the exact method your course teaches takes precedence.

1. Check that the scene is safe, tap the child's shoulder or the infant's foot and shout; if there is no response, shout for help.
2. If someone else is with you, send them to call emergency services and fetch an AED. If you are alone and have a phone, put it on speaker and call while you begin. If you are alone with no phone, give about 2 minutes of CPR first, then call and fetch an AED.
3. Place the child on a firm surface and check for breathing for no more than 10 seconds.

4. For a child, place the heel of one hand on the lower half of the breastbone. For an infant, either encircle the chest with both hands and press with both thumbs side by side, or use the heel of one hand, on the center of the chest just below the nipple line.
5. Press down at least one third of the depth of the chest (about 2 inches [5 centimeters] in a child and about 1.5 inches [4 centimeters] in an infant), at 100 to 120 a minute, and let the chest come fully back up after each push.
6. If you are trained, give 2 breaths after every 30 compressions; with 2 rescuers, give 2 breaths after every 15 compressions.
7. Use an AED as soon as it arrives, with child pads or a child setting if available (for a child under about 8 years). If none is available, use the AED with adult pads rather than no AED.
8. Continue until help takes over or the child responds.

> ⚠ **SAFETY**
>
> Do not press on the soft tissue under the chin or neck when you lift the jaw, and keep your hands on the breastbone, away from the upper abdomen. Keep an infant's head in a neutral position; tilt a child's head back only slightly past neutral.

STOP AND CALL FOR HELP IF a child or an infant is unresponsive and not breathing normally, or if you cannot clear an obstruction.

The steps differ from the adult sequence because cardiac arrest in children most often follows respiratory failure or shock and not a primary heart problem. Early breaths and oxygen matter more. Giving medicines is outside many roles; follow your scope and

medical direction. An alone rescuer with no phone gives about 2 minutes of CPR before leaving to call. Compression depth is stated as a fraction of the chest so that it fits any size of child.

NURSING AND RESPONDER PRIORITIES. Match the technique to the age, and call early for any child who is not breathing normally.

See also: Entry 2.2 (choking); Entry 3.4 (CPR and AED); Entry 6.1 (child assessment).

6.3 Pregnancy Emergencies and Imminent Birth

In pregnancy the care you give the patient is also care for the baby. The principles of Parts 1 to 5 still apply, with a few adaptations.

CHANGES OF NORMAL PREGNANCY. In pregnancy the blood volume rises and the heart rate is higher at rest, so a pregnant patient can lose a good deal of blood before the usual signs of shock appear. The blood pressure often runs a little lower in the middle months. These changes mean that you cannot rely on the vital sign ranges of Entry 1.4. Treat the patient's own signs of distress as serious and tell the clinicians that she is pregnant and how far along she is.

POSITIONING. In the later months of pregnancy, the enlarged uterus can press on the large vessel that returns blood to the heart when the patient lies flat on her back, causing faintness and a fall in blood pressure.

Figure 25 shows left-lateral positioning; notice the rolled blanket behind her back that holds the position.



Figure 25. A pregnant patient lying on her left side.

PROCEDURE: POSITION A PREGNANT PATIENT ON THE LEFT SIDE

Purpose: to take the weight of the uterus off the large vessel that returns blood to the heart. Scope: follows your training level and local protocol.

1. Make the scene safe, explain what you will do, and ask permission.
2. Check airway and breathing, and call for help if she is unwell.
3. If you suspect a neck or back injury, do not turn her; call for help and follow your spinal motion restriction protocol (Entry 4.3).

4. Help her roll onto her left side, supporting her shoulder and hip.
5. Put a pillow under her head and a rolled blanket behind her back to hold the position.
6. Recheck her breathing, color and responsiveness often, and note the time she was placed on her side.
7. If she becomes unresponsive and is not breathing normally, lay her on her back on a firm surface and start CPR (Entry 3.4).

> ⚠ **SAFETY**
>
> Do not leave her flat on her back in late pregnancy. Do not tilt her for CPR, because compressions need a firm, flat surface.

STOP AND CALL FOR HELP IF she bleeds heavily, has severe pain, a seizure, a severe headache with vision changes, faints, or the baby is crowning.

The enlarged uterus can squeeze the large vein that carries blood back to the heart when she lies flat. Lying on her left side lifts that weight away, so more blood returns to the heart and the blood pressure improves.

EMERGENCIES. Call for help for any pregnant patient with vaginal bleeding, severe abdominal pain, a severe headache, changes in vision, swelling of the face and hands, a seizure, reduced movements of the baby, a fall or injury, or loss of fluid. A seizure in late pregnancy can be a sign of eclampsia, a dangerous complication of high blood pressure, and a severe headache with vision changes can be an early warning of it. Heavy bleeding is an emergency, and the signs of shock may appear late.

CPR IN PREGNANCY. Entry 3.4 described adult CPR. In advanced pregnancy, the same compressions apply, and the AED is used as usual. If a second rescuer who is trained in the method is available, that person moves the uterus gently to the patient's left by hand while compressions continue. Call for help early, because the patient and the baby both depend on fast advanced care.

IMMINENT BIRTH. Birth is imminent when the patient feels an urge to push, the baby's head is visible at the vaginal opening (crowning), or contractions are very close together. Call emergency services and do not try to stop the birth or delay it. Make the patient comfortable, wash your hands or put on gloves, and have clean towels and a blanket ready. Let the baby emerge on its own and do not pull on the baby or the cord; if you must receive the baby, hold a clean towel under the body. This is basic help until trained responders arrive, so follow your protocol and the dispatcher's instructions. Once the baby is born, dry the baby with a towel, keep the baby warm against the patient's skin under a blanket, and check that the baby is breathing. Do not cut the cord unless you have been trained and protocol directs it, and do not pull on the cord to deliver the placenta. If the cord is visible before the baby, or the baby is not breathing after birth, tell the dispatcher at once and follow the instructions you are given.

RED FLAGS

Heavy bleeding, severe pain, a seizure, a severe headache with vision changes, a baby that is not breathing or is limp after birth, or a cord that comes out before the baby. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Position on the left side, call early, and watch for shock.

See also: Entry 3.4 (CPR); Entry 3.5 (shock); Entry 6.2 (infant breathing).

6.4 Older Adults

Older adults are a large share of emergency patients, and the same illness often looks different in them. The core habits do not change. What changes is how much weight you give to the vague complaint, the quiet patient and the small change.

WHAT CHANGES. The body of an older adult has less reserve, so a patient may be seriously ill with few signs. A heart attack may bring breathlessness, weakness or tiredness with little or no chest pain. An infection may bring confusion and no fever. A fall may break a hip or bleed inside the skull with few signs at first. Older adults often take several medicines, and some of them blunt the body's response, for example a medicine that slows the heart may hide a fast pulse. Entry 3.2 gave the classic chest pain pattern. In older adults, and in women of any age, a heart attack may instead show as breathlessness, nausea, sweating or unusual tiredness, so treat unexplained breathlessness or sudden weakness as possible heart trouble and call early. A stroke may show as sudden confusion or a fall rather than a drooping face (Entry 5.2), and a serious abdominal problem may bring little pain, no fever and a soft belly (Entry 5.6), so the red flags of those entries apply even when the picture looks mild.

CONFUSION. New confusion is not a normal part of aging. When it comes on suddenly it is called delirium, and it is a sign of illness or another cause: infection, low blood glucose, low oxygen, medicines, pain, dehydration or a head injury. A patient with dementia may be confused all the time, so ask family or the caregiver what the usual state is and treat a change from that as an emergency.

FALLS. A fall is both an injury and a clue. Ask why the patient fell, whether they blacked out, whether they hit their head, and whether they take blood-thinning medicines. Entry 4.1 described the risk of bleeding after a minor fall.

MEDICINES. Ask for the list of medicines or look for it. Some medicines thin the blood, some lower the blood pressure, some blunt the heart rate, and some cause drowsiness or confusion on their own. A patient who has recently started a new medicine or changed a dose may be reacting to it. Do not stop or change any medicine yourself. Giving medicines is outside many roles; follow your scope and medical direction.

HEAT AND COLD. Older adults sense heat and cold less well. Entry 4.6 described the signs; check the temperature of the home.

RED FLAGS

New confusion or drowsiness, a fall with a head strike on blood-thinning medicine, breathlessness with no clear cause, a fast or slow pulse with faintness, and any change from the usual state reported by family. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Take vague symptoms seriously, ask what is usual for the patient, and call early.

See also: Entry 3.2 (chest pain); Entry 4.6 (heat and cold); Entry 5.1 (altered mental status); Entry 5.4 (sepsis).

6.5 Behavioral and Mental Health Emergencies

A person in a mental health crisis needs what every patient needs: a safe scene, a calm responder and a prompt call for help. The skills here are mostly skills of speaking and listening, and they

protect both the patient and you.

WHAT IT IS. A behavioral emergency is a situation in which a person's thoughts, feelings or actions put them or others at risk. It may involve severe anxiety, panic, agitation, confusion, hearing or seeing things that are not there, thoughts of suicide, or aggression. Physical illness can cause the same behavior, such as low blood glucose, low oxygen, a head injury, infection or poisoning, so check the reversible causes of Entry 5.1 first. Do not assume that unusual behavior is only a mental health problem.

WHY PEOPLE BECOME AGITATED. Fear, pain, intoxication, withdrawal and an acute psychiatric illness can each cause agitation on top of the physical causes above, and a person may have more than one at once. A calm approach helps whatever the cause. Ask about medicines and substances, and about recent injury, as part of the history.

PANIC AND HEART TROUBLE. A panic attack can cause chest tightness, a racing heart and breathlessness, and several of these also occur in heart attacks and asthma. Do not assume that a patient with these symptoms is only anxious; check the vital signs and call for help if there is chest pain or the symptoms do not settle.

SAFETY FIRST. Assess the scene before you approach. Look for weapons, for exits, and for anyone else at risk. Keep a clear route out for yourself and the patient, stay at a safe distance, and do not stand between the patient and the door. If you feel threatened, leave and call for help. Do not try to restrain a patient; this book does not teach restraint, and it carries serious risks for both of you.

DE-ESCALATION. Speak slowly and quietly, introduce yourself, and say what you want to do. Keep your hands visible and your posture relaxed. Listen more than you talk, and acknowledge the

person's feelings without arguing about what they believe. Offer simple choices, such as where to sit. Avoid sudden movements, a crowd of onlookers and loud noise. Give the person time.

THOUGHTS OF SUICIDE. Ask directly and kindly whether the person is thinking of suicide. Asking does not put the idea into someone's head. If the answer is yes, stay with the person, take the concern seriously, and call for help. Use language that treats suicide as a health matter, such as "died by suicide", and avoid phrases that blame or glamorize. Do not leave the person alone, and remove means of harm from the area if you can do so safely. If the person has already harmed themselves, call 911 and give first aid for the injury. In the United States, people in crisis can call or text 988 to reach the national crisis line, and 911 is for immediate danger.

RED FLAGS

A person who talks about wanting to die or has a plan or the means, a weapon, threats to others, severe agitation with confusion, chest pain or breathing trouble with panic, or signs of poisoning. Call emergency services at once.

SAFETY

Do not argue with, shame or promise secrecy to a person at risk. Do not leave a person who may harm themselves alone. Leave the scene if it becomes unsafe and call for help.

STOP AND CALL FOR HELP IF the person has a weapon, is threatening to harm themselves or others, or you cannot keep the scene safe.

WHEN THE PERSON REFUSES HELP. A person in crisis may refuse care. Entry 1.5 set out consent. Stay calm, keep talking and call for help, because decisions about refusal rest with clinicians and the law of your state.

NURSING AND RESPONDER PRIORITIES. Check the scene, rule out medical causes, ask directly about suicide, and stay with the person.

See also: Entry 5.1 (altered mental status); Entry 5.5 (poisoning and overdose); Entry 6.6 (violence).

6.6 Abuse, Neglect and Interpersonal Violence

Responders see injuries and illnesses, and some of them are not accidents. The patient may be a child, an older adult or a partner who cannot ask for help, and you may be the only professional who sees the signs.

WHAT IT IS. Abuse is harm done to a person by someone with power over them, and it can be physical, sexual, emotional or financial. Neglect is the failure to provide care a person needs. Interpersonal violence includes violence between partners and other family members. These affect people of every age, sex and background.

SIGNS. Be alert when the injuries do not fit the explanation, when the story changes, when the delay in seeking care is long, when injuries are in different stages of healing, or when a patient seems afraid of the person with them. In children, watch for bruising in an infant who cannot yet move around, bruises in unusual places, burns with clear outlines, a child who is withdrawn or very

afraid, and signs of neglect such as poor hygiene or untreated illness. In older adults, watch for unexplained injuries, pressure sores, dehydration and poor nutrition.

WHAT TO DO. Treat the injuries first. Speak to the patient privately if you can do so safely, and listen without judging. Record what you see and what the patient says in their own words, with the times, and do not interrogate. Do not confront the suspected abuser. Report your concern through the channels your service or school requires. In many states the legal duty to report may belong to you personally, so telling a supervisor may not be enough; rules differ by state, role and the age of the patient. A report is a request for protection, and you do not need proof.

RED FLAGS

An injury that does not fit the story, injuries of different ages, a frightened patient or a controlling companion, delay in seeking care, and signs of neglect. Report your concern and call for help if anyone is in immediate danger.

SAFETY

Do not question the patient about abuse in front of the person you suspect, promise to keep a secret you cannot keep, or confront the suspected abuser. Your first aim is the patient's safety now.

RECORDING WHAT YOU SEE. Write the patient's own words, the time, who was present, and each injury by place and size, for example a round bruise about 1 inch (2.5 centimeters) across on the left upper arm. Describe and do not conclude, so that the next clinician can judge.

NURSING AND RESPONDER PRIORITIES. Treat the injuries, listen privately, record facts, and report as your role requires.

See also: Entry 4.1 (injury patterns); Entry 6.1 (children); Entry 6.4 (older adults).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Sam was a nursing student on a placement in an emergency department triage area when a mother carried in her 18-month-old, Noor, who had been hot and fussy for two days and was not feeding. Noor lay against her mother's shoulder. Sam looked from the doorway before going near. Noor's eyes opened when her mother spoke, but she did not look around, and she let herself be handled without protest. Her breathing was fast, and the skin between her ribs pulled in with each breath. Her lips were pink, but her hands and feet looked pale and a little mottled.

Sam told the triage nurse what she had seen. The nurse took Noor's temperature and her breathing and heart rates, noted a slow return of color to Noor's fingertip, and moved her to the front of the line. Sam stayed with the mother, kept her calm, and asked when Noor had last drunk and wet a diaper. The mother said it had been many hours.

A doctor saw Noor within minutes. Afterward the triage nurse asked Sam what she had noticed first, and Sam answered in the order she had looked: how Noor seemed, how she breathed, and the color of her skin. She also gave the nurse the times the mother had mentioned, because Sam had written them down as she heard them.

Care Priorities

CARE PRIORITIES

Patients who do not fit the adult model share one trap: the usual numbers and the usual story mislead. Ask what is normal for this patient before you decide what is abnormal, and ask the person who knows, whether that is a parent, a caregiver or a relative. Look before you touch, because a frightened child, a person in crisis and a patient with dementia all react to how you approach, and that first reaction is part of your assessment. Give extra weight to the quiet patient, since a child who has stopped crying or an older adult with no complaint can be the sickest person present. When the picture does not make sense, such as an injury that does not fit the story, write down what you saw and what you were told in the patient's own words, and pass it to the person who can act. Finally, check your scope before you act: equipment sizes and legal duties differ by age and by state, and when you are unsure the right move is to call early and stay.

Review Questions

1. A 2-year-old is held by her mother in the doorway. She is limp, does not look at you and has stopped crying. Which part of the pediatric assessment does this describe?

   A) Circulation to the skin.

   B) Work of breathing.

   C) Appearance.

   D) Airway, pulse and temperature.

2. How does choking relief in a conscious infant differ from that in an adult?

   A) It uses chest thrusts in place of abdominal thrusts.

   B) It uses abdominal thrusts with less force.

   C) It uses blind finger sweeps.

   D) It begins with CPR.

3. An alone rescuer finds an unresponsive child who is not breathing normally, with no phone to hand. According to Entry 6.2, which is correct?

   A) Leave at once to call for help and return later.

   B) Give 5 rescue breaths only and wait.

   C) Wait until help arrives.

   D) Give about 2 minutes of CPR, then call for help and fetch an AED.

4. Why is a pregnant patient in late pregnancy who feels faint helped onto her left side?

   A) It lowers the risk of vomiting.

   B) It moves the weight of the uterus off the large vessel that returns blood to the heart.

   C) It slows labor.

   D) It turns the baby head down.

5. A baby is born before help arrives. Which action is correct?

   A) Pull on the cord to deliver the placenta.

   B) Cut the cord at once with clean scissors.

   C) Dry the baby, keep the baby warm against the patient's skin and check breathing.

   D) Hold the baby upside down to clear fluid.

6. An 82-year-old who usually chats freely is drowsy and confused, and her daughter says this is new. What does this most suggest?
   - **A)** An illness, such as infection, low glucose or low oxygen, that needs a cause to be sought.
   - **B)** Normal aging.
   - **C)** Dementia, so nothing more is needed.
   - **D)** Tiredness from a poor night's sleep.

7. A person in crisis says they want to die. Which response matches Entry 6.5?
   - **A)** Change the subject to avoid upsetting them.
   - **B)** Promise to keep it secret.
   - **C)** Tell them how much they have to live for.
   - **D)** Ask directly and kindly about suicide, stay with them and call for help.

8. Short case. A 3-month-old is brought in by a parent. The infant has a bruise on the cheek and is unusually quiet and feeding poorly. The parent says the baby rolled off a sofa. List, in order, the first four things you would do.

REFERENCE

Appendix A: Answers to Review Questions

Part 1

1. B. Scene safety comes before patient contact because an injured responder becomes a second patient.

2. D. Triage orders patients by how soon they must be seen to stay safe, not by arrival, volume of complaint or expected stay.

3. A. A level is a starting point; a patient who deteriorates is re-assessed, moved to a higher level and escalated.

4. C. Blood pressure can stay normal for a time in shock, so cool, pale skin and fast breathing still call for urgent attention.

5. D. A patient who responds only to a loud voice is rated V (voice).

6. A. A pulse oximeter does not show how well a patient is breathing out carbon dioxide, so a normal reading can coexist with poor ventilation.

7. C. Findings in survey order, the vital signs and what you did belong in Assessment.

8. Model answer: (1) Check the scene for hazards and protect yourself. (2) Take the first look and assess responsiveness. (3) Call for help early, because fast breathing and a reduced response to

voice suggest the patient may worsen. (4) Assess airway and breathing and count the breathing rate. Then continue through circulation, disability and exposure.

Part 2

1. B. No air at the mouth with opposite chest and abdomen movement means complete obstruction with strong effort to breathe.

2. A. The jaw thrust moves the neck less than the head-tilt chin-lift.

3. D. Suction is used only when fluid is seen or heard, in short bursts.

4. C. A conscious adult with a severe blockage needs back blows and abdominal thrusts in the order your training and protocol teach; compressions begin only if the person becomes unresponsive.

5. A. Over-squeezing or giving breaths too fast can push air into the stomach and cause vomiting.

6. D. A patient who has stopped struggling but remains blue and unable to speak may be tiring and failing.

7. B. Call emergency services and help the patient use their own epinephrine auto-injector; delay is dangerous. Giving medicines is outside many roles; follow your scope and medical direction.

8. Model answer: (1) Make the scene safe and call for help early. (2) Check airway and breathing and assess breathing rate, effort, words per breath and mental state. (3) Place the patient in the position that is easiest for breathing, usually upright. (4) Give oxygen only as your scope and medical direction allow. Oxygen needs care because in some patients with long-term lung disease a high concentration of oxygen can make breathing less effective, so the tar-

get is set by protocol or medical direction; do not withhold oxygen from a patient who needs it. Drowsiness is a warning of failing breathing. Giving medicines is outside many roles; follow your scope and medical direction.

Part 3

1. C. In a heart attack, blood flow to part of the muscle is cut off; in cardiac arrest the heartbeat has stopped being effective, and a heart attack may or may not lead to it.

2. A. Serious chest pain is treated as cardiac until a clinician shows otherwise, so you call early and keep the patient still; driving delays the call.

3. B. A flat line in a patient who is awake and talking is most likely a loose lead or an equipment problem, so you read the patient first.

4. D. The recommended adult rate is 100 to 120 compressions a minute.

5. A. Everyone must be clear of the patient before the shock so that nobody receives it.

6. C. A fast heart rate with cool, pale skin is an early sign; warm hands with a strong pulse are not.

7. D. The body compensates early, so blood pressure may be normal while the patient is already in shock.

8. Model answer: (1) Check that the scene is safe, tap the shoulders and shout; he does not respond. (2) Call for help or have someone call emergency services and fetch the AED. (3) Look for breathing for no more than 10 seconds; gasping is not normal breathing, so treat it as cardiac arrest. (4) Begin compressions on the lower half of the breastbone at 100 to 120 a minute, at least 2

inches (5 centimeters) deep, letting the chest recoil. (5) When the AED arrives, switch it on, attach the pads, follow its prompts, make sure everyone is clear, and resume compressions at once after the shock or if none is advised. Compressions come first because they keep a little blood moving to the brain and heart while the AED is fetched.

Part 4

1. B. Heavy bleeding can empty the circulation within minutes, so it is brought forward before the airway; your protocol sets the exact order.

2. D. A fall from a height is a high-risk mechanism; the others are usually low risk.

3. A. Pressure acts on the vessel directly and is available to everyone.

4. C. A tourniquet goes between the wound and the heart, a few inches above the wound and not over a joint.

5. D. Spinal motion restriction means keeping the head, neck and trunk in line and limiting movement.

6. B. A limb that loses any of the three after splinting may have a splint that is too tight.

7. C. The palm of the patient's hand with the fingers is about 1 percent of the body surface.

8. Model answer: (1) Make the scene safe and call for help early, because confusion with a hot body in the heat suggests heat stroke, which is an emergency. (2) Move him to a cool, shaded place, remove excess clothing and check his airway and breathing. (3) Cool his body quickly by the method your protocol directs, such as cool water or wet cloths with fanning, and keep watching

his airway and responsiveness. (4) Give nothing to drink while he is confused, record the times and his responsiveness, and stop cooling when he is cool enough as your protocol directs. This is an emergency because the body temperature has risen to a dangerous level and the brain is affected.

Part 5

1. B. Low blood glucose is a reversible cause that can be checked in minutes.

2. D. Swallowing is the safety check, and the recheck shows whether the treatment worked.

3. A. Treatment windows run from the last time the patient was well, not from when the symptoms were found.

4. C. Some strokes are bleeds, and aspirin could make bleeding worse. Giving medicines is outside many roles; follow your scope and medical direction.

5. B. Protect the head, do not restrain, and note the time.

6. D. Infection with new confusion, fast breathing and poor circulation points to sepsis.

7. A. This is the opioid overdose pattern. Support breathing and call for help first. Giving medicines is outside many roles; follow your scope and medical direction.

8. Model answer: (1) Make the scene safe and call for help early, saying you suspect a serious abdominal emergency with possible bleeding. (2) Check airway, breathing and circulation. (3) Let her find the position of comfort and give nothing by mouth, because she may need surgery or may vomit. (4) Take a full set of vital signs with the times, watch for shock, keep her warm and record what you see.

Part 6

1. C. Appearance reflects tone, interaction and cry.

2. A. Infant choking relief uses back blows and chest thrusts; abdominal thrusts and blind sweeps are avoided.

3. D. This is the alone-rescuer sequence in Entry 6.2; follow your local protocol.

4. B. Left-side positioning moves the uterus off the large vessel.

5. C. Dry the baby, keep the baby warm and check breathing; do not pull on the cord.

6. A. New confusion is a sign of illness.

7. D. Ask directly, stay with the person and call for help; do not promise secrecy.

8. Model answer: (1) Look from a distance with the three-part assessment (how the infant seems, how the infant breathes, the color of the skin) and call for help early if any part is abnormal. (2) Complete the primary survey in the usual order, check the glucose if your role allows, and treat the injury. (3) Speak calmly to the parent and record the explanation in the parent's own words with the time, without accusing anyone. (4) Report your concern through the channels your service requires, because a bruise in an infant who cannot yet roll over may not fit a fall from a sofa; describe it, report your concern, and leave the judgment to those trained to make it, since you do not need proof to report.

REFERENCE

Appendix B: Abbreviations

ABCDE. Airway, Breathing, Circulation, Disability, Exposure.

ACS. Acute coronary syndrome.

AED. Automated external defibrillator.

AVPU. Alert, Voice, Pain, Unresponsive.

BVM. Bag-valve-mask.

COPD. Chronic obstructive pulmonary disease.

CPR. Cardiopulmonary resuscitation.

ED. Emergency department.

EMT. Emergency medical technician.

FAST. Face, Arm, Speech, Time (a stroke screening reminder).

GCS. Glasgow Coma Scale.

MI. Myocardial infarction (heart attack).

RR. Respiratory rate (breathing rate).

SpO2. Oxygen saturation from a pulse oximeter.

TBSA. Total body surface area.

REFERENCE

Appendix C: Reference Values

These tables are a reference copy of values given in the Parts. Normal values change with age, sex, weight, fitness and health, and your service may use different ranges. Follow your local protocol.

Adult resting ranges (Entry 1.4)

Measure	Typical range
Pulse (heart rate)	60 to 100 beats per minute
Breathing rate	12 to 18 breaths per minute
Blood pressure	between 90/60 and 120/80 mm Hg
Temperature	97.7 to 99.1 degrees F (36.5 to 37.3 degrees C)

Children's resting ranges, reference copy (Entry 6.1)

Age	Breathing rate (breaths per minute)	Heart rate awake (beats per minute)
Infant (1 month to 1 year)	30 to 53	100 to 190
Toddler (1 to 2 years)	22 to 37	98 to 140
Preschool (3 to 5 years)	20 to 28	80 to 120
School age (6 to 11 years)	18 to 25	75 to 118
Adolescent (12 to 15 years)	12 to 20	60 to 100

Sources differ, and newborns in the first month have their own ranges; follow the values in your protocol.

Glasgow Coma Scale layout (Entry 1.3)

Response	Score and description
Eyes	4 opens spontaneously; 3 opens to sound; 2 opens to pressure; 1 none
Verbal	5 oriented; 4 confused; 3 words only; 2 sounds only; 1 none
Motor	6 obeys commands; 5 localizes pain; 4 normal flexion; 3 abnormal flexion; 2 extension; 1 none

The three scores are added to give a total from 3 to 15. A score of 15 is the highest.

Adult burn size by the rule of nines (Entry 4.5)

Region	Share of body surface
Head and neck	9 percent
Each arm	9 percent
Front of trunk	18 percent
Back of trunk	18 percent
Each leg	18 percent
Genital area	1 percent

The palm of the patient's hand, with the fingers, is about 1 percent of the body surface.

REFERENCE

Glossary

A

Abdominal thrusts. Quick inward and upward presses on the upper abdomen used to dislodge an object from the airway of a conscious adult or child over 1 year who is choking.

Acute coronary syndrome (ACS). A group of conditions in which blood flow to part of the heart muscle falls suddenly; the heart attack is the best known.

Airway. The passage that carries air from the nose and mouth to the lungs.

Altered mental status. A change in alertness, thinking or behavior compared with the person's usual state.

Anaphylaxis. A severe, whole-body allergic reaction that can narrow the airway and lower the blood pressure within minutes.

Angina. Chest discomfort that appears when the heart muscle needs more oxygen than a narrowed artery can supply, and that usually eases with rest.

Asthma. A long-term lung condition in which the airways narrow and swell in response to triggers, causing wheezing, cough and breathlessness.

AVPU. A quick scale of alertness: Alert, responds to Voice, responds to Pain, Unresponsive.

B

Bag-valve-mask (BVM). A face mask, one-way valve and squeezable bag used to move air into the lungs of a patient who is not breathing adequately.

C

Capillary refill time. The time for color to return to a fingertip after it has been pressed; a long time suggests poor blood flow to the skin.

Cardiac arrest. The sudden stop of effective heartbeat; the patient is unresponsive and not breathing normally.

Central cyanosis. A blue or gray tint to the lips and tongue caused by a low level of oxygen in the blood; a late sign.

Compensated shock. The early stage of shock, in which the body keeps the blood pressure up by speeding the heart and narrowing vessels.

Concussion. A brain injury caused by a blow or jolt that can cause confusion, headache or memory loss, with or without loss of consciousness.

COPD. Chronic obstructive pulmonary disease, a long-term lung disease with narrowed airways and damaged lungs.

CPR. Cardiopulmonary resuscitation: chest compressions, with or without rescue breaths, to keep blood moving when the heart has stopped.

Crowning. The moment during birth when the baby's head is visible at the vaginal opening.

D

De-escalation. Calm words and actions used to reduce a person's agitation and the risk of harm.

Defibrillation. A shock delivered to the heart to correct a rhythm that prevents effective pumping.

Delirium. A sudden change in thinking and attention caused by illness, medicines or injury; a sign that the patient is unwell.

Direct pressure. Firm pressure on a bleeding wound with a clean dressing or cloth; the first step in controlling bleeding.

Dislocation. A joint forced out of its normal position.

E

Epiglottis. A flap of tissue that covers the entrance to the larynx when swallowing.

Epinephrine (adrenaline). The first-line medicine for anaphylaxis, carried by some patients in a prefilled auto-injector. Giving medicines is outside many roles; follow your scope and medical direction.

Exposure. The last step of the primary survey: exposing the body as needed to look for injuries while protecting dignity and warmth.

F

First look. A rapid general look at the patient, lasting about 30 seconds, made before the lettered checks of the primary survey.

Fracture. A break in a bone.

Frostbite. Freezing of skin and tissue, most often in fingers, toes, ears and nose.

G

Glasgow Coma Scale (GCS). A score from 3 to 15 based on eye opening, verbal response and motor response, used to describe level of consciousness.

H

Heat exhaustion. Illness from heat with heavy sweating, tiredness, headache and nausea.

Heat stroke. A dangerous rise in body temperature with confusion, collapse or seizure; an emergency.

Hemorrhage. Severe bleeding.

Hypoglycemia. Low blood glucose; for many people with diabetes, below 70 mg/dL (3.9 mmol/L).

Hypothermia. An abnormally low body temperature that begins with shivering and clumsiness and can progress to drowsiness.

Hypoxia. A low level of oxygen in the tissues.

J

Jaw thrust. Lifting the jaw forward with the fingers behind its angles to open the airway while moving the neck as little as possible.

L

Larynx. The voice box, at the top of the windpipe.

Local protocol. The written procedures of your service, school or hospital that set what you do in a given situation.

M

Mechanism of injury. How an injury happened, which suggests which injuries to look for.

Medical direction. Guidance from a physician or other authorized clinician that sets or approves what responders do.

N

Naloxone. A medicine that reverses the effects of an opioid overdose. Giving medicines is outside many roles; follow your scope and medical direction.

Non-rebreather mask. An oxygen mask with a reservoir bag that delivers a high concentration of oxygen to a breathing patient.

O

Opioid. A drug class that includes some pain medicines and illegal drugs; overdose slows or stops breathing.

Oxygen saturation (SpO2). The percentage of hemoglobin carrying oxygen, estimated by a pulse oximeter.

P

Pharynx. The throat, behind the nose and mouth.

Pneumonia. An infection of the lungs.

Primary survey. The fixed sequence of checks (airway, breathing, circulation, disability, exposure) that finds immediate threats to life.

Pulse oximeter. A clip-on device that estimates oxygen saturation and pulse rate.

R

Recovery position. A position on the side that keeps the airway clear in a breathing patient who is unresponsive.

Respiratory distress. Difficult or labored breathing.

Respiratory failure. The stage at which breathing can no longer meet the body's needs.

Rule of nines. A method of estimating the share of the body surface burned, using regions of 9 percent or multiples of 9 percent, with 1 percent for the genital area.

S

Scope of practice. The tasks you are legally authorized and trained to perform.

Secondary survey. A slower head-to-toe examination and focused history carried out after immediate threats have been treated.

Sepsis. A life-threatening response to infection that causes organ dysfunction.

Shock. Failure of the circulation to deliver enough blood and oxygen to the organs.

Spinal motion restriction. Keeping the head, neck and trunk in line and limiting movement when a spinal injury is possible.

Splint. A support that holds an injured limb still.

Standard precautions. The basic infection-control measures used with every patient, including hand hygiene and protective equipment.

Stridor. A high-pitched sound, most often heard on breathing in, that suggests a significant partial blockage of the airway.

Stroke. Damage to part of the brain from a blocked or burst blood vessel.

Suction. Removal of liquid or small particles from the mouth and throat with a device.

T

Tourniquet. A band tightened around a limb to stop blood flow to a severely bleeding wound.

Trachea. The windpipe, running from the larynx into the chest.

Transient ischemic attack. A brief blockage of blood flow to the brain with stroke-like signs that clear on their own; a warning of stroke.

Triage. Sorting patients by how soon they must be seen.

V

Vital signs. Heart rate, breathing rate, blood pressure and temperature, usually with oxygen saturation added.

W

Wound packing. Pushing gauze into a deep wound onto the bleeding source and pressing over it.

REFERENCE

Index

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